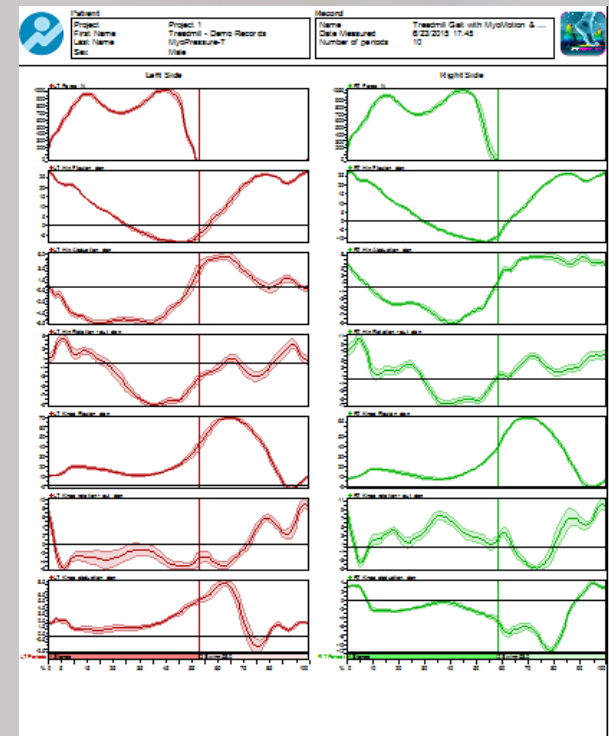
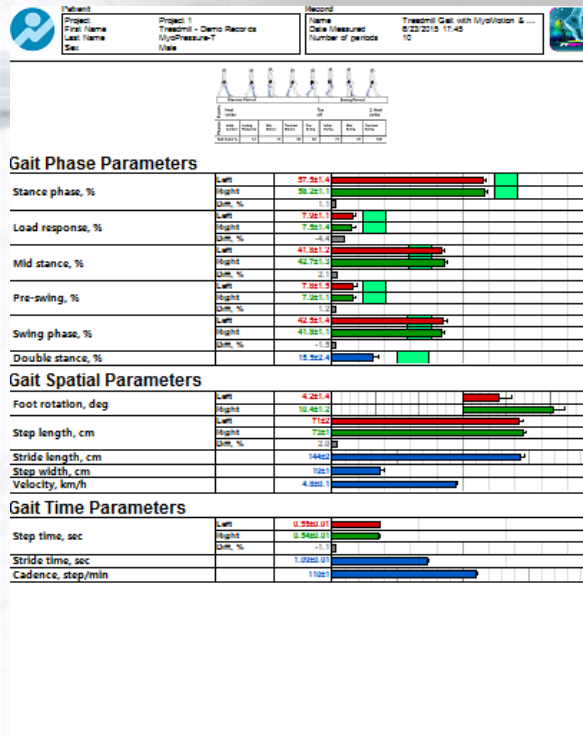
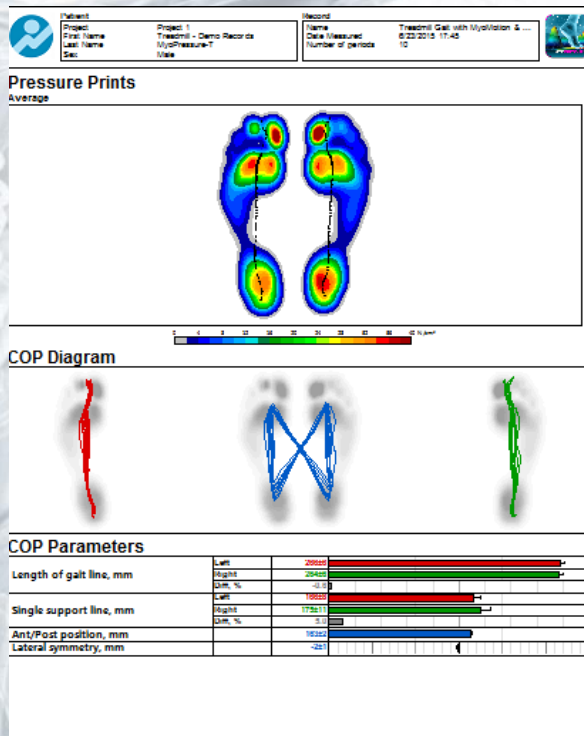
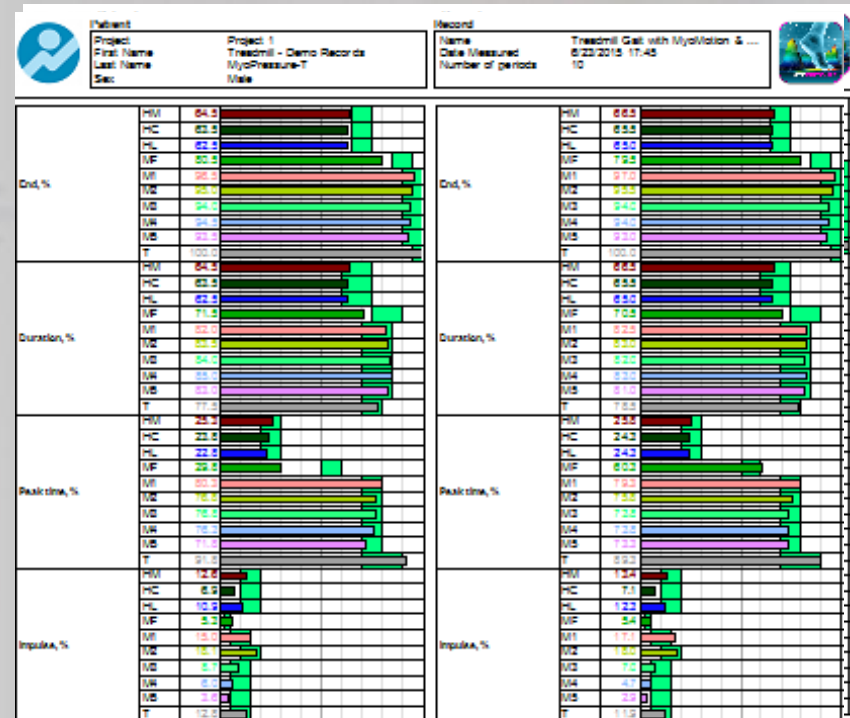
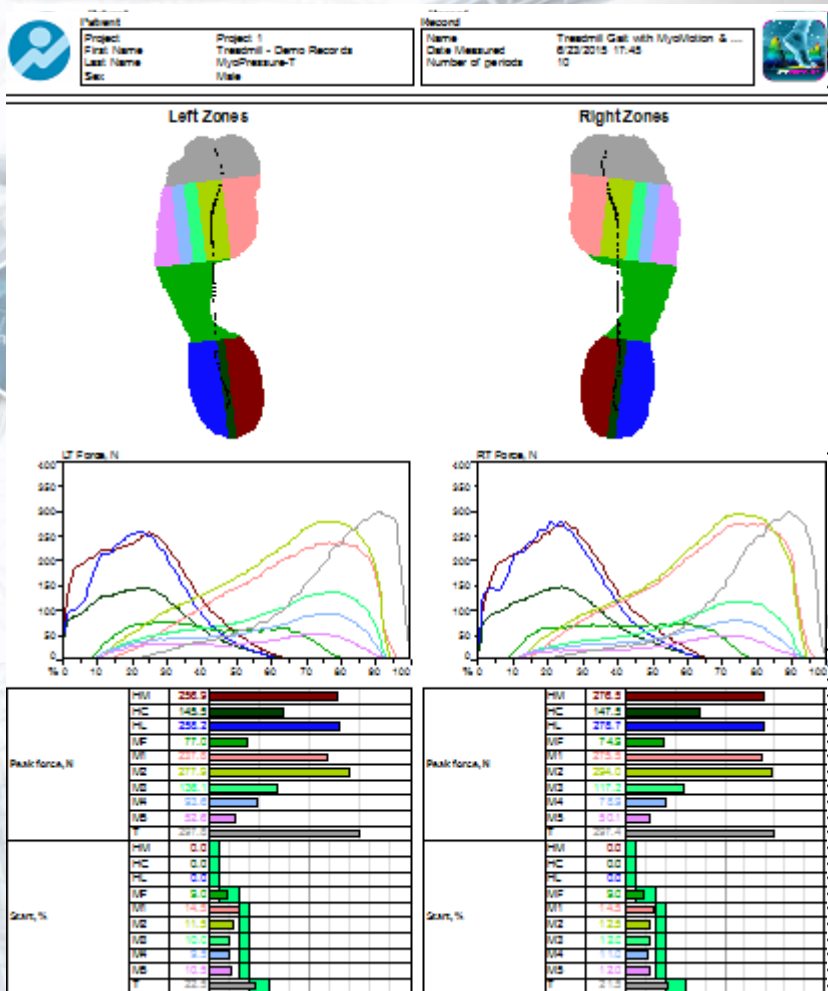


What is in the Reports and Why Should We Care?





What is in the Advanced Reports?

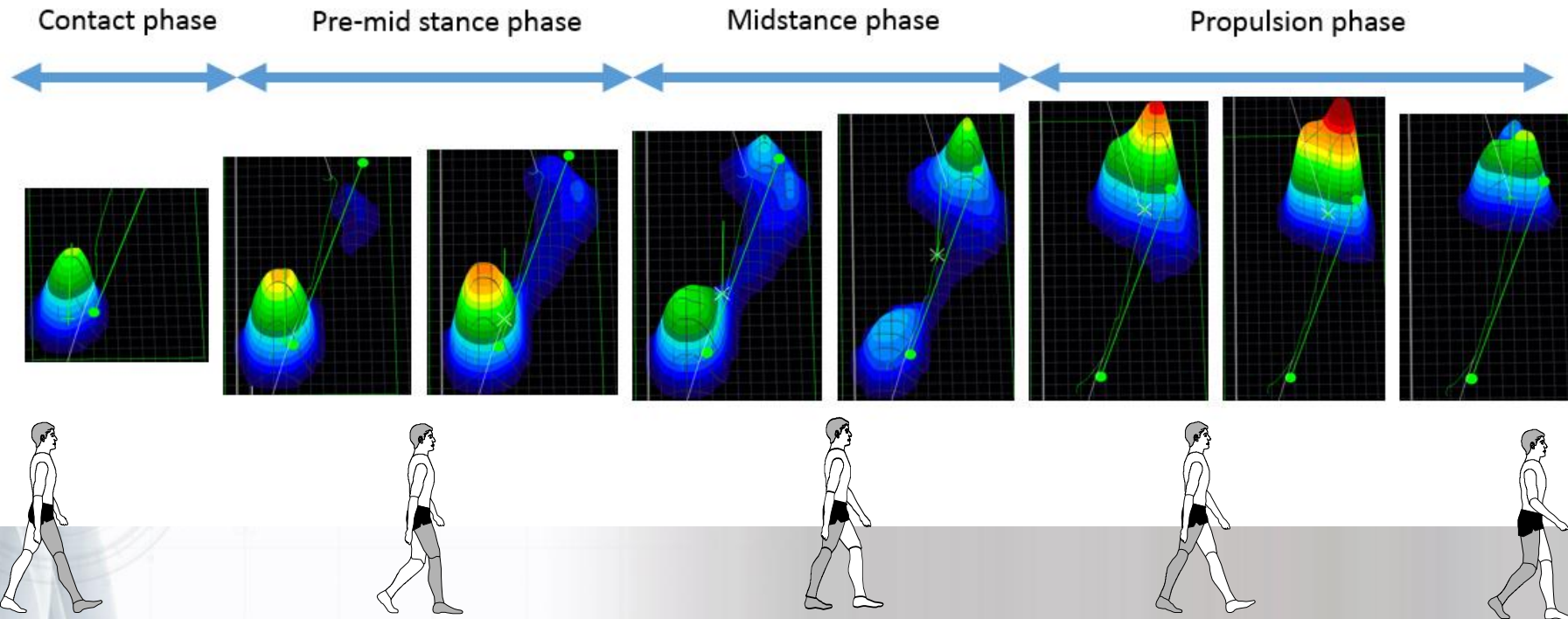
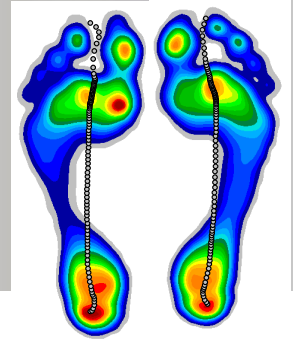




DYNAMIC BILATERAL GAIT

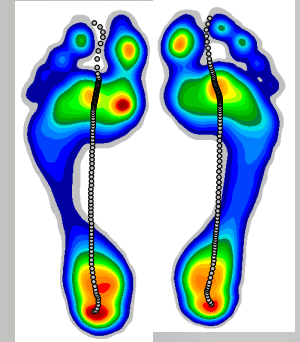
Average Pressure Plots with Summed Gait line Center of Pressure Transition

Normal Gait Sequence:



Average Pressure Plots with Summed Gait line Center of Pressure Transition

Normal Gait Sequence:

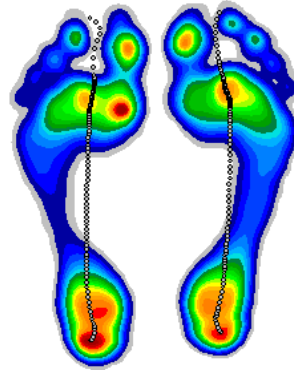


- ☐ Check for pressures of the 1st metatarsal head.
- ☐ Ideally, you want moderated pressures at the metatarsal heads 1-5, with good pressures sub 1st met head bilateral.
- ☐ Equal loss of pressure at the heel bilaterally is ideal
- ☐ If one foot is accelerating faster than the other foot, you will often see a loss of heel pressure on the faster foot. This is often (but not always) an indication of a short limb.

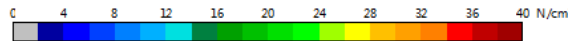
Average Max Pressure Plot

Pressure Prints

Average



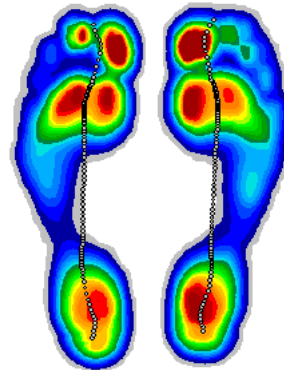
Maximum pressure values for every step are averaged together.



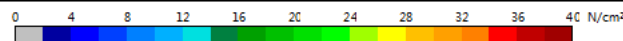
Peak Max Pressure Plot

Pressure Prints

Maximum



Highest pressure values from any step during recording are compiled.

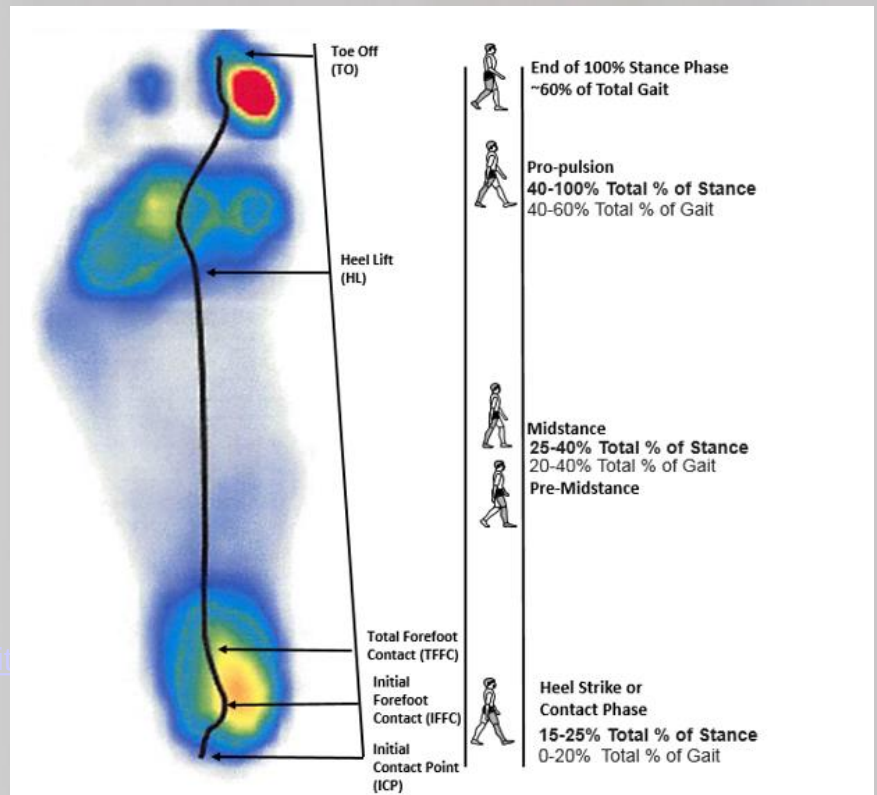


Center of Pressure Overlay

Biomechanics is all about physical events happening at certain times and in a specified order. We have a suite of tools that provide clinicians the opportunity to gather, integrate, and compare relevant patient data to aid in diagnosis, treatment planning, and proof of treatment efficacy needed to meet the principals of evidence based medicine.

The **COP display** and the **Average Pressure Print** combined give insight to this story so you have information about your patient and the normative.

<http://www.golfmed.net/orthotics/gait>



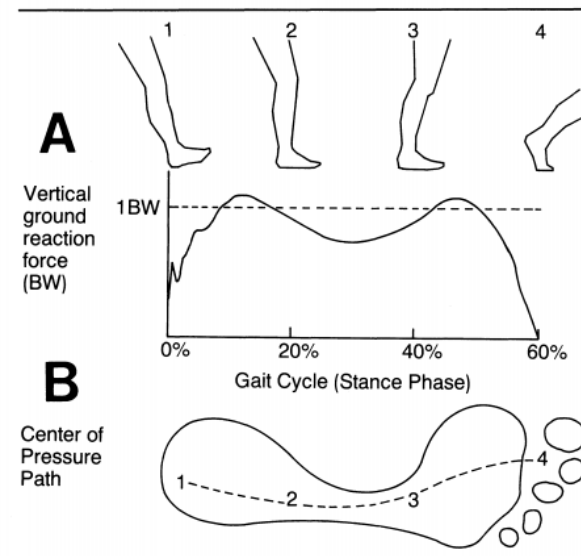
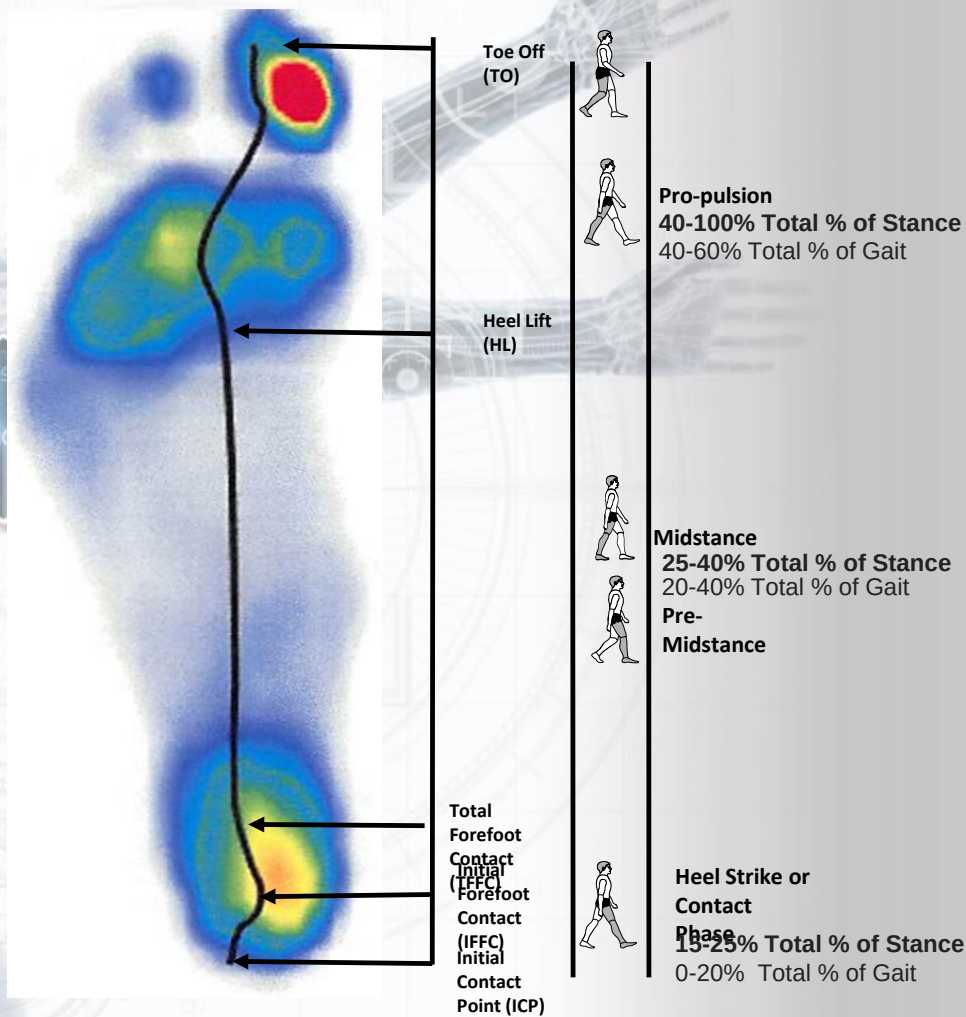
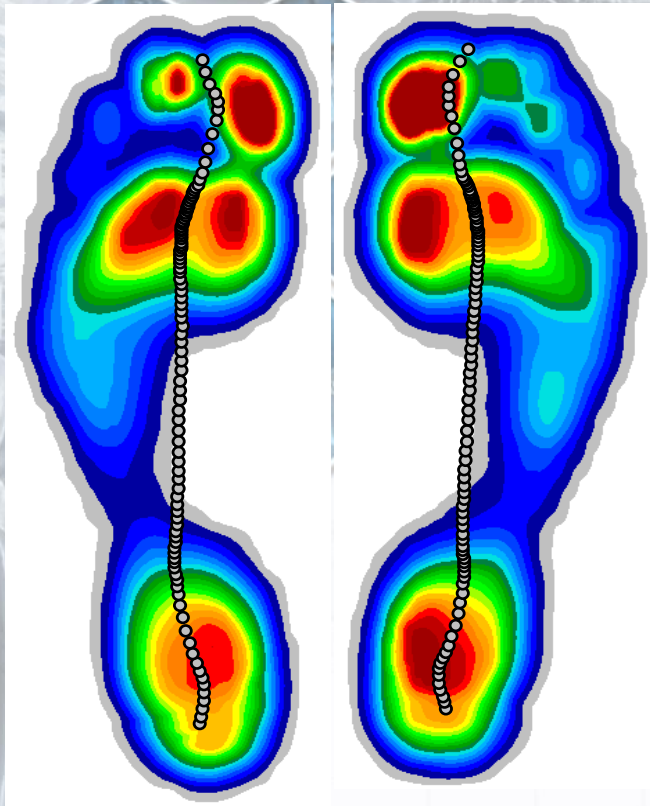


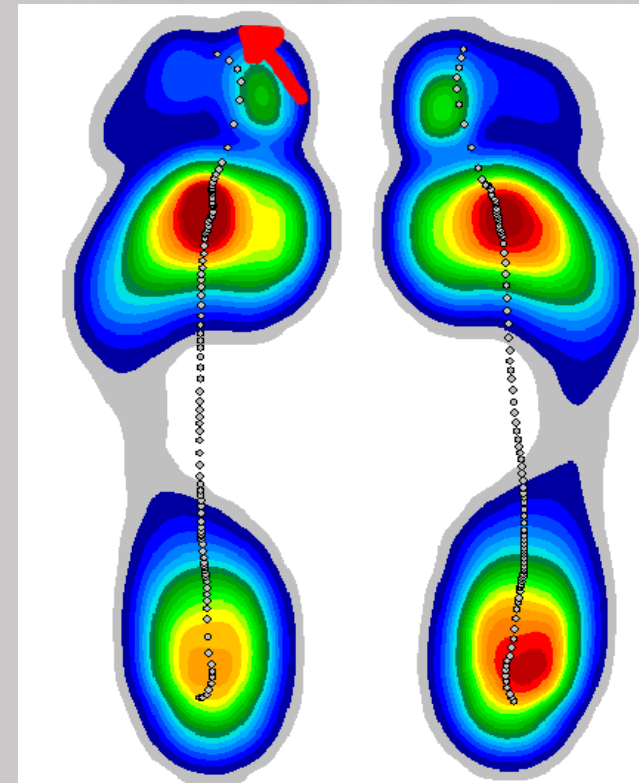
Fig. 3. Ground reaction forces (GRFs) beneath foot during walking: (A) Graph of classic vertical GRF during stance phase of gait cycle (BW = body weight; 1 = heel-strike; 2 = foot flat; 3 = midstance; 4 = toe-off); (B) path of the center of pressure, which represents series of instantaneous centroids of GRF during walking.

How do I use the COP?

- ❑ Progression of the Center of Force (black dotted and trailing line).
- ❑ The ideal progression: midline on the foot, from the heel to the 2nd metatarsal head, or slightly lateral of this point.



Last 4-5 contact COP points determine where the last moment is at the medial distal portion of the hallux. With a bunion, the hallux moves laterally at that last moment, exaggerating a bunion (hallux abductor valgus).

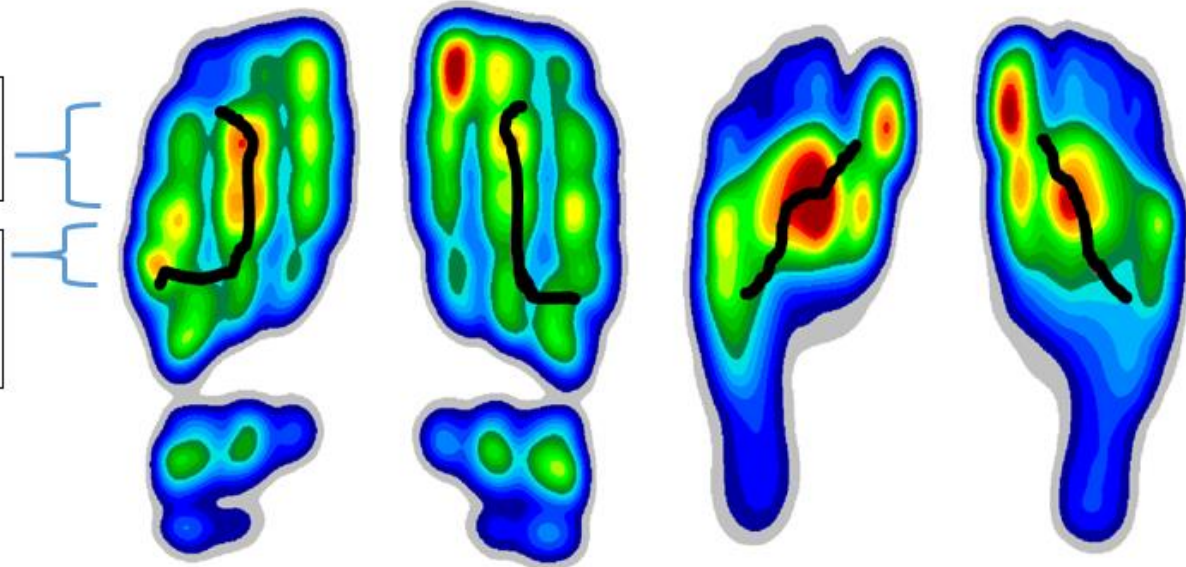


Average Pressure Plots with Summed Gait line Center of Pressure transition

Running Sequence

Late and retrograde supination or lateral, outwards movement of the COP – where is the energy going if it is not forwards?

Exaggerated and accelerated pronation or medial, inwards movement of the center of pressure – what is the ankle and in turn the knee and abductors forced to do here?



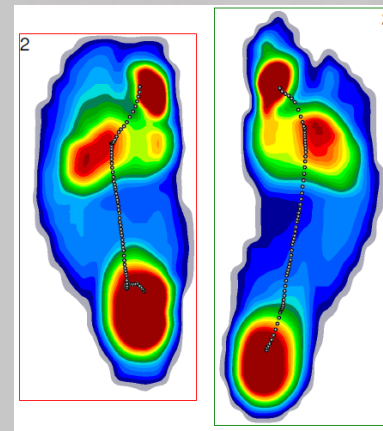
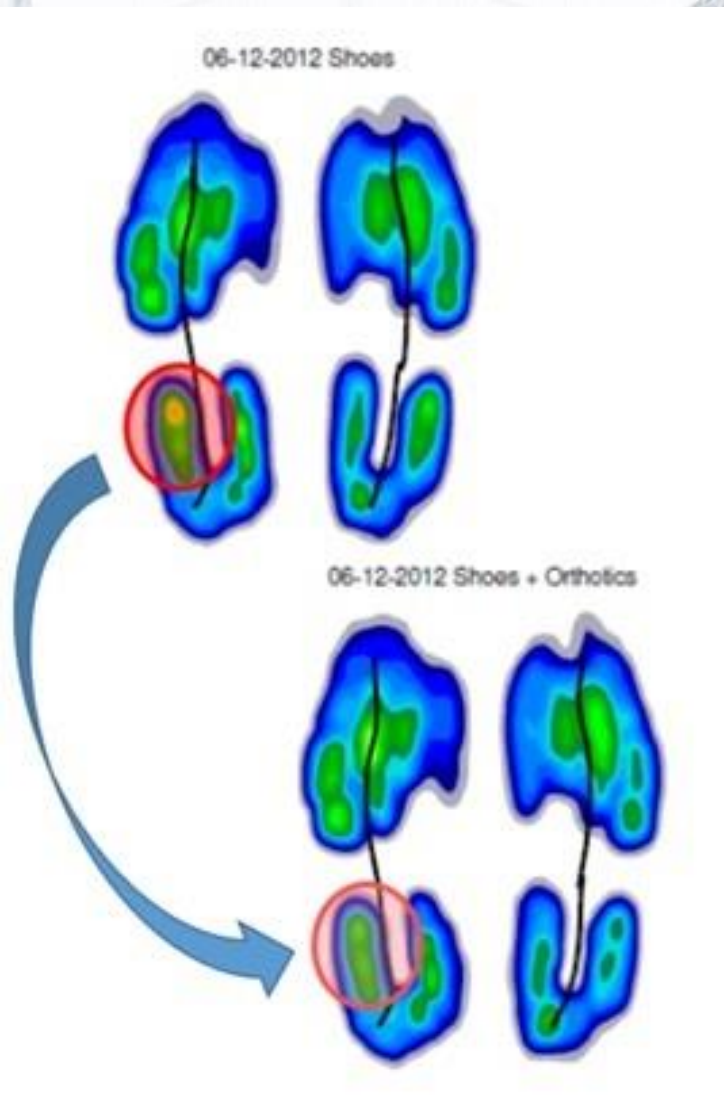
Shod Run 12mph

Barefoot Run 12mph

Average and Peak Max Norms

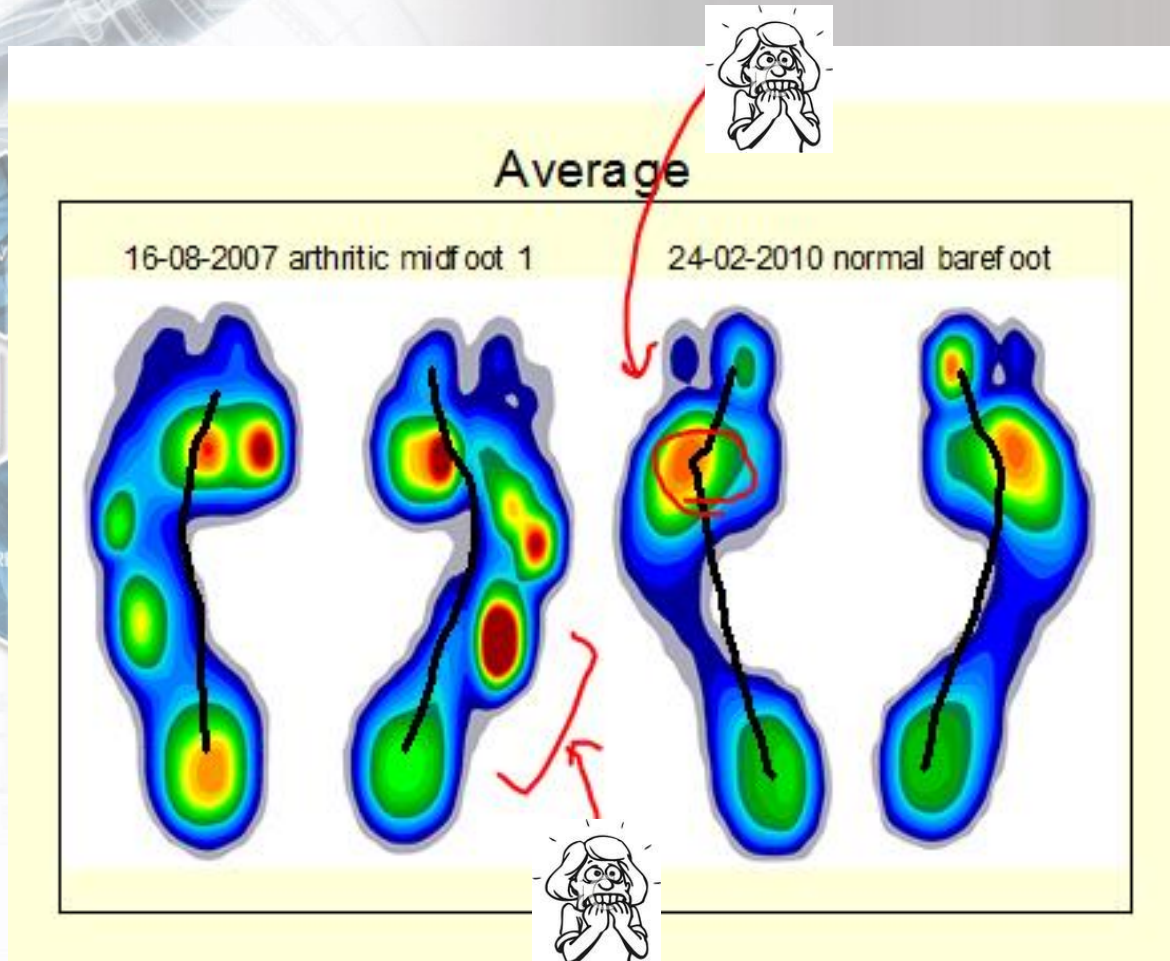
Pressures during walking should not exceed

- 40 N/cm² under heel
- 55 N/cm² under front part of foot.
- The force exerted on foot should be supported all the toes.

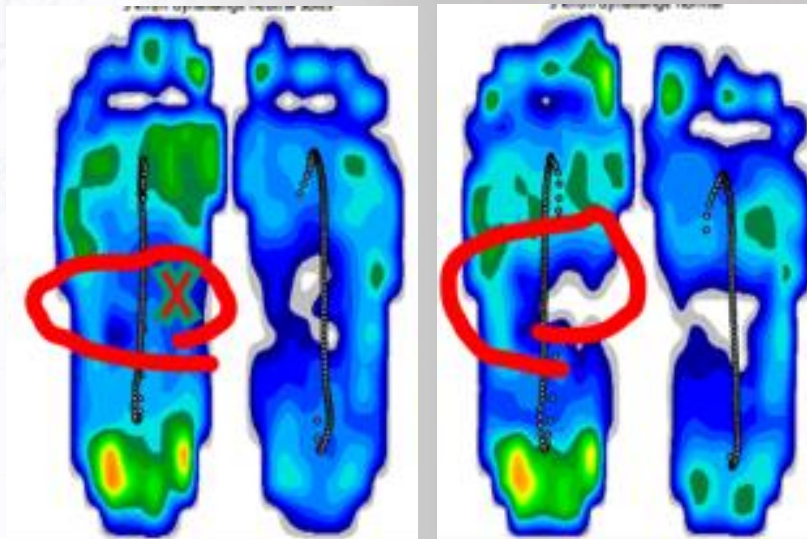
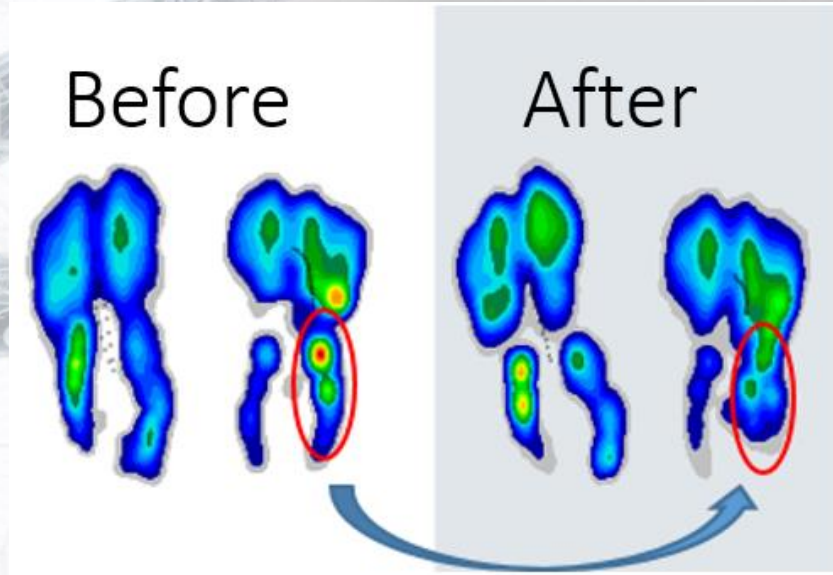


Deviations? Surgery required?

Severe change of direction at the 2nd met? Forefoot Varus issue?

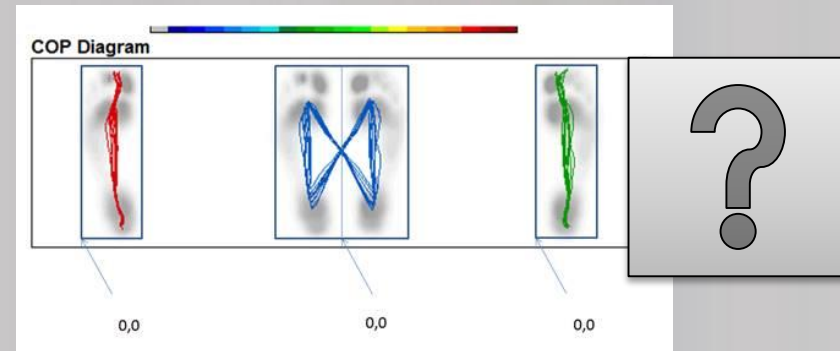
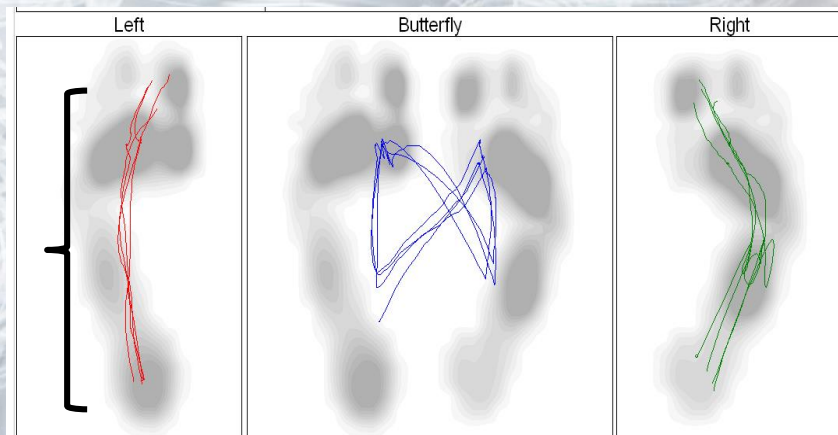


Orthotic Intervention?



COP Diagram - Stance Phase and Single Support Phase – Butterfly

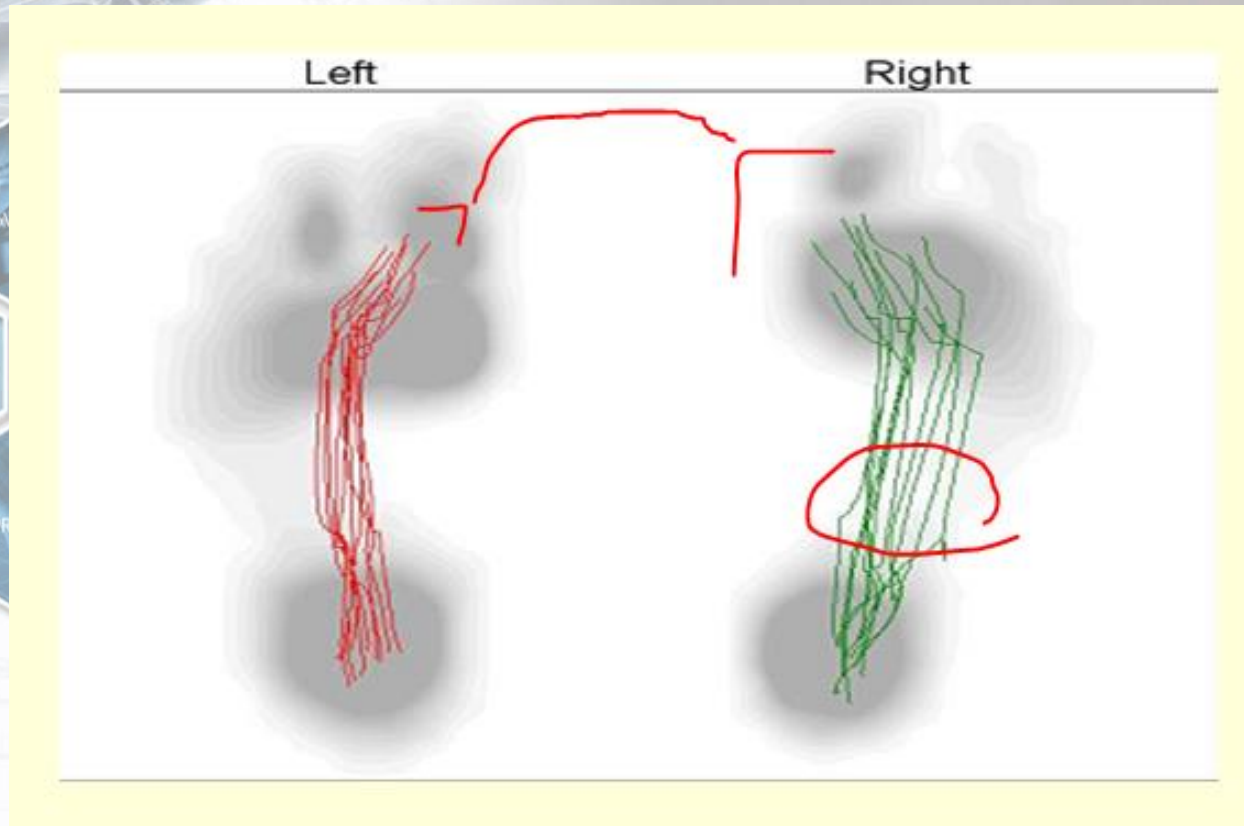
Gait Line Length



For advanced users: the export displays coordinates of COP using above axes.

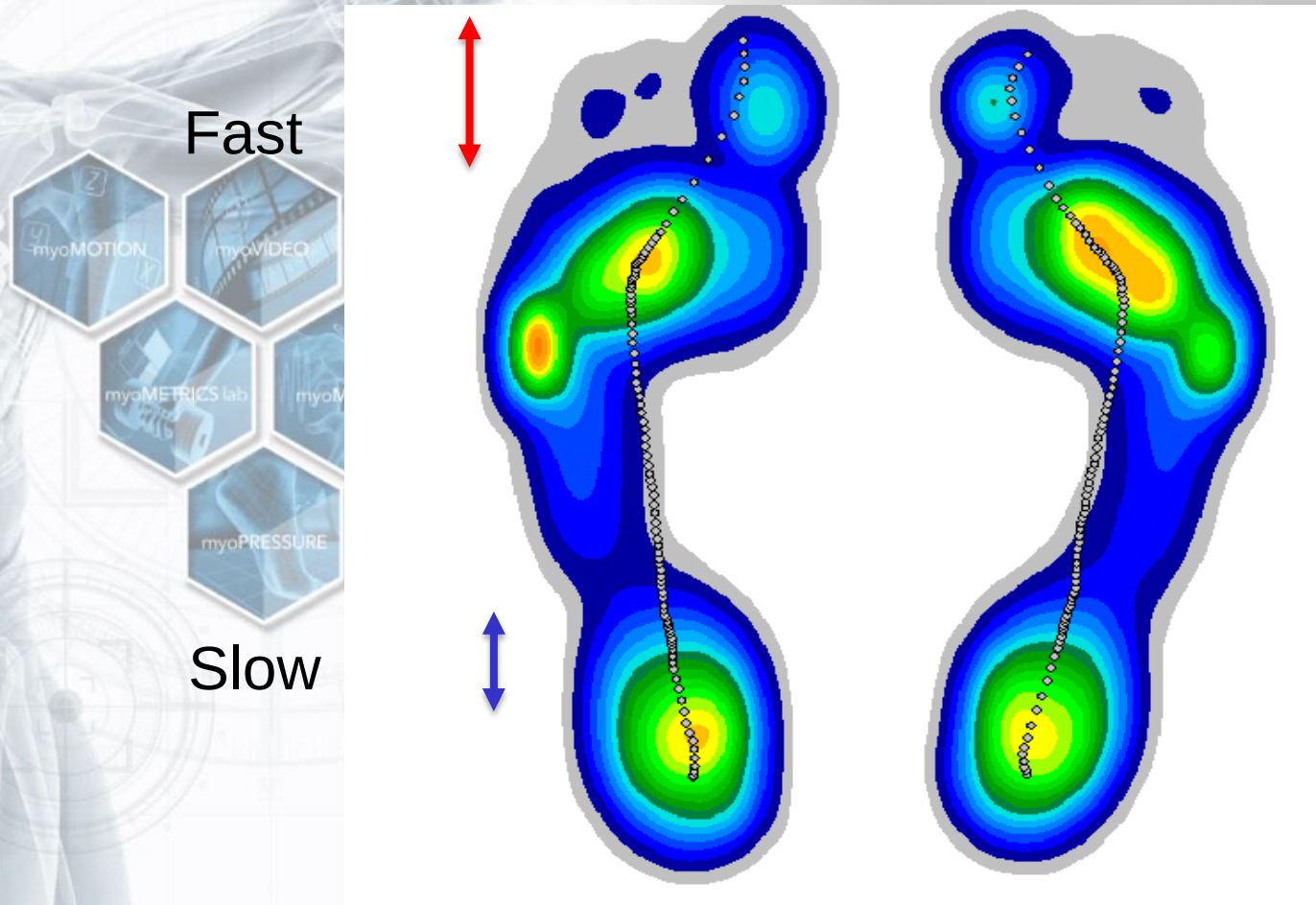
- ❑ Length of the line describes the course of the pressure center only when individual ground contacts of one side of the body are taken into consideration.
- ❑ Comprises the course of the Maximum Pressure Plot (MPP) for all steps recorded on one side of the body.

Stance Phase COP: Symmetrical? Compensation?



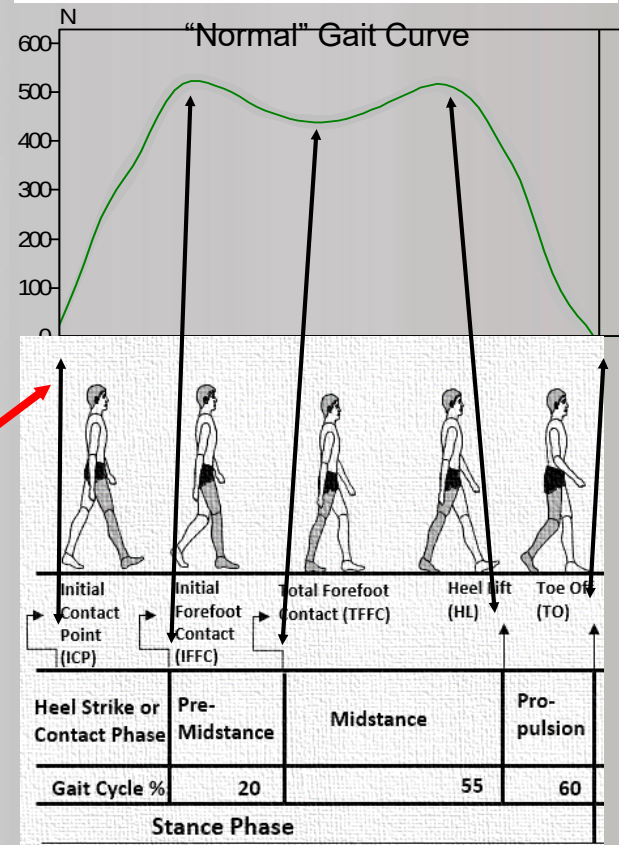
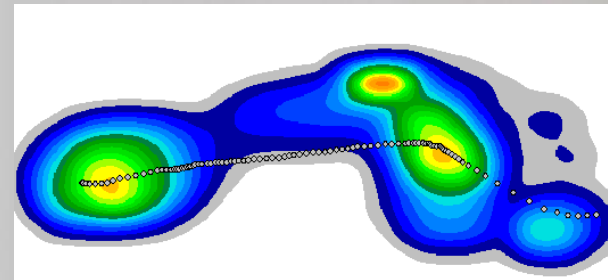
The lateral variability parameter quantifies this.

Stance Phase COP: Speed and Movement of the COP Gait Line

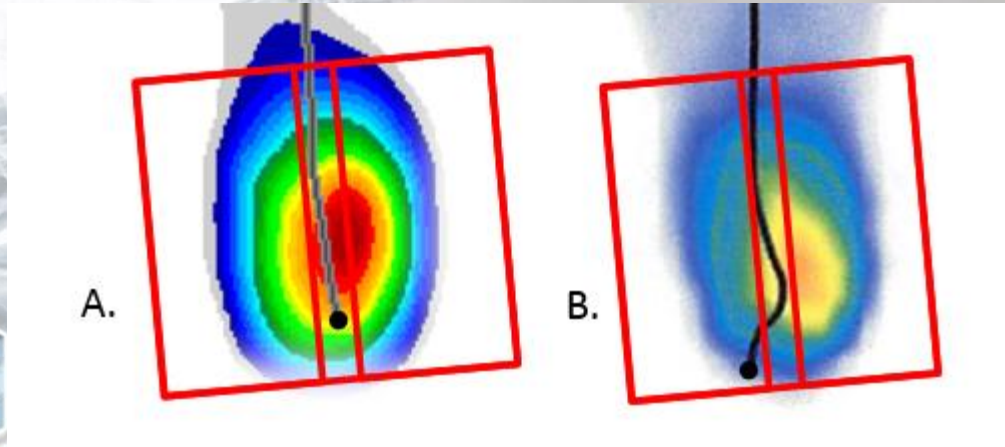


Cross check the influences of COP speed by looking at impulse values.

Where is the Initial Contact?



Where is the Initial Contact?



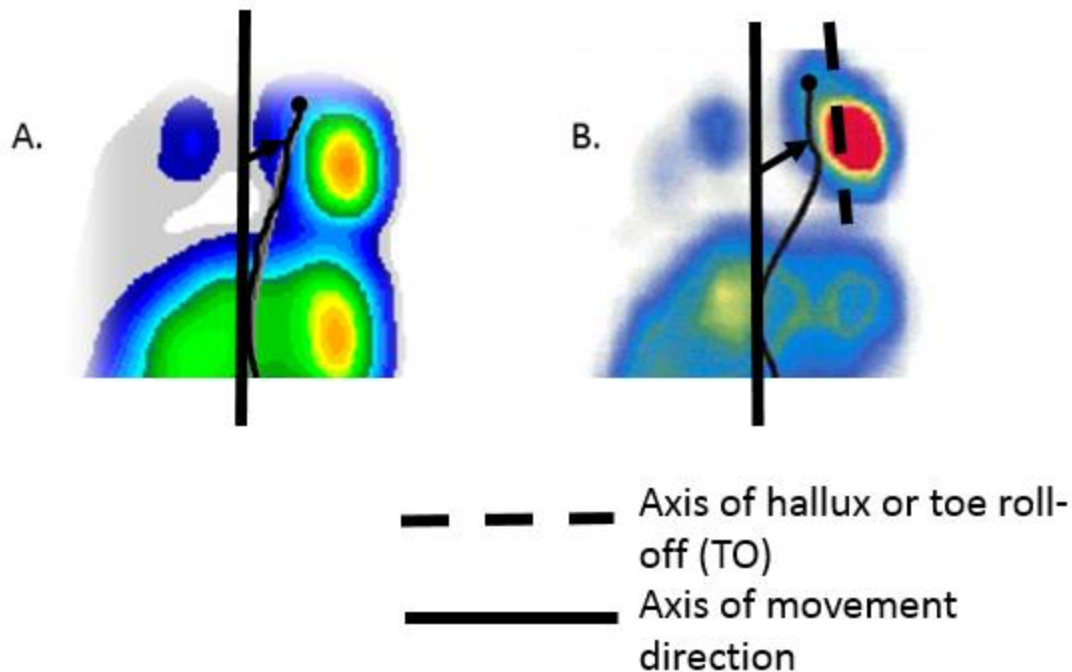
Importance of time/impulse data:

To the clinician, this means that I have 3 ways (or combination thereof) to fix a patient's gait concerns,

1. **Decrease the pressure (dampen),**
2. **Decrease the "assault" time (canting, arch height, increase efficiency, phasic adjustment) &**
3. **Broaden the area of that impulse (off-load)**

FDM-T helps me make these choices

Where is Toe-Off?



- ❑ What are the key importance's of Frontal plane motion?
- ❑ Prepping for ground clearance and Limb progression?
- ❑ What muscles prep this movement?

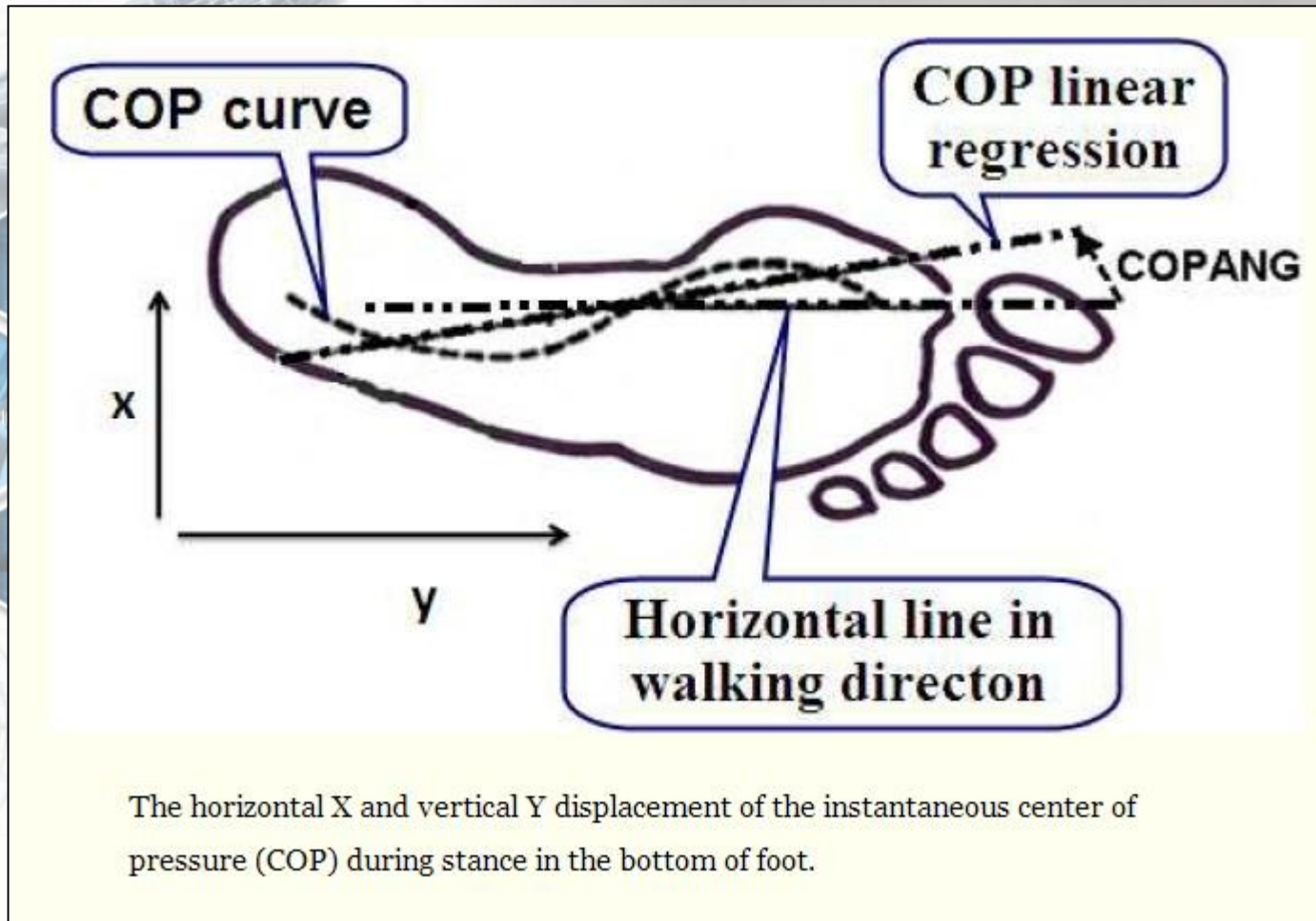
What else can I do with COP?

Gait Line Norms and Symmetry

- ❑ The +/- 1.96 SD limit suggested that non-pathological gait falls within 12-16% asymmetry for gait-lines.
 - ❑ plantarflexor weakness = 30-50% of foot length to toe-off
 - ❑ dorsiflexor weakness = initial contact to 25-40% of foot length, and after 55-80% of foot length.
young, healthy able-bodied subject populations
 - ❑ used to suggest non-pathological gait-line asymmetry pattern limits
-
- http://www.researchgate.net/publication/23681105_Expressing_gait-line_symmetry_in_able-bodied_gait

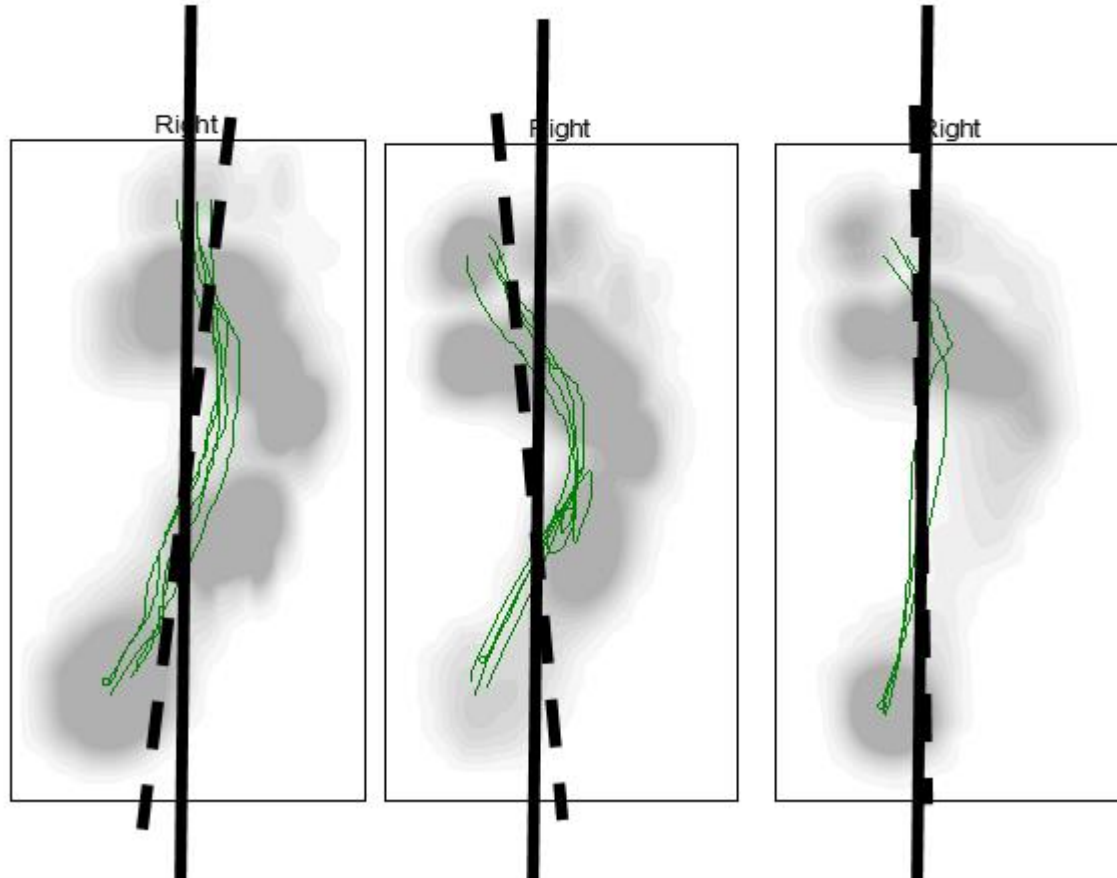
What else can I do with COP?

COP Shape



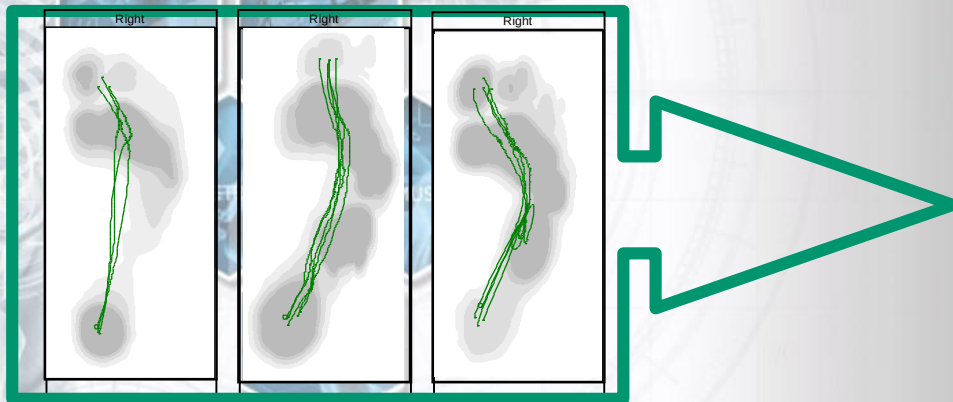
COP linear regression angle between forward direction and linear regression line.

Shape defines shoe last



Foot progression angle:

The angle of deviation of the long axis of the foot from the line of progression (typically 7-10 degrees).



- ✓ The larger this angle, the more lasted your shoe type should be.

COP Linear Regression line and Pathologies

A. The horizontal angle measured from center of pressure linear regression was positive in the control group (4.6 ± 2.4)

B. But negative in the patient group (-2.3 ± 1.6)

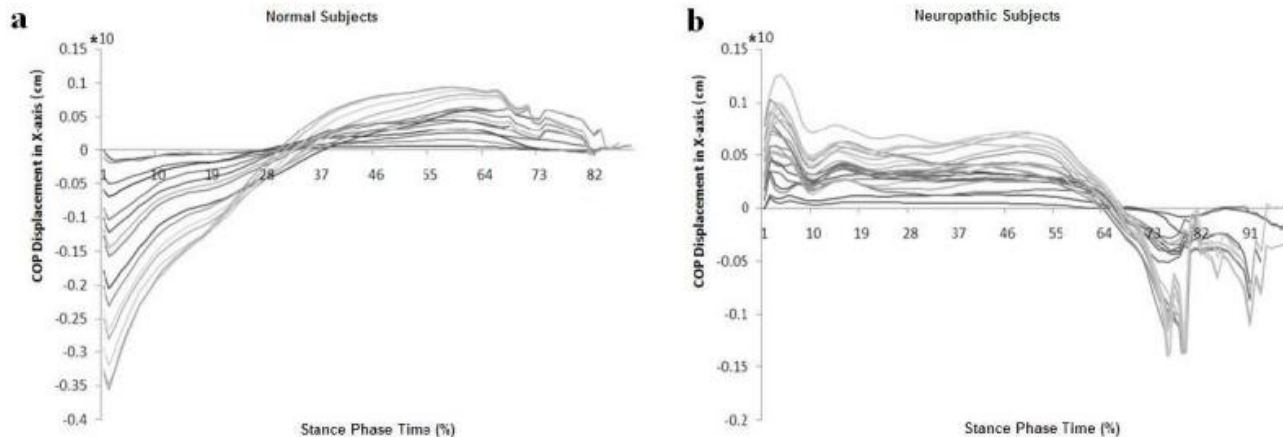


Diagram shows center of pressure patterns during a normal stride in X-axis direction during a normal stride among (a) normal and (b) neuropathic subjects with drop foot

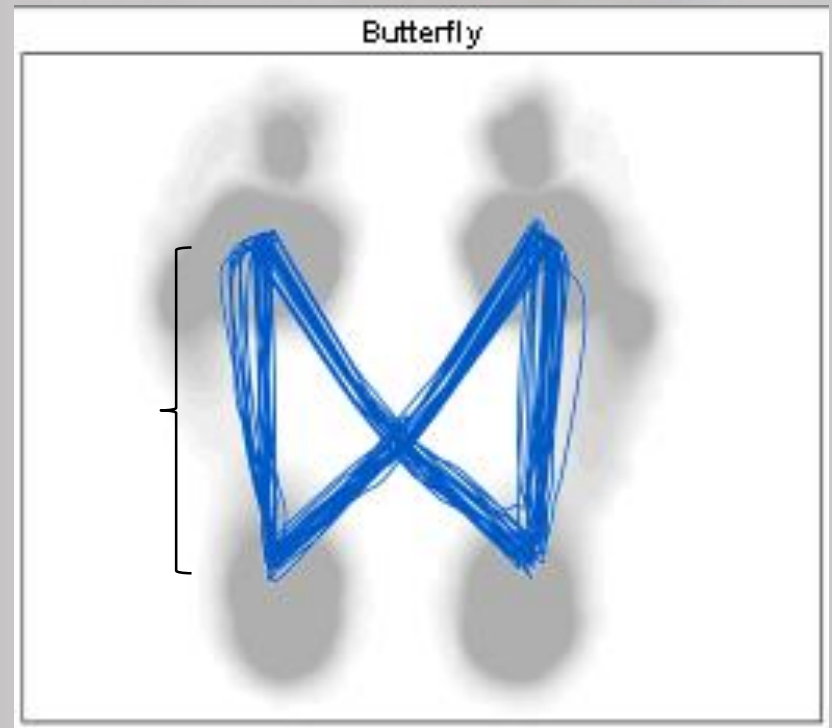
- ✓ A 4.6 degree line (external from axis of foot) was measured for normal patients but patients with drop foot has a -2.3 linear regression line in this study (internal from axis of foot).

Butterfly Diagram: Single Support Line

The Butterfly or Cyclogram shows the COP for both feet combined. It is very useful for judging gait symmetry.

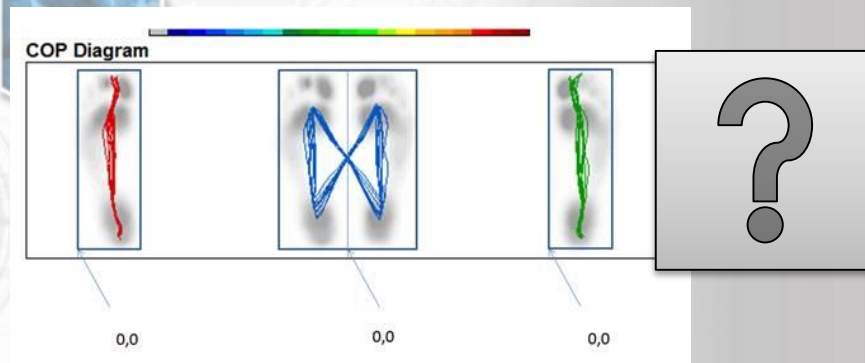
❑ Average length of the lines that describe the course of the pressure center, when all ground contacts are taken into consideration.

✓ 4% difference or lower is optimal

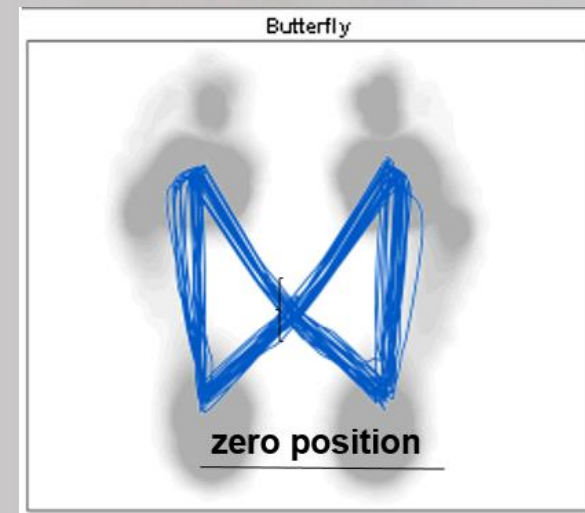


Butterfly Diagram: Anterior / Posterior Position

Describes the shift (forwards), of the intersection point of the course of the COP in the cyclogram display taking all the steps into consideration.

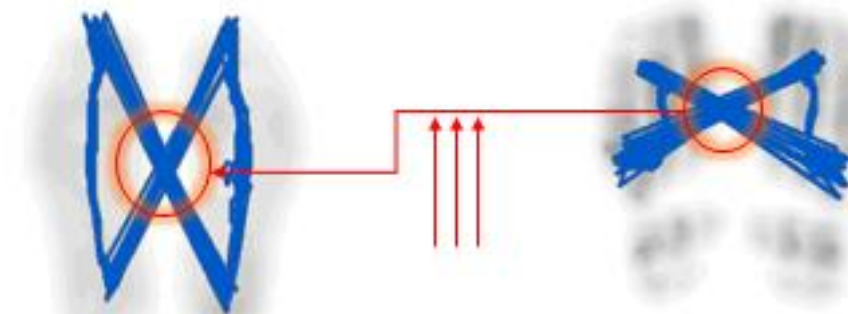
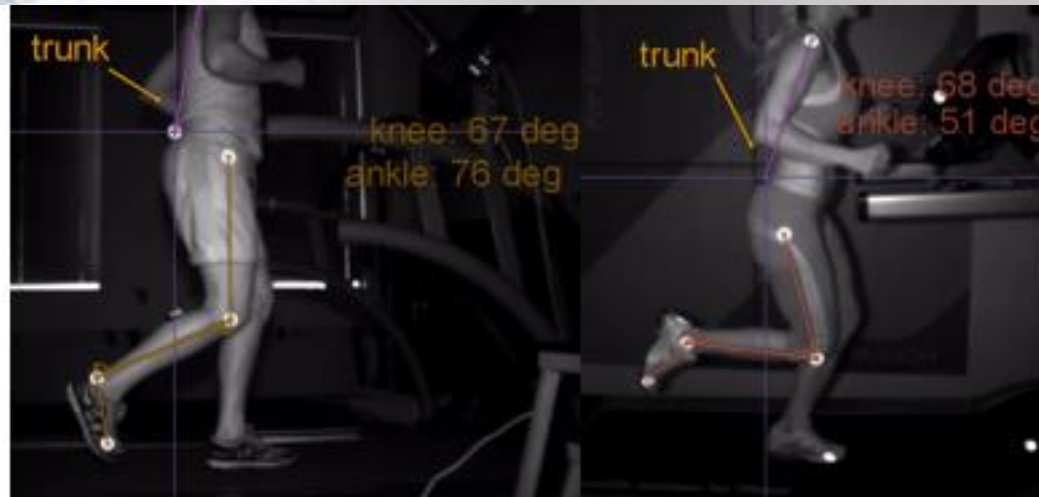


For advanced users: the export displays coordinates of COP using above axes



Why does it matter?

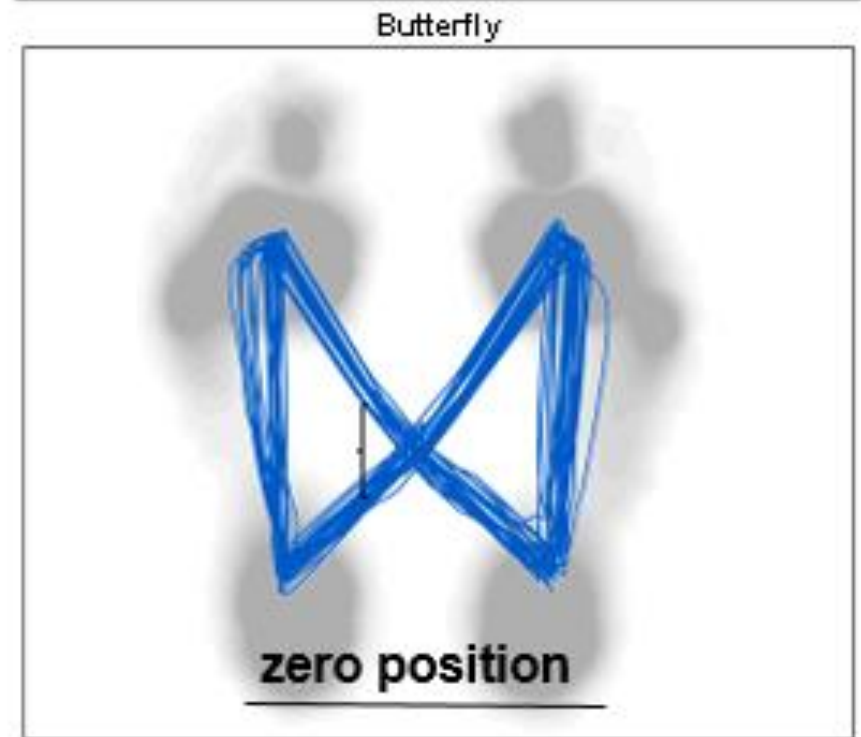
- ❑ Where does the COG/COP crossover occur?
- ❑ Where is the rest of the kinematic chain?
- ❑ Does it follow?



COP moves forward:
Center of Mass maintains momentum – and possibly minimizes loading rate.

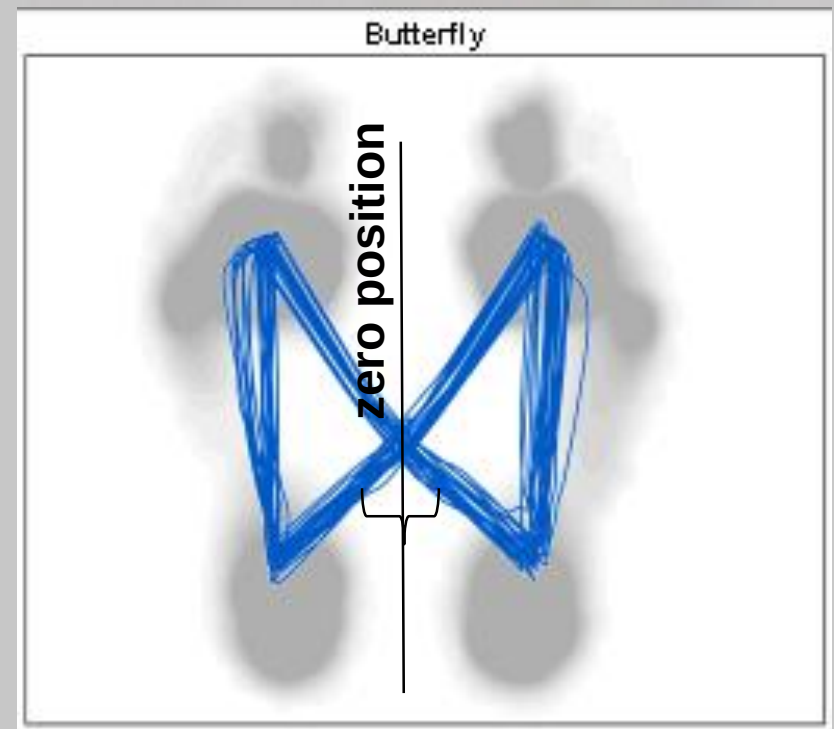
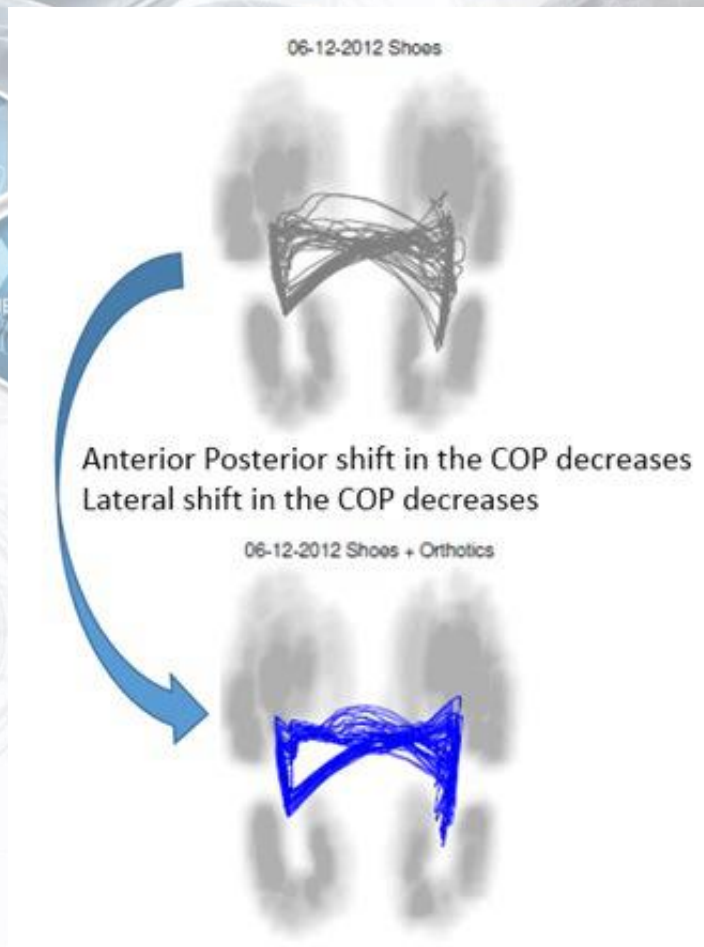
Anterior / Posterior Variability

Standard deviation in the anterior / posterior position when taking all steps into consideration.



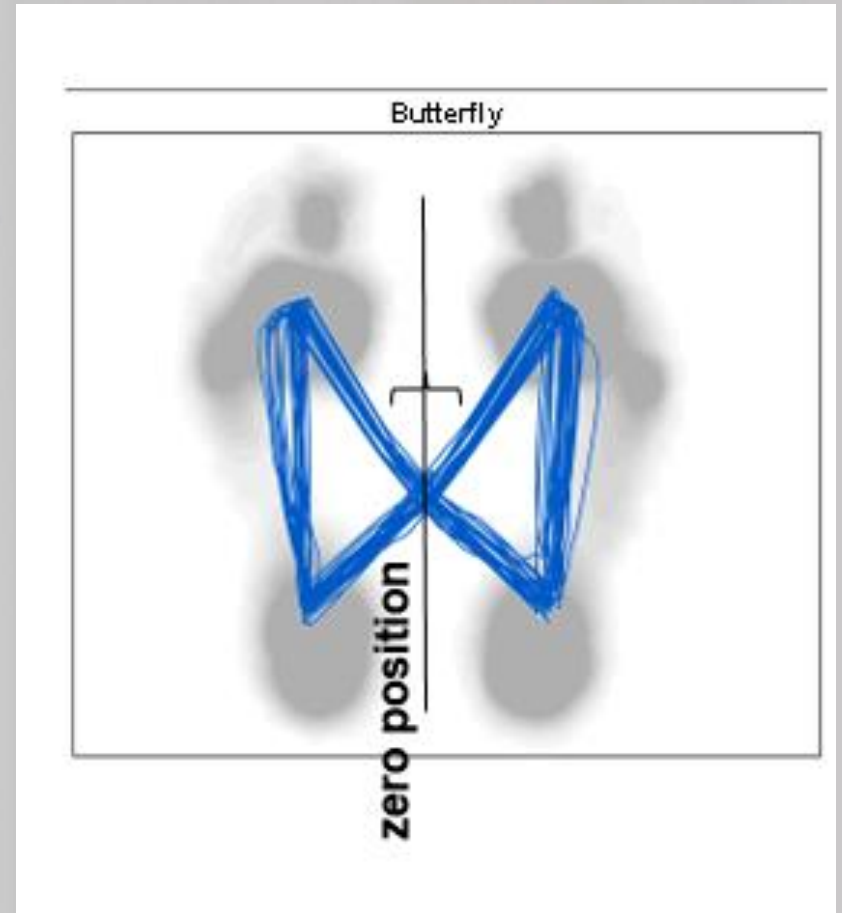
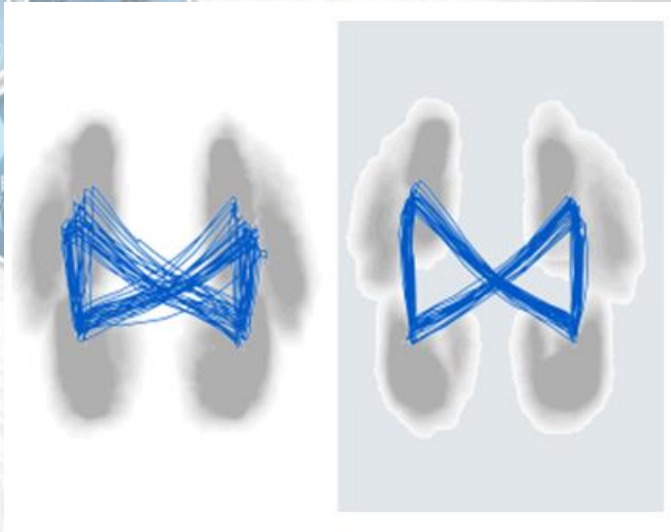
Lateral Symmetry

- ❑ Describes the shift to the left / right of the intersection point of the course of the CoP in the cyclogram display taking all the steps into consideration.
- ❑ A negative value indicates a shift to the left side, and a positive value a shift to the right side.



Lateral Variability

Standard deviation in the lateral symmetry calculated from all steps.

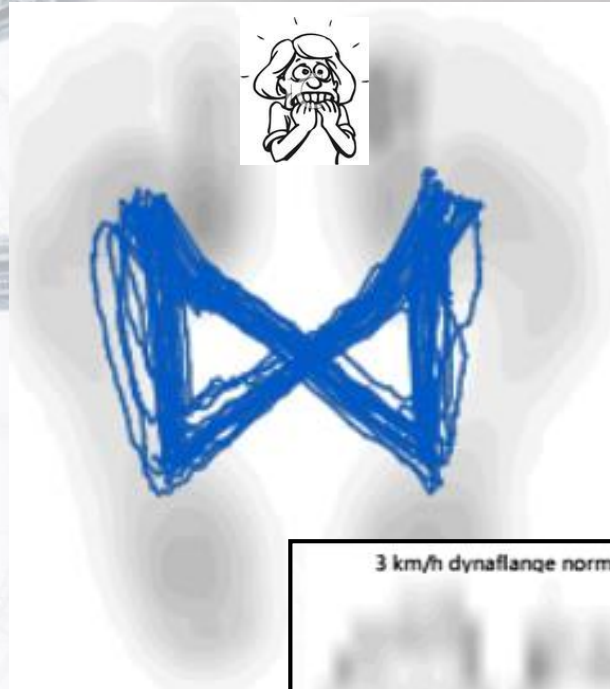


Is there enough neuromuscular strength to support this instability?

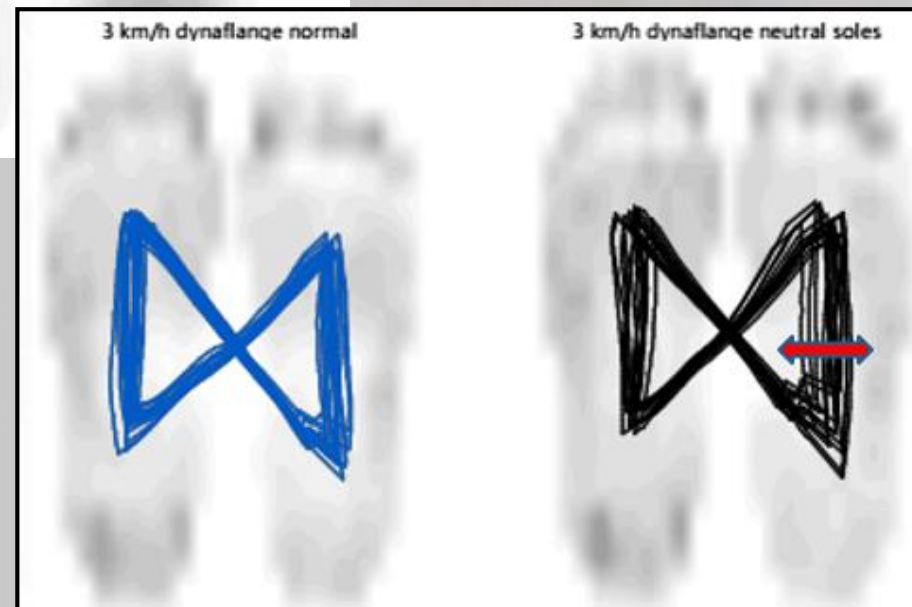
Why does it matter?

steady gait

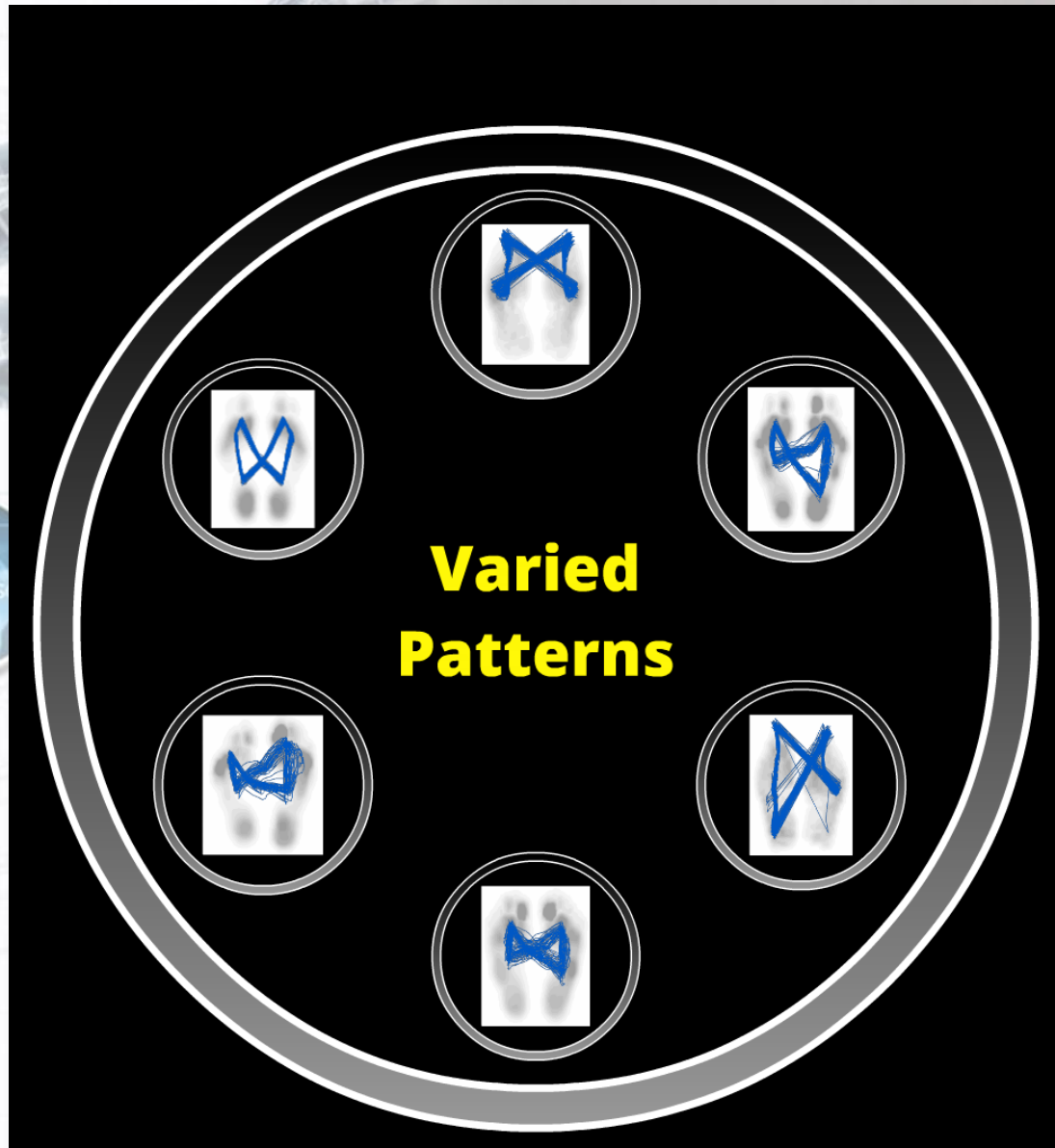
unsteady gait

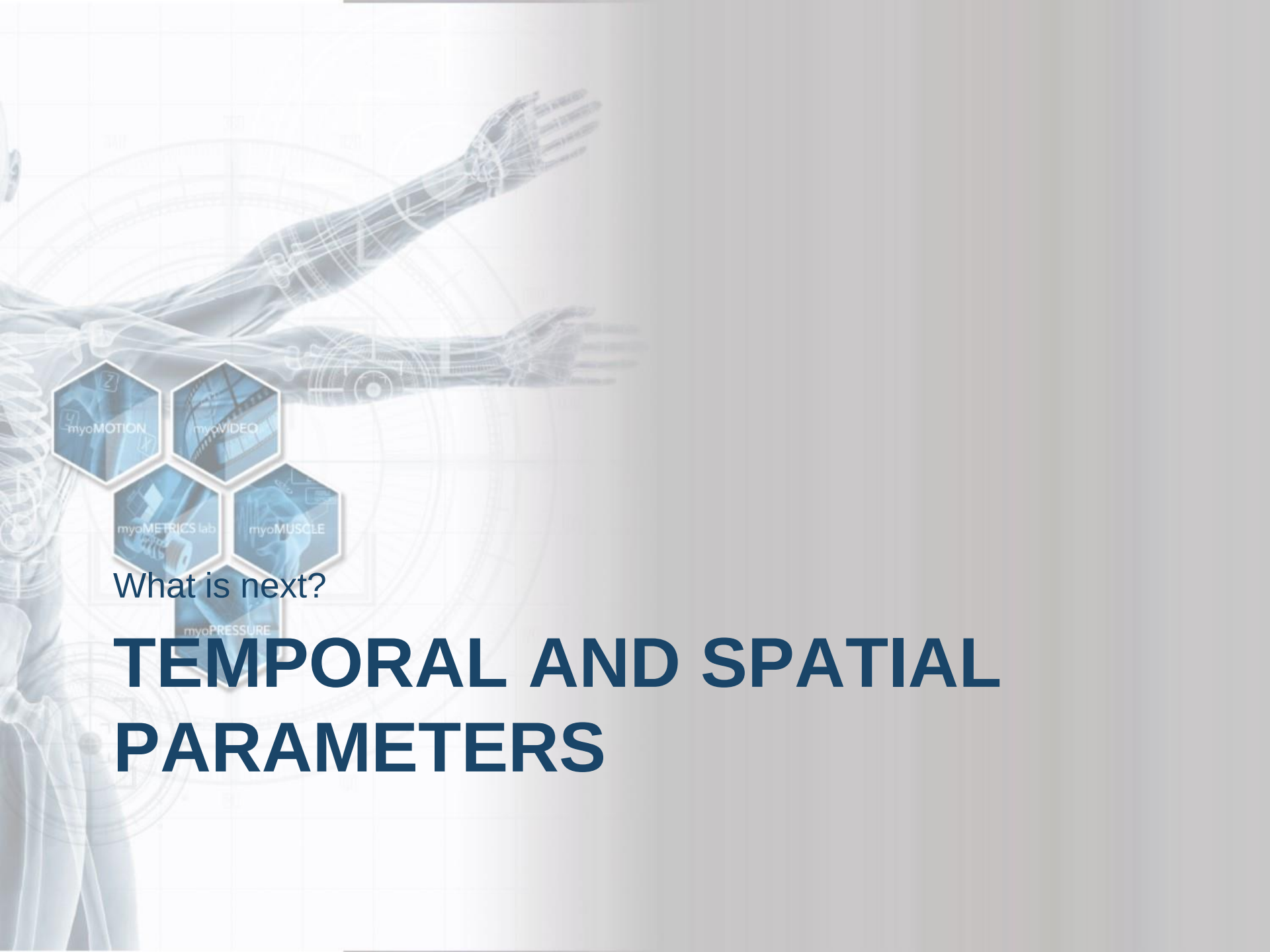


What do these patterns suggest in terms of base of gait?



Why does it matter?





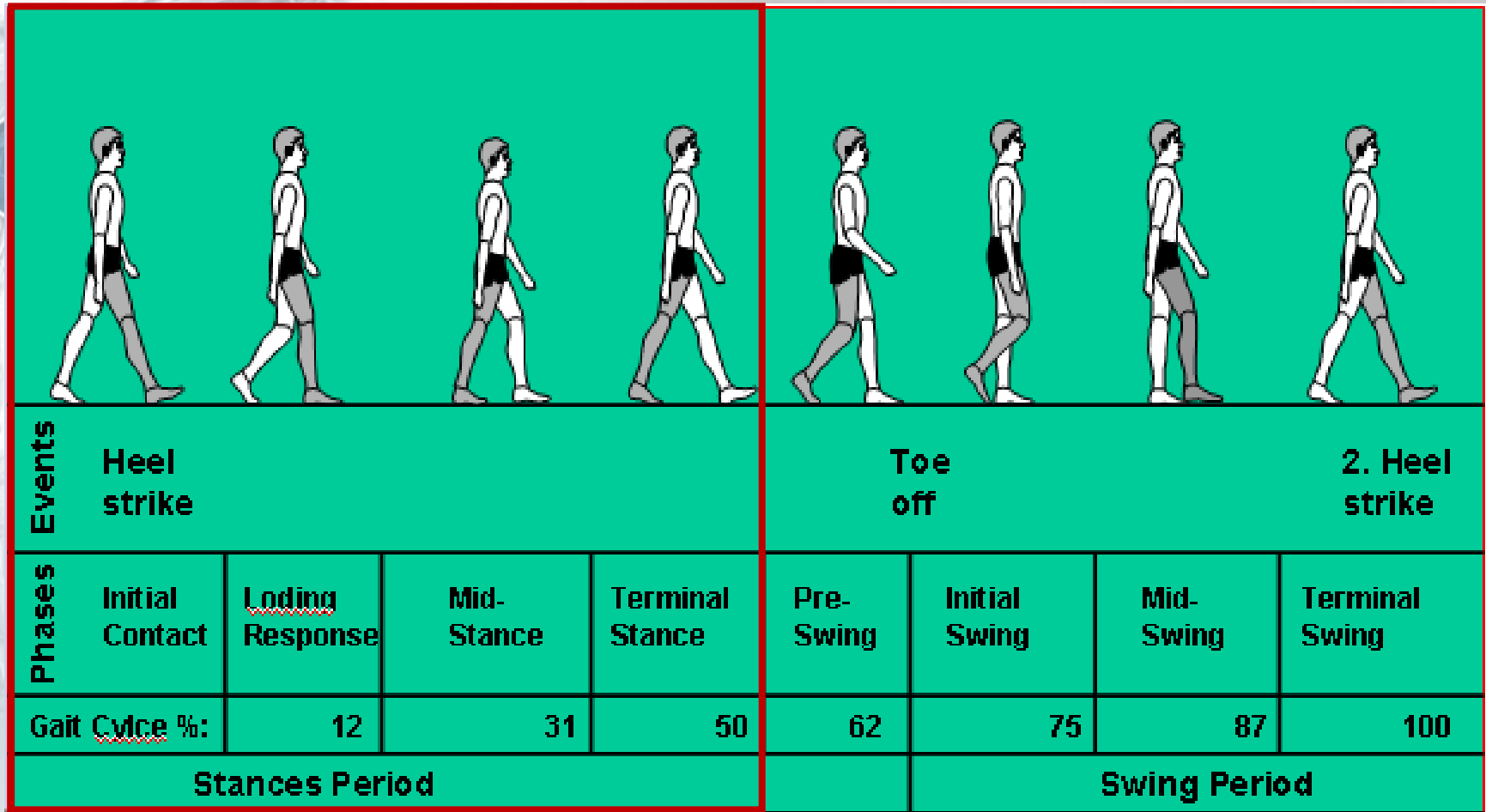
What is next?

TEMPORAL AND SPATIAL PARAMETERS

Stance Time

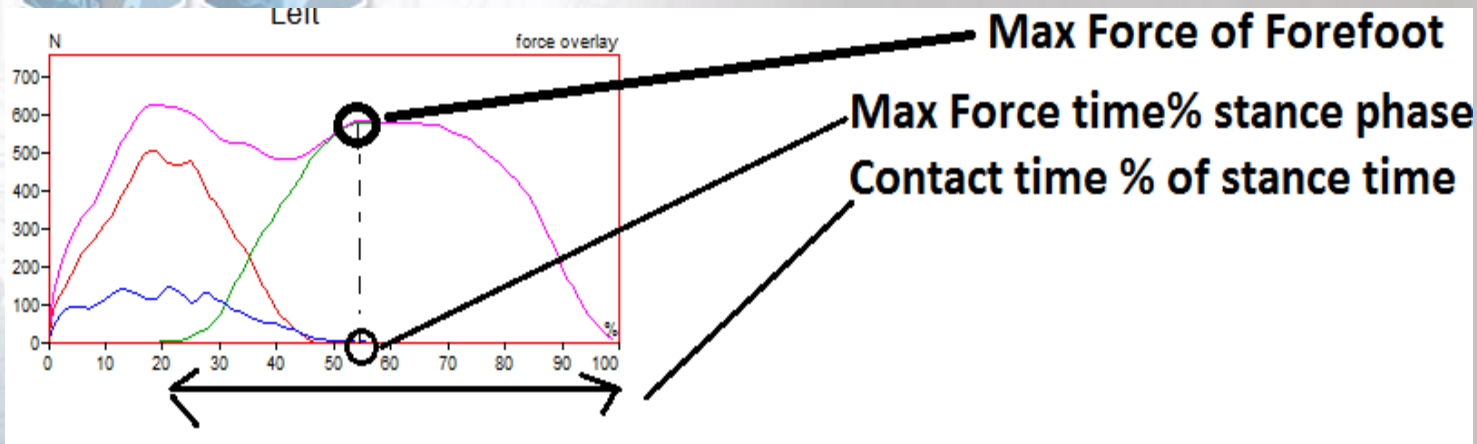
(percent of stride time)

Period within a gait cycle during which the foot has contact with the ground.



Loading Response

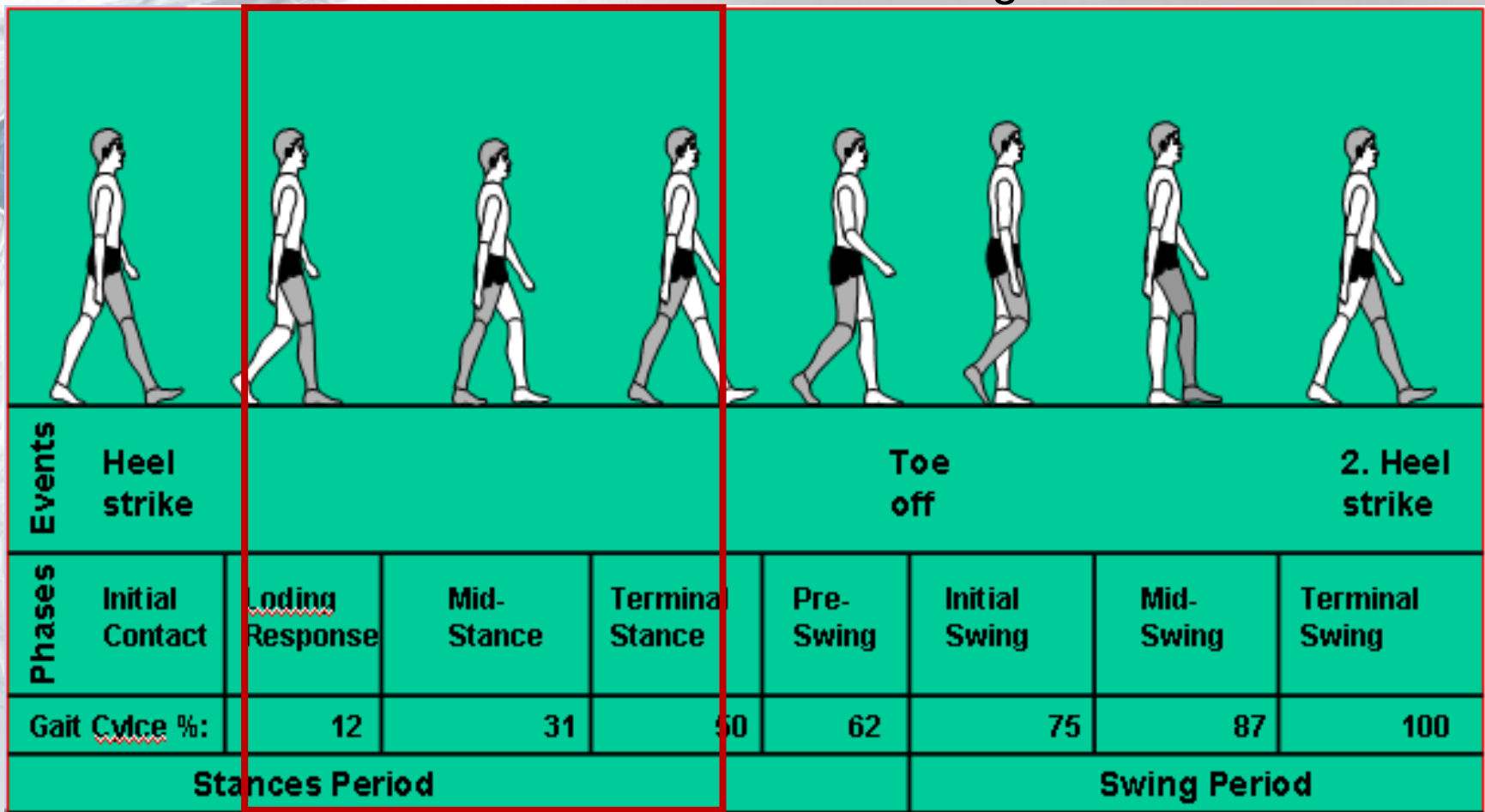
- ❑ Load Response (percent from gait cycle): starting when forefoot contacts=flatfoot phase
- ❑ Describes the period Load Response within a gait cycle



Single Support Time

(percent of stride time)

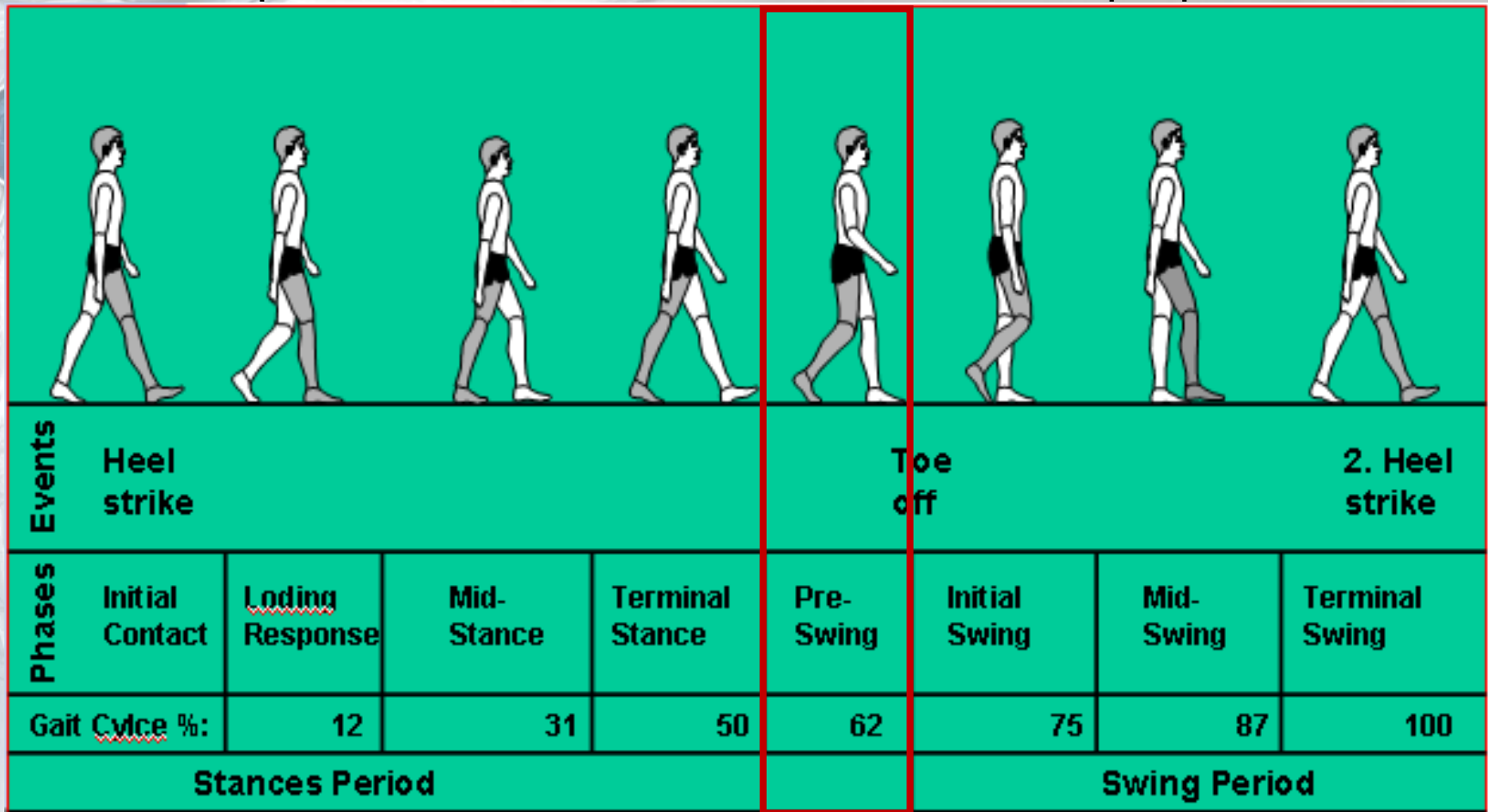
Period within a gait cycle during which the contralateral foot has no contact with the ground.



Pre-swing

(percent of stride time)

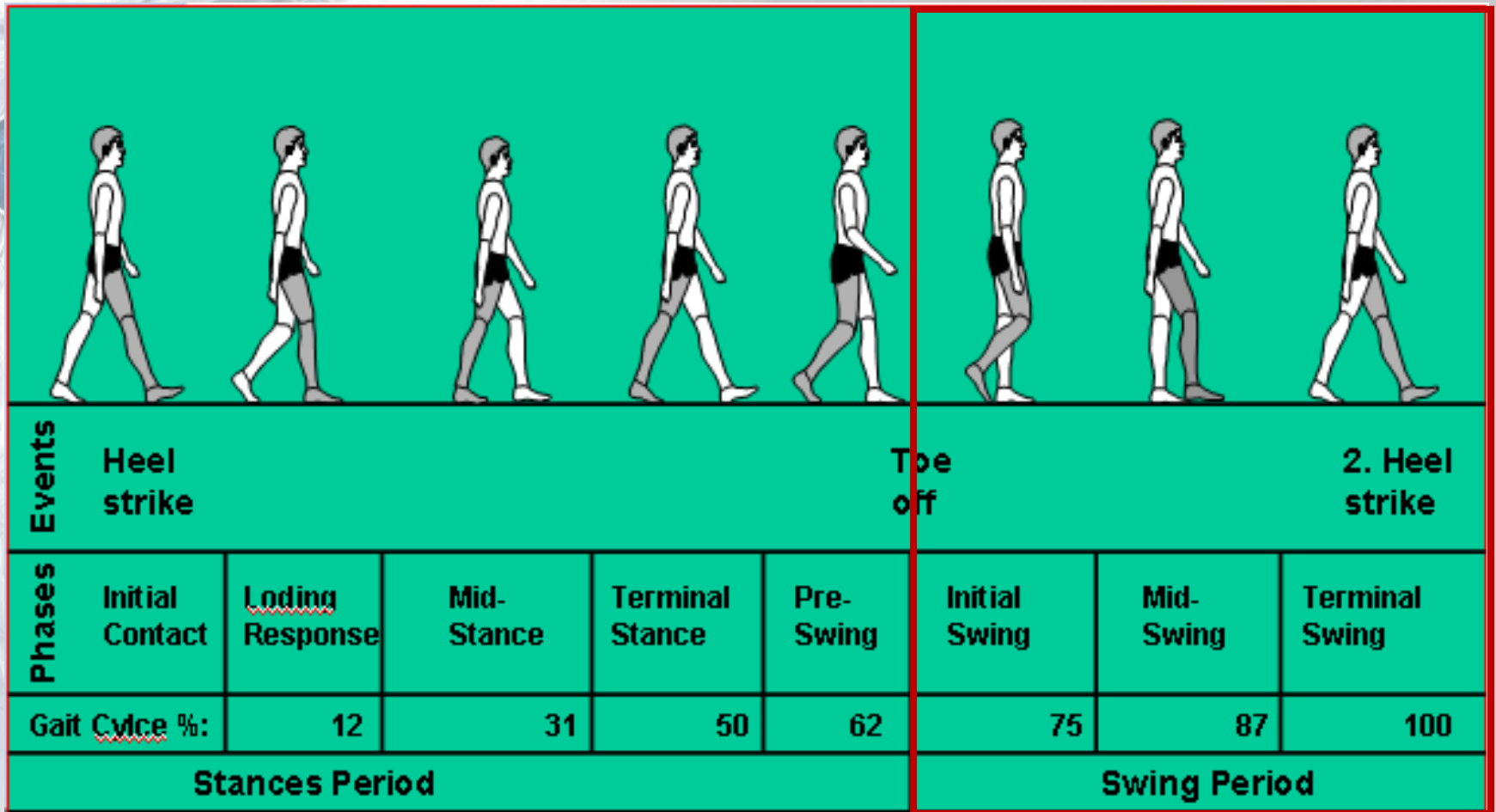
Period Pre-swing within a gait cycle is usually based on the portion after the heel lifts. Also called propulsion.



Swing Time

(percent of stride time)

Period within a gait cycle during which the foot has no contact with the ground.



Foot Rotation (external or internal rotation)

Foot rotation (degrees)

- Angle between longitudinal foot axis and walking direction



(+) plus = Outside (laterally)

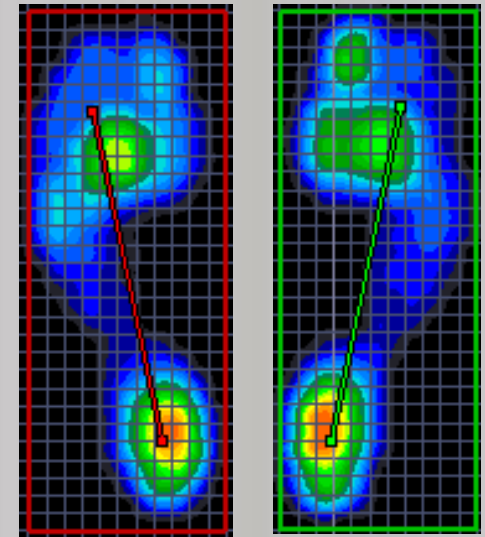
(-) minus = Inside (medially)

- ☐ Positive rotation on one side = externally rotated limb/foot and could be compensation for a same sided weak gluteus medius

- ☐ Too Negative = excessive supination

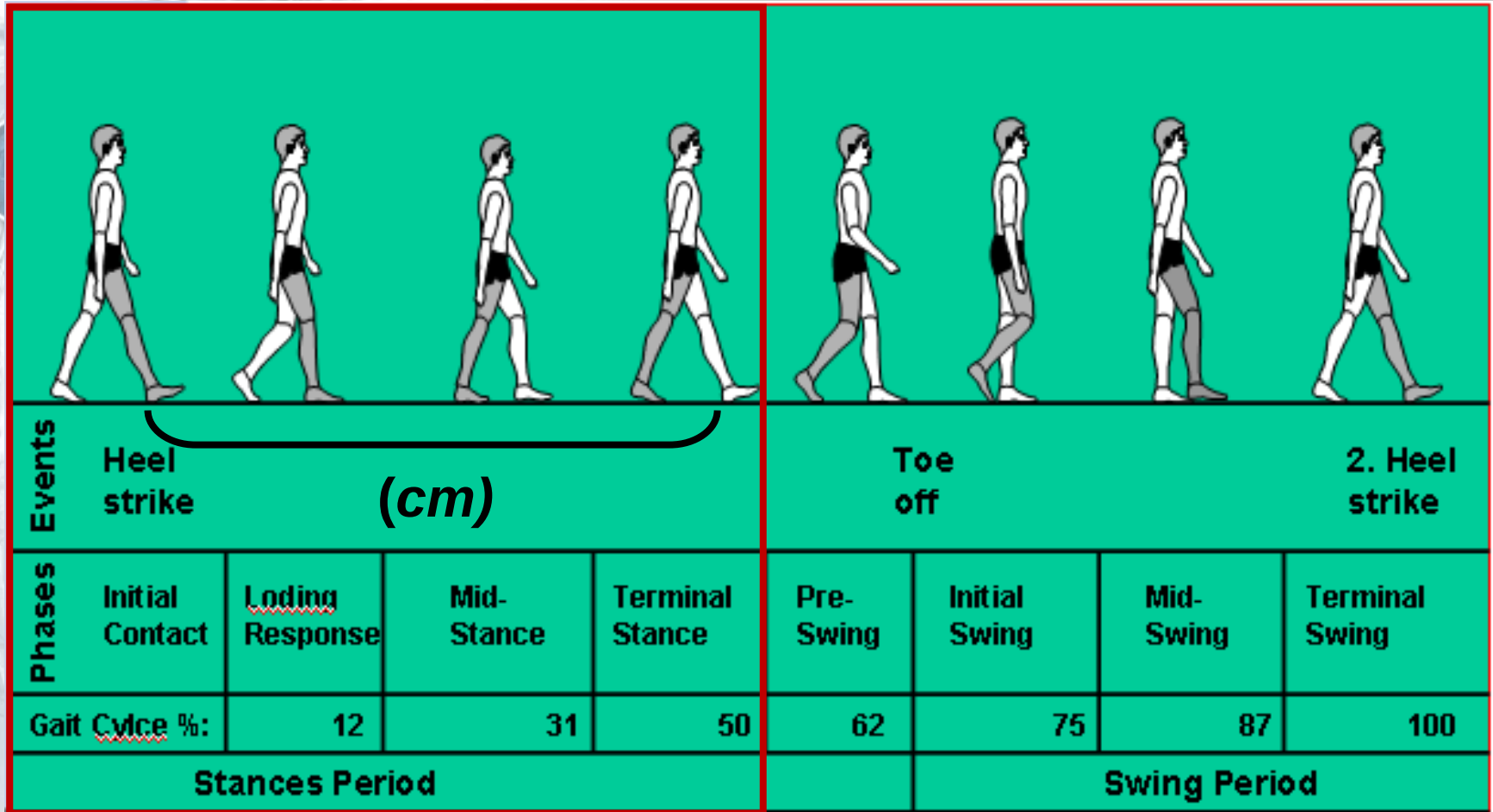
- ☐ Too Positive = excessive pronation

- ☐ Difference can differentiate whether it is a functional kinematic chain issue or more local musculoskeletal issue



Step Length (cm)

The distance between the heel contact of one side of the body and the heel contact of the contralateral side.

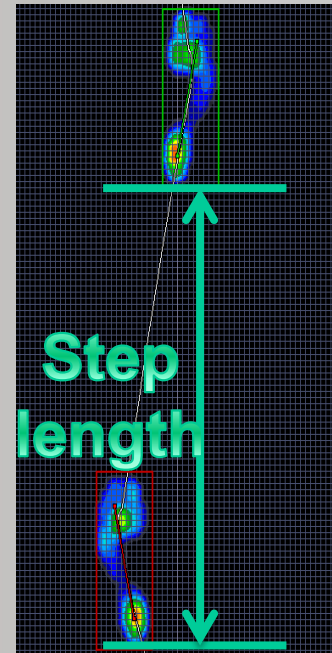


Step Lengths

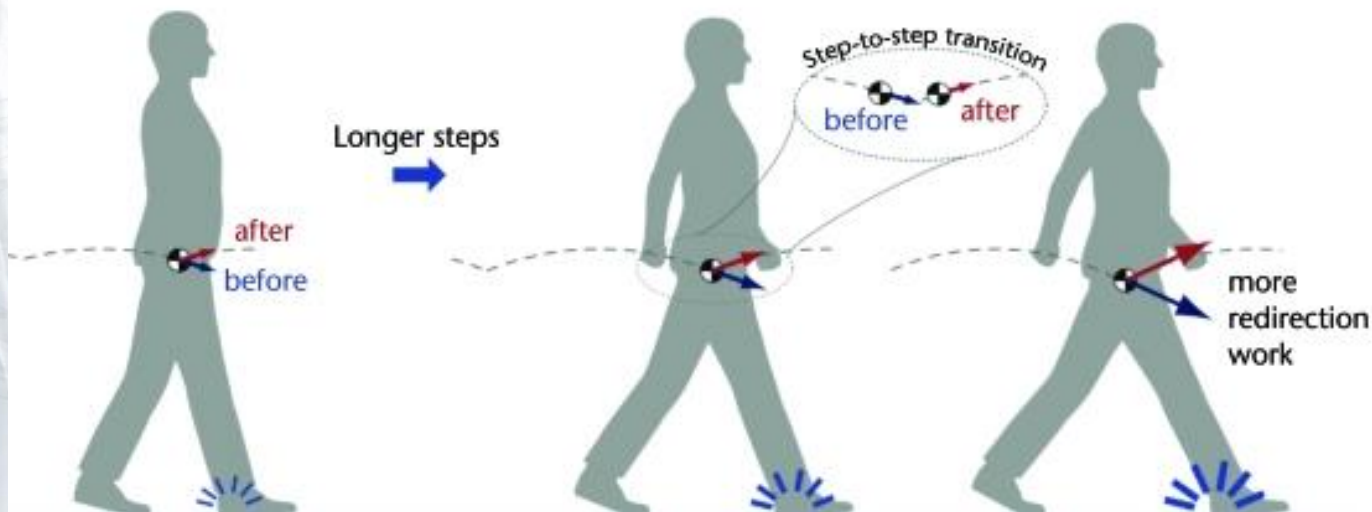
Step Length (cm)

- ❑ Shows the distance between the heel contact of one side of the body and the heel contact of the contralateral side

Step length, cm	L	67±3		200 cm
	R	50±1		
Stride length, cm	L	69±4		
	R	50±1		
Stride length, cm	L	136±4		
	R	99±1		

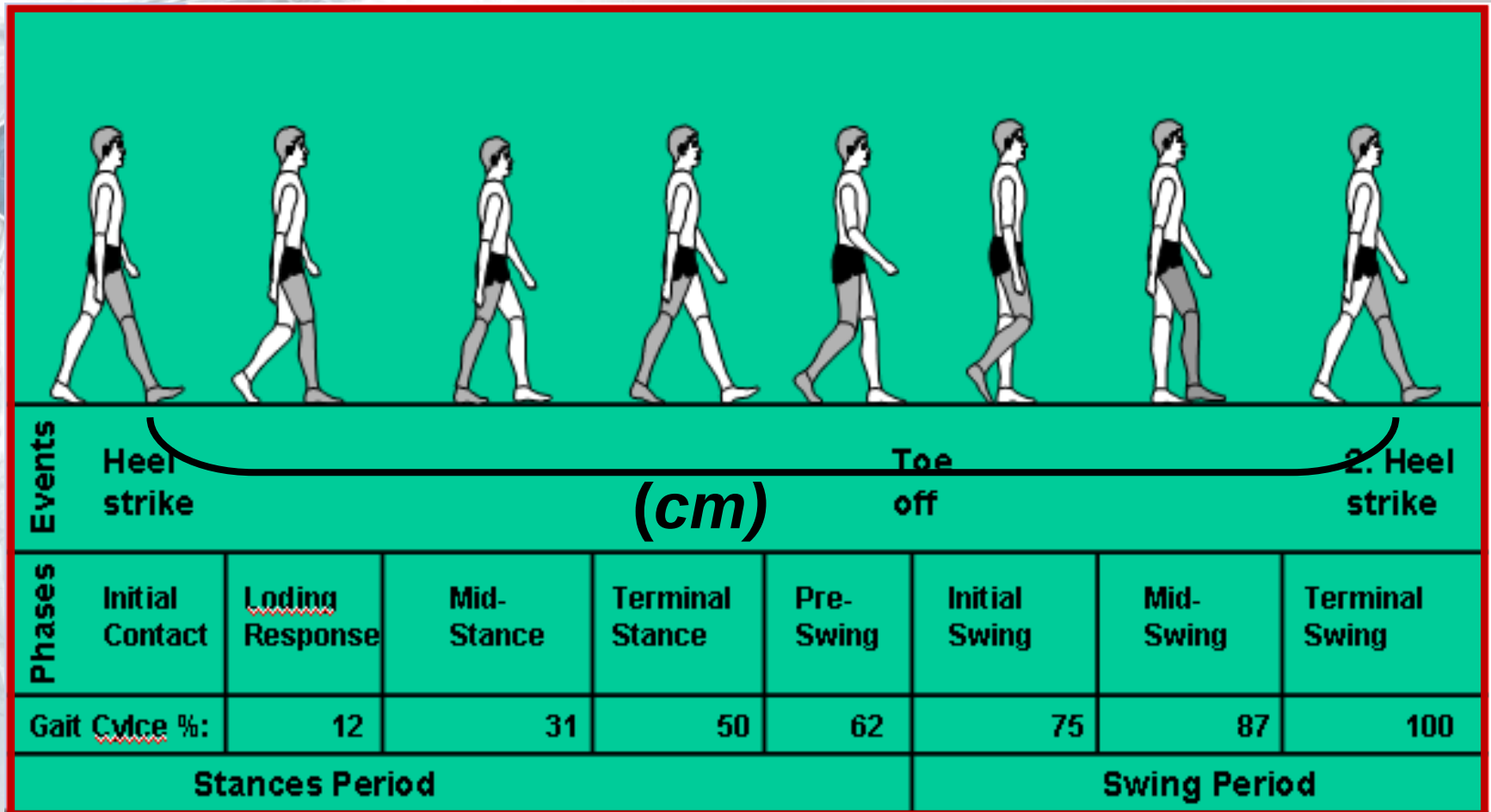


A Collision costs with longer steps (fixed step frequency)



Stride Length (cm)

The distance between two heel contacts on the same side of the body.



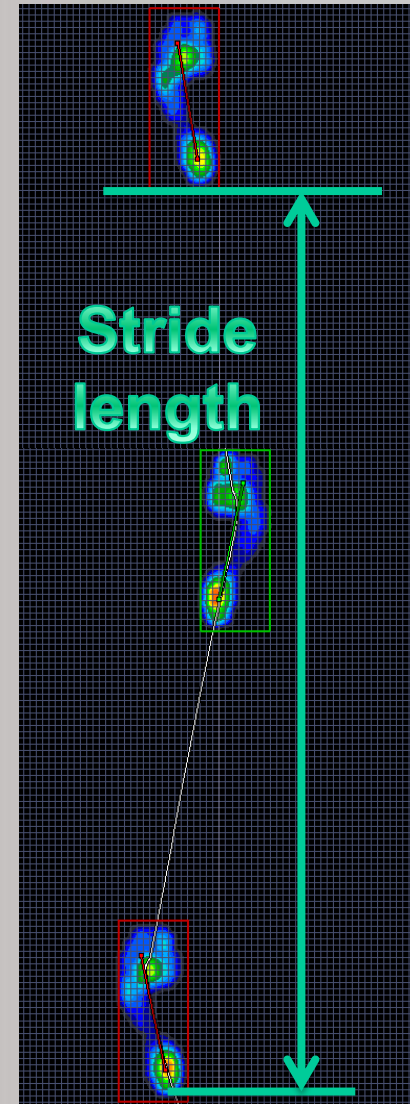
Stride Lengths

Stride Length (*cm*)

Shows the distance between two heel contacts on the same side of the body

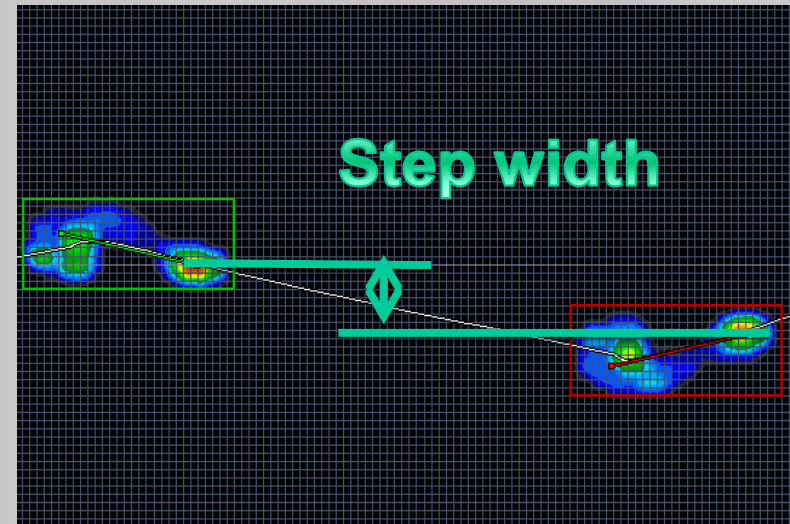
Stride length, cm	09-12-2007 shoe 1	158+/-3
	09-12-2007 shoe 2 no ortho	158+/-2

- ☐ Decreasing stride length is a potential mechanism of strain reduction during running.
- ☐ If stride length is decreased, the number of loading cycles will increase for a given mileage

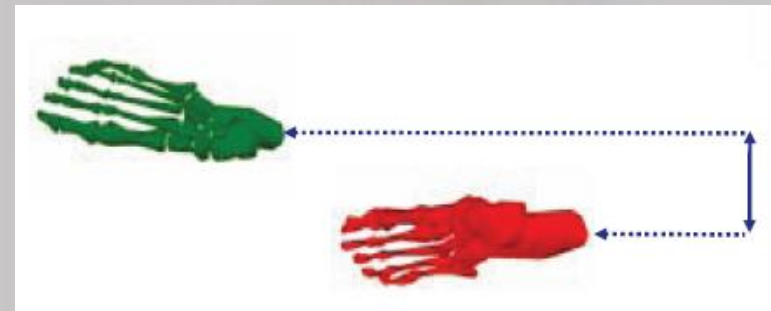
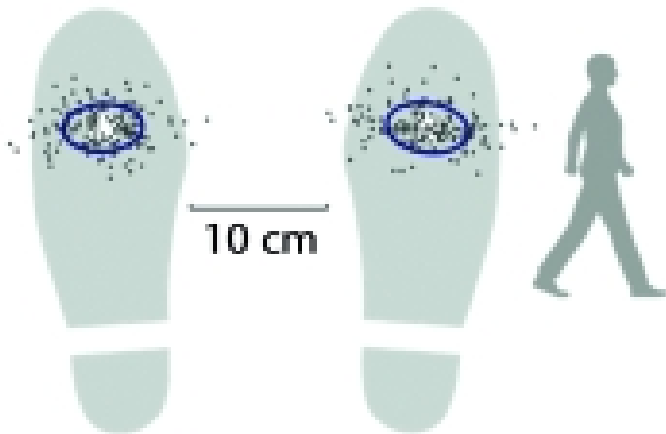


Step Width

- ❑ Step width (cm) Step width (base of gait) is the lateral distance between the heel centers of two consecutive foot contacts (this typically measures 6-10 cm).



B Step Variability (Normal)



Step Width Norms

Step and Stance Time Norms

- ❑ A wide base of support (BOS) has long been believed to be a hallmark of unsteady gait¹⁻⁵ Static standing stability increases with increases in BOS.
- ❑ Extreme (either too little or too much) step width variability is associated with falls

From: [J Neuroengineering Rehabil. 2005; 2: 21.](#)

Published online 2005 July 26. doi: 10.1186/1743-0003-2-21

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Table 3

Mean (SD) gait characteristics stratifying by gait speed and past year fall history.

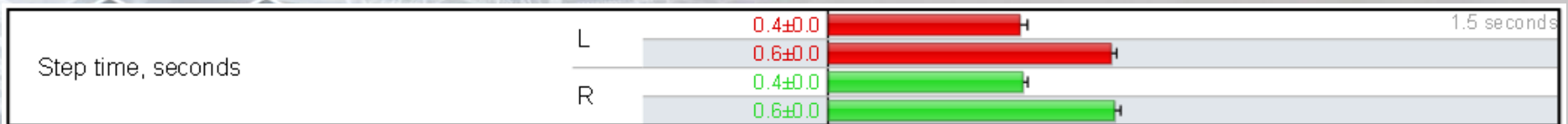
	Gait Speed < 1.0			Gait Speed ≥ 1.0 m/s		
	No falls n = 185 ≥ 1 fall n = 37 p-value			No falls n = 237 ≥ 1 fall n = 44 p-value		
Gait Variability CV* (%)						
Step width	15.6 (15.9)	15.7 (7.7)	.95	19.6 (16.6)	26.8 (29.1)	.02
Step length	7.5 (3.4)	7.7 (3.7)	.77	5.4 (2.3)	5.5 (2.2)	.77
Stance time	5.7 (2.3)	6.0 (2.3)	.47	4.3 (1.5)	4.5 (1.5)	.55
Step time	5.4 (1.7)	5.3 (2.1)	.83	4.1 (1.6)	4.2 (1.3)	.79
Gait speed (m/s)	.85 (.11)	.82 (.14)	.11	1.18 (.13)	1.18 (.14)	.98

*CV = coefficient of variation

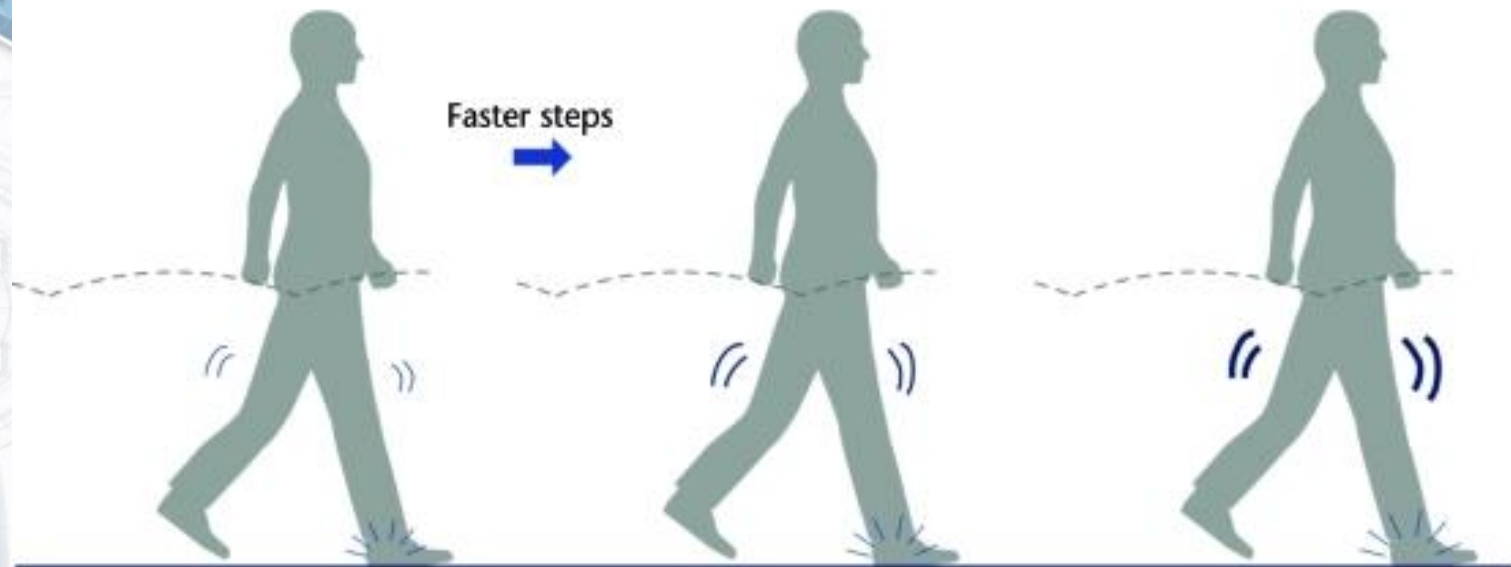
Step Time

Step Time (sec)

- ❑ Describes the period in sec. within a gait cycle between the heel contact of one side of the body and the heel contact of the contralateral side

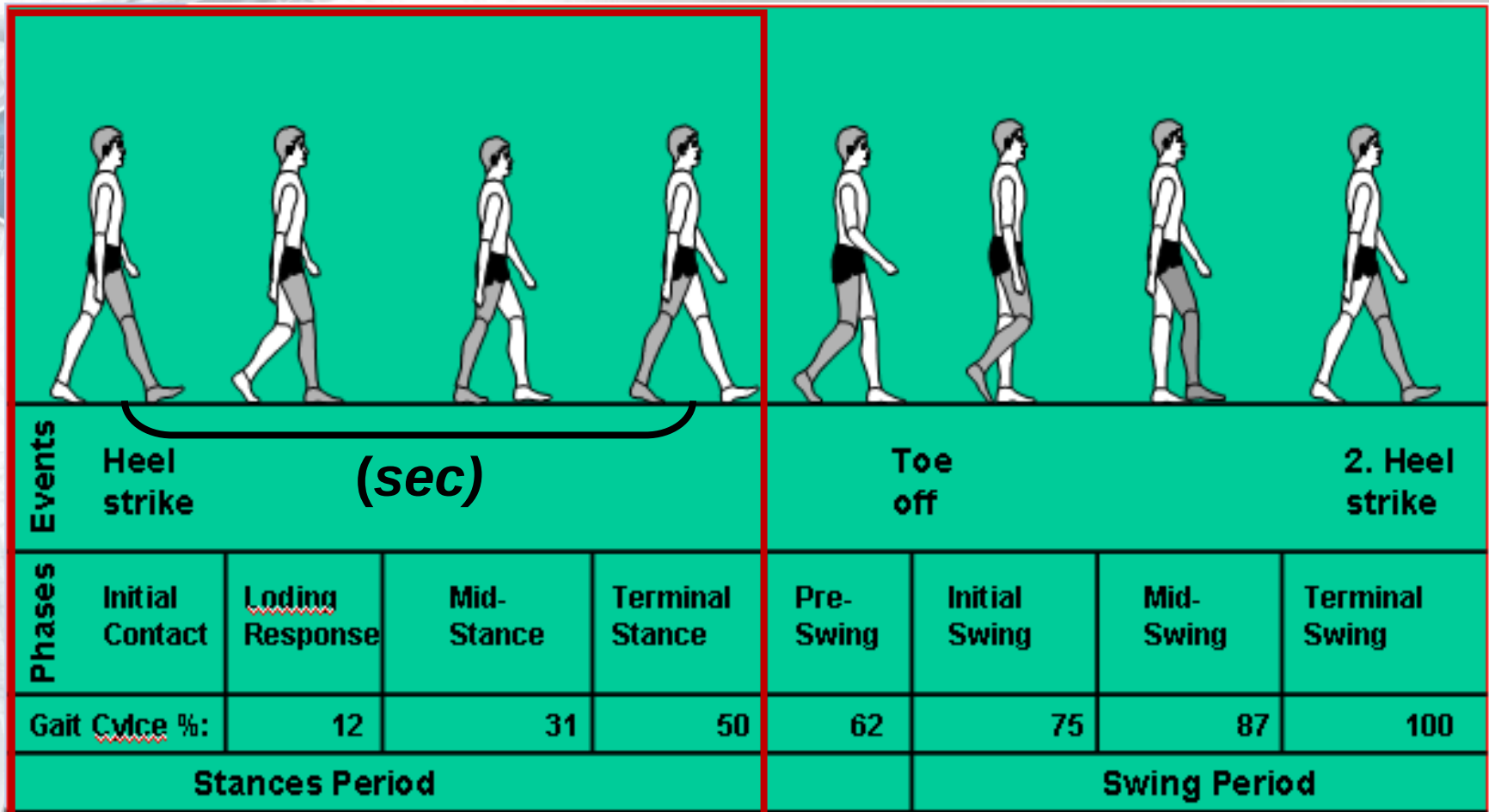


B Limb swing costs with faster steps (fixed step length)



Step Time (seconds)

Period within a gait cycle between the heel contact of one side of the body and the heel contact of the contralateral side.



Step Time Walking and Running

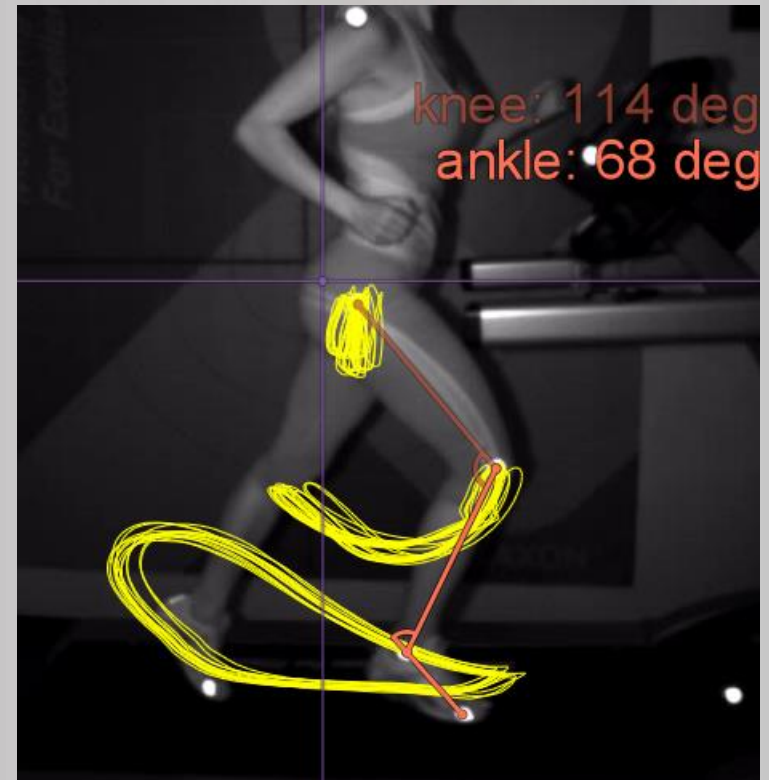
- ❑ With competitive runners a 0.001 second difference in step time separates gold, silver and bronze. Making certain they are symmetrical is key especially when training and assessing vertical oscillation

- ❑ Walking:

- Normative Values (Kirtley, 2006, p.23)

- Dependant on Age, Leg length, Gender, Pathology and Environment

	Speed (m/s)	Cadence (steps/min)	Step Length (m)
Men	1.3-1.6	110-115	0.7-0.8
Women	1.2-1.5	115-120	0.65-0.75



Timing Norms

Cadence (*steps/min*)

Measurement of steps/second and is calculated from the reciprocal of the Stride Time.

❖ *Normal 110-115*

Speed of the gait cycle

❖ *Comfortable walking speed = 80 m/min or 3 mph*

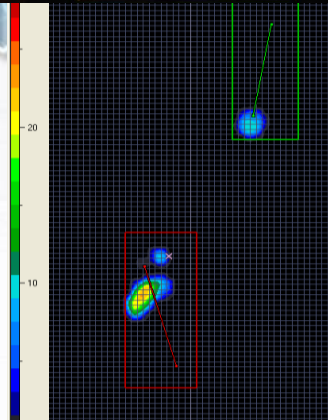
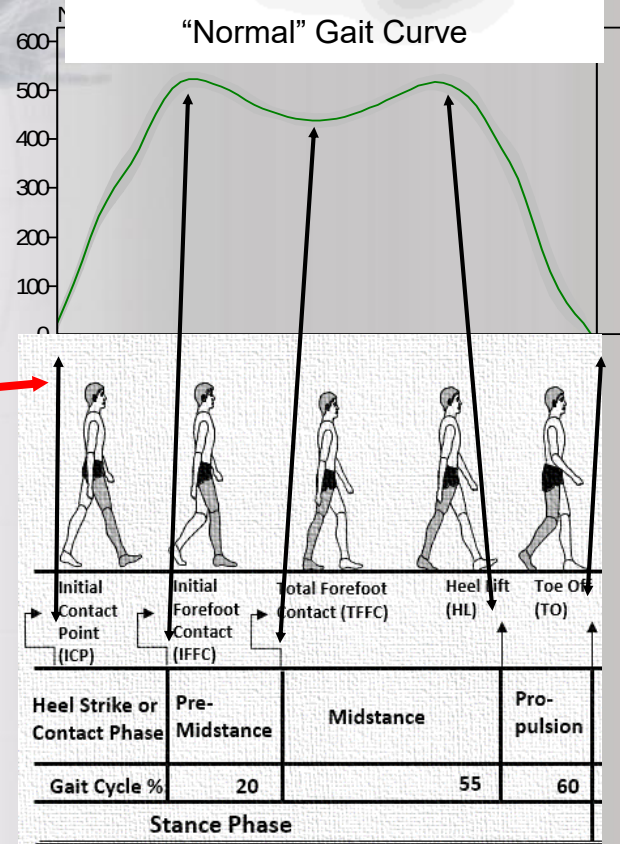
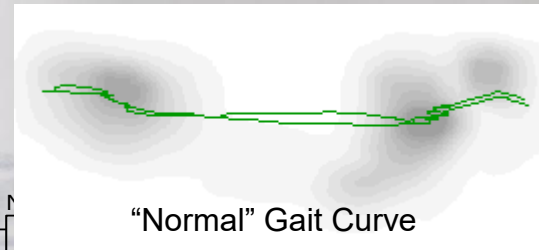
Velocity Variability (percent of velocity)

❖ *Variability of the speed of the respective stride.*

[..\MyoPressure FDM-TNormative reference values Part 1 Gait.pdf](#)

[..\MyoPressure FDM-TNormative Gait - Mary Rogers.pdf](#)

Average Ground Reaction Force Curve for Walking

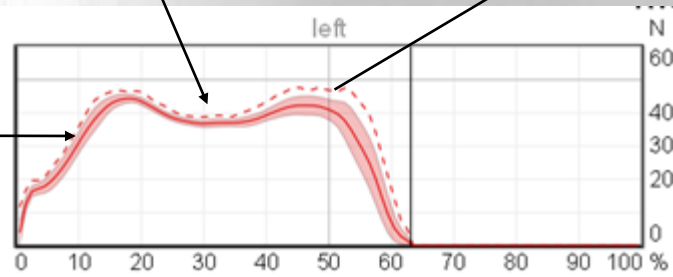


Average Ground Reaction Force Curve for Walking

- ❑ Mid-foot
- ❑ Lateral front foot
- ❑ Central front foot
- ❑ Medial front foot

Toes (I and/or II/III)

Heel

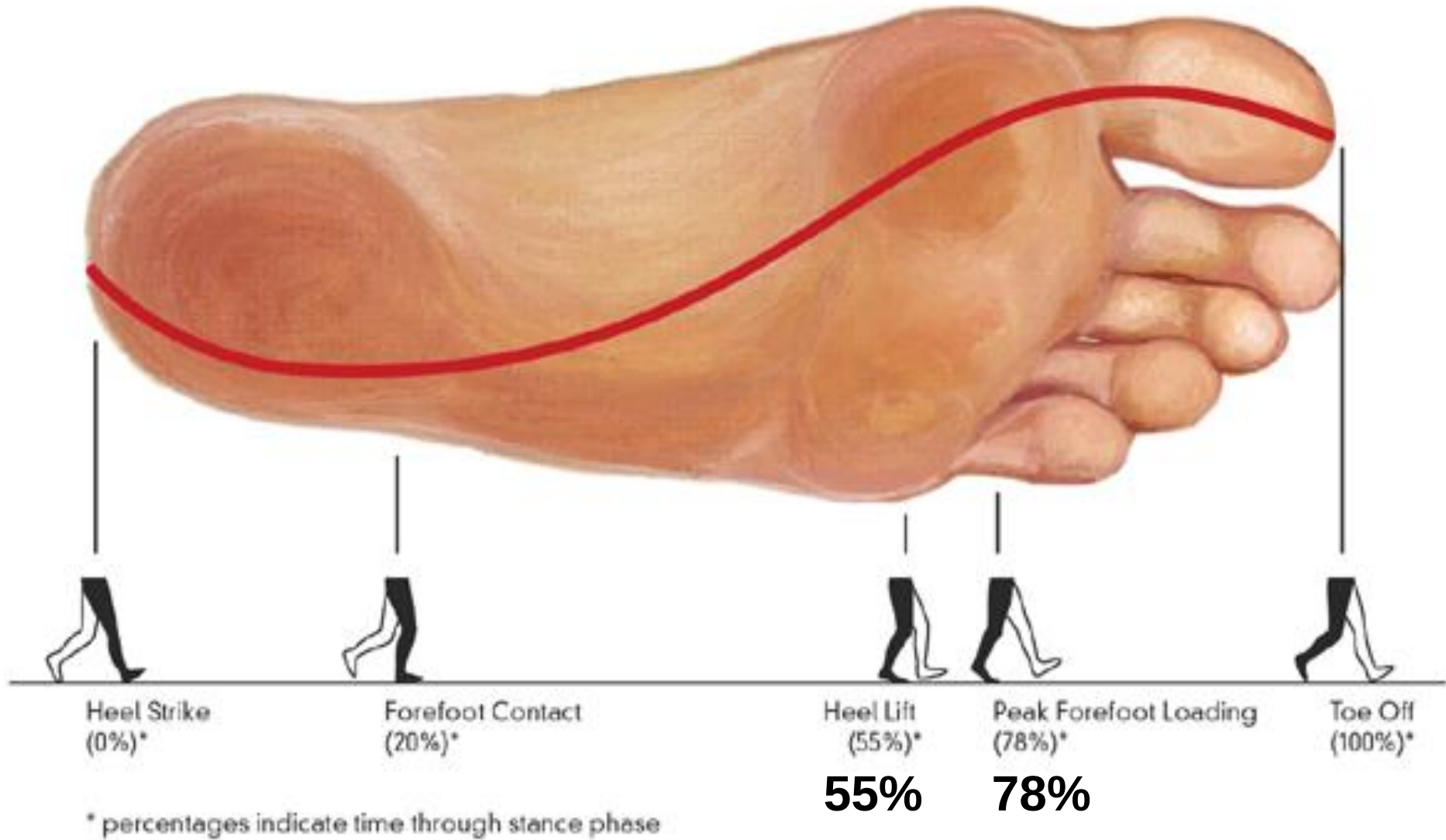


Percent of Stance phase

- First line to 1st peak/ approx 25%
- 1st peak to 2nd peak approx 50%
- 2nd Peak to Termination approx 25%

Normal COP Curve

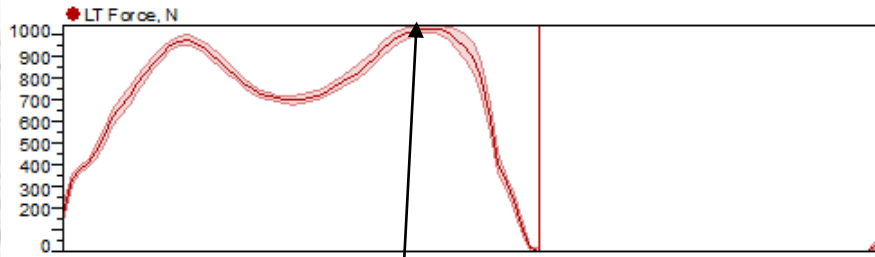
- ❑ Weight is first received by the lateral side of the heel because the foot is slightly supinated as it contacts the ground. The foot immediately begins to pronate and in normal gait, pronates until about 22% of the stance phase. Weight is then received by the medial side of the heel, then the 5th metatarsal head, and then the forefoot loads from a lateral to medial direction. As the foot unloads, the heel unloads first, then the metatarsals and toes begin to unload from lateral to medial, and the hallux is the last to unload at toe-off.
- ❑ Pronation shifts weight to the medial side of the foot, and supination shifts weight toward the lateral side. There is usually an initial amount of supination due to the contact position of the foot at heel strike. Pronation then shifts weight medially in a curve. Supination of the foot during propulsion then reverses the weight thrust, and weight is finally directed down the long axis of the hallux at toe-off.



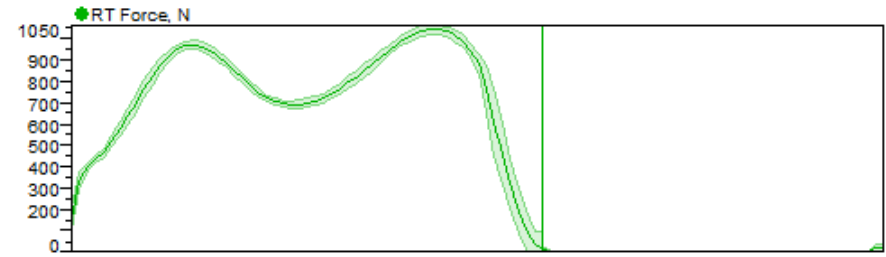
NOTE where heel lift is!

Force over % Gait curve

Left Side

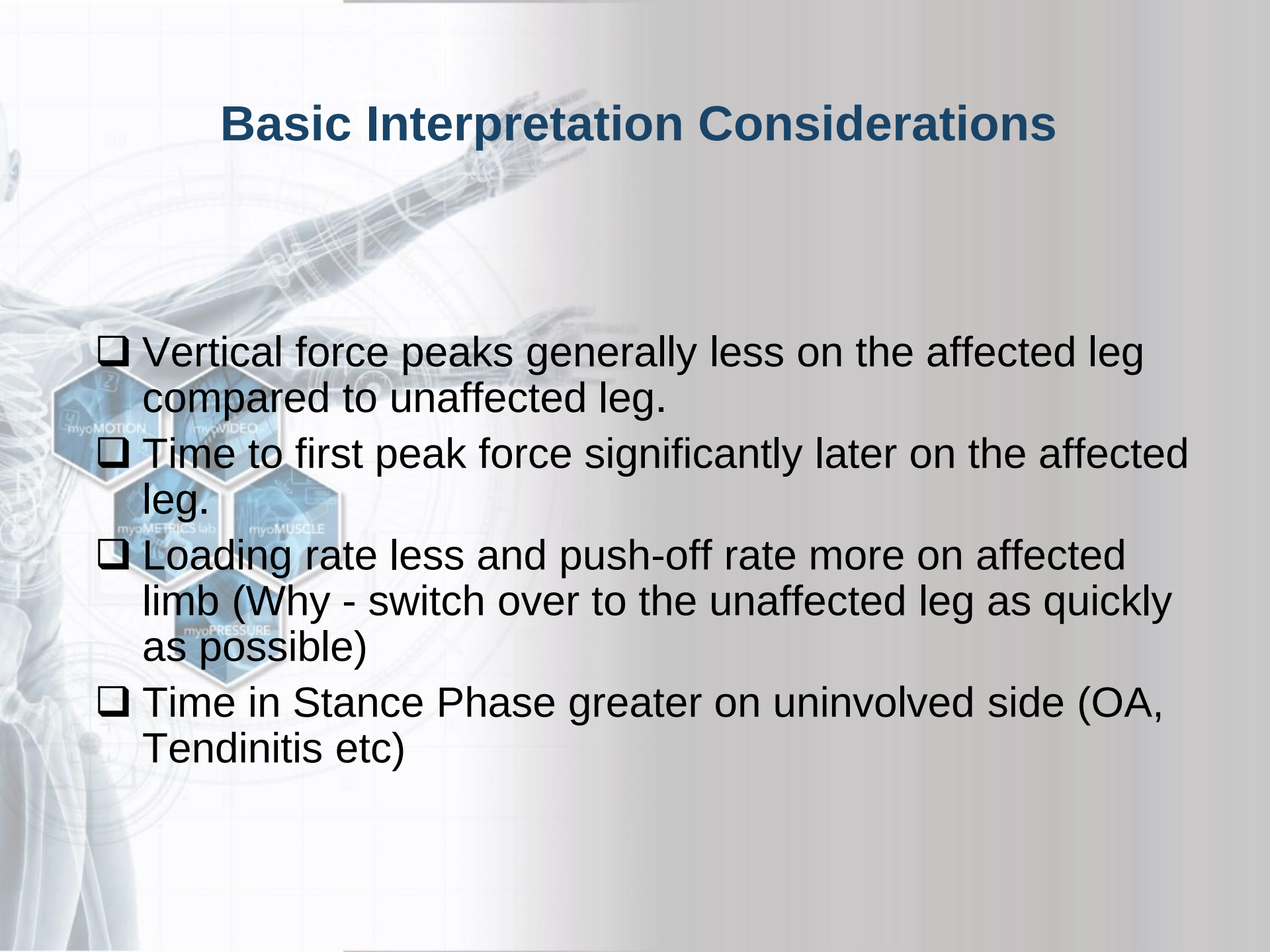


Right Side



- Should be M-shaped
- **Sensitive feet**, e.g. (diabetics, etc.) - local pressure peaks should be avoided and maximum pressure load less than approx. 25 N/cm², in order to avoid damage to the soles of the feet.

Basic Interpretation Considerations

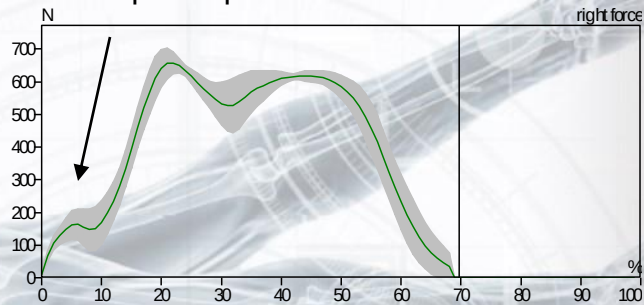
- 
- ☐ Vertical force peaks generally less on the affected leg compared to unaffected leg.
 - ☐ Time to first peak force significantly later on the affected leg.
 - ☐ Loading rate less and push-off rate more on affected limb (Why - switch over to the unaffected leg as quickly as possible)
 - ☐ Time in Stance Phase greater on uninvolved side (OA, Tendinitis etc)

Basic Interpretation Considerations

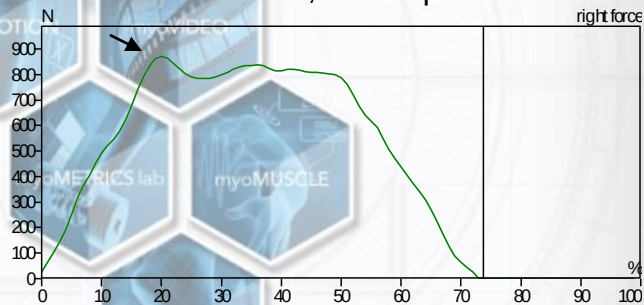
- ☐ Heel strike lateral to neutral – rear-foot supination.
- ☐ Heel strike medial to neutral – rear-foot pronation (valgus presentation).
- ☐ Often accompanied by involved SI issue ipsilaterally (internally rotated femur, weak glut med etc.).
- ☐ Lateral gastroc occasionally painful to palpation.
- ☐ Reduced time in Stance phase.

Many asymmetries will resonate

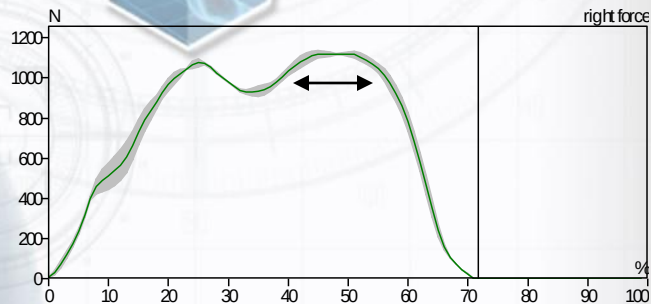
Heel spur or pain



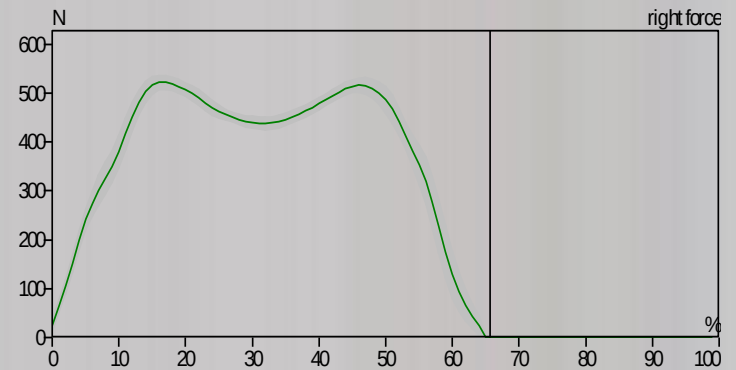
Possible tendonitis, or compensation in heel



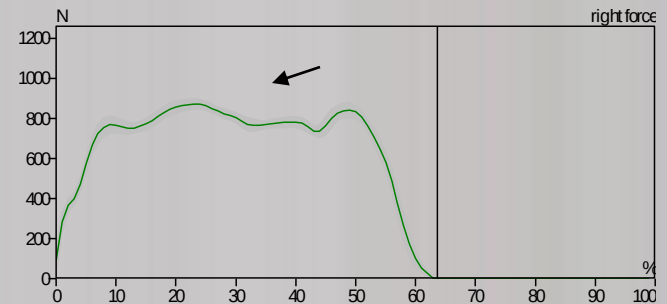
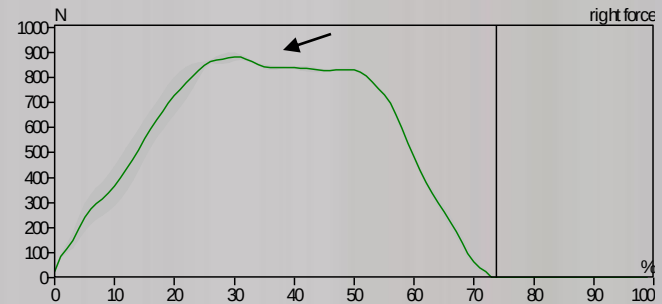
Possible Back Pain



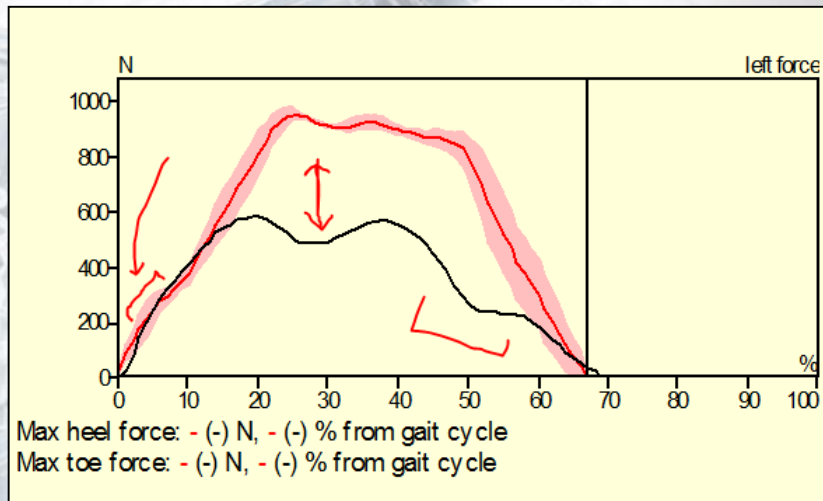
"Normal" Gait Curve



Possible back or upper body pain, minimal propulsion through the mid foot

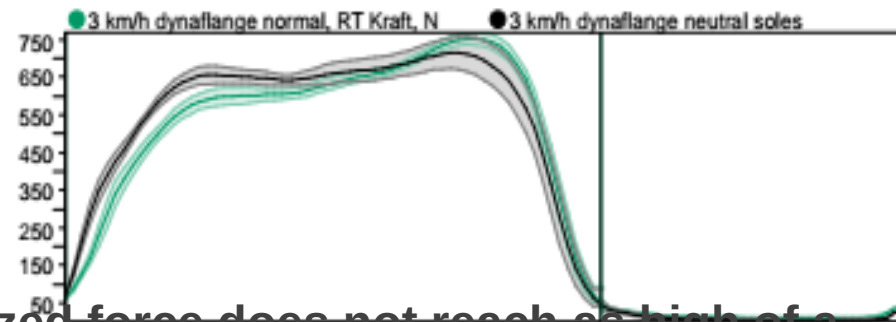
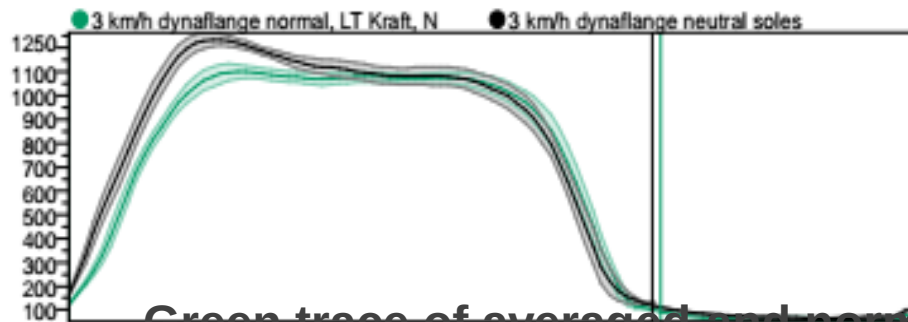


Reduced Forces? Reduced Pressures?



Left Side

Right Side



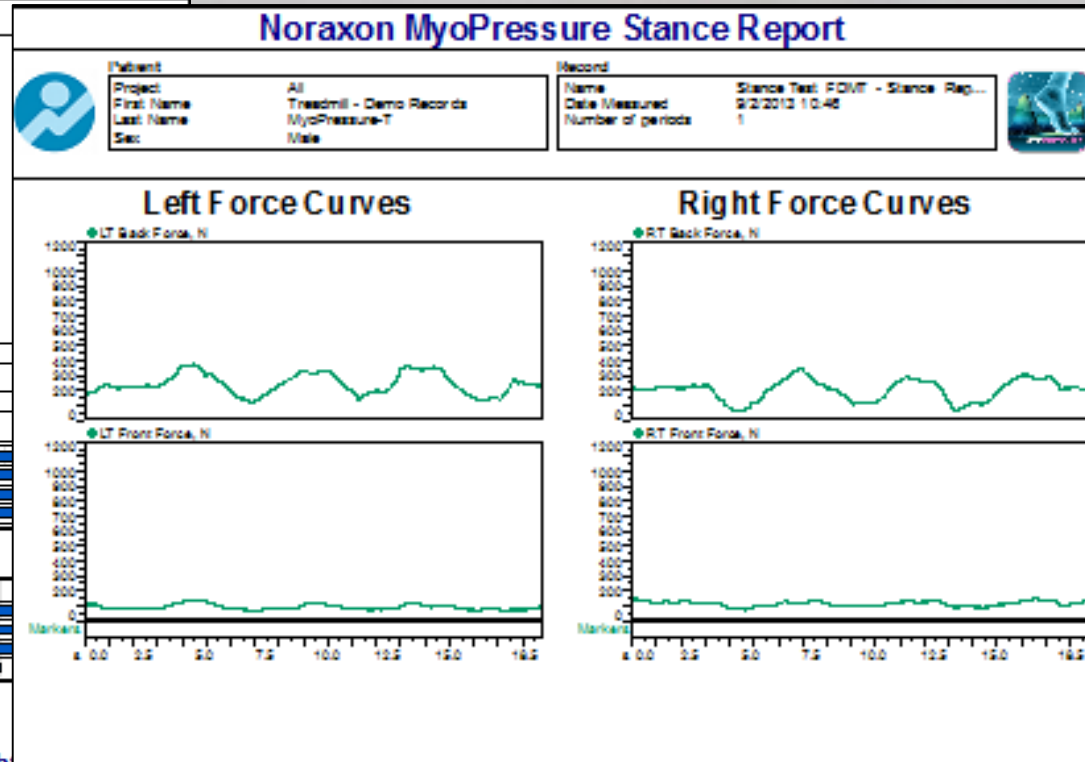
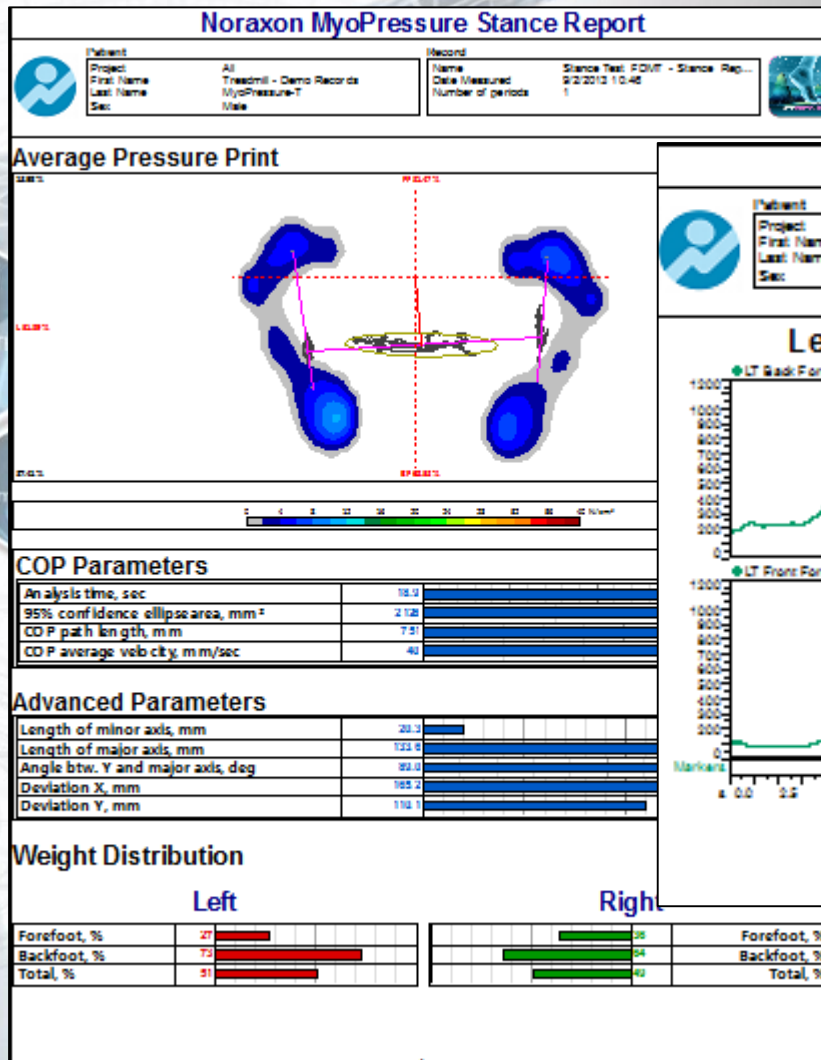
Green trace of averaged and normalized force does not reach as high of a peak value as the un-treated side, in grey. Nor does it reach its peak at the same accelerated rate.



What is next?

STANCE AND BALANCE

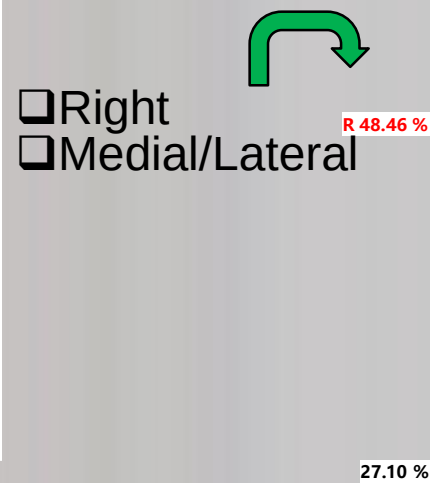
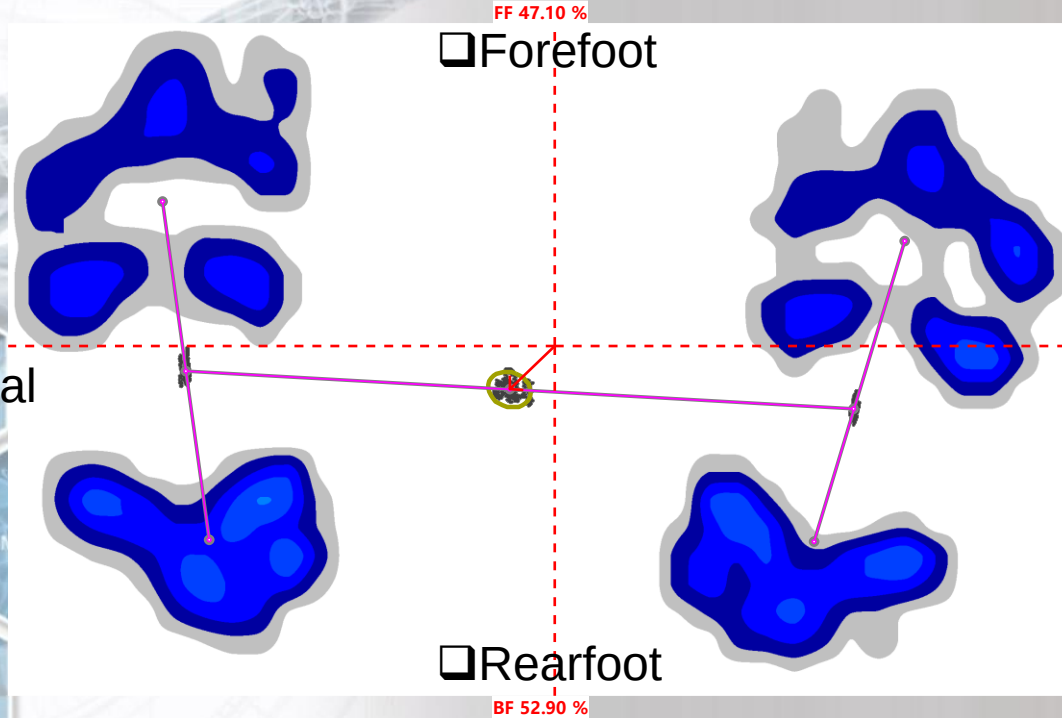
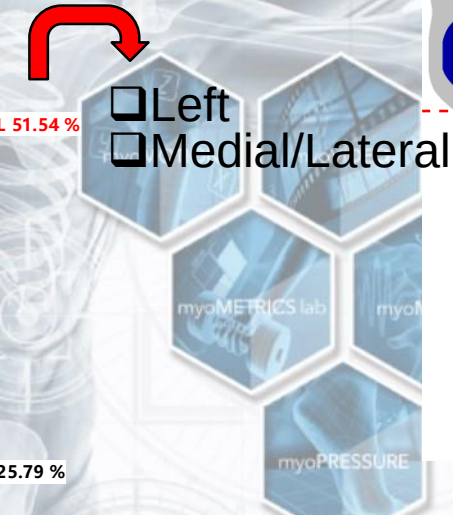
Stance Report



Average Load

25.75 %

21.36 %

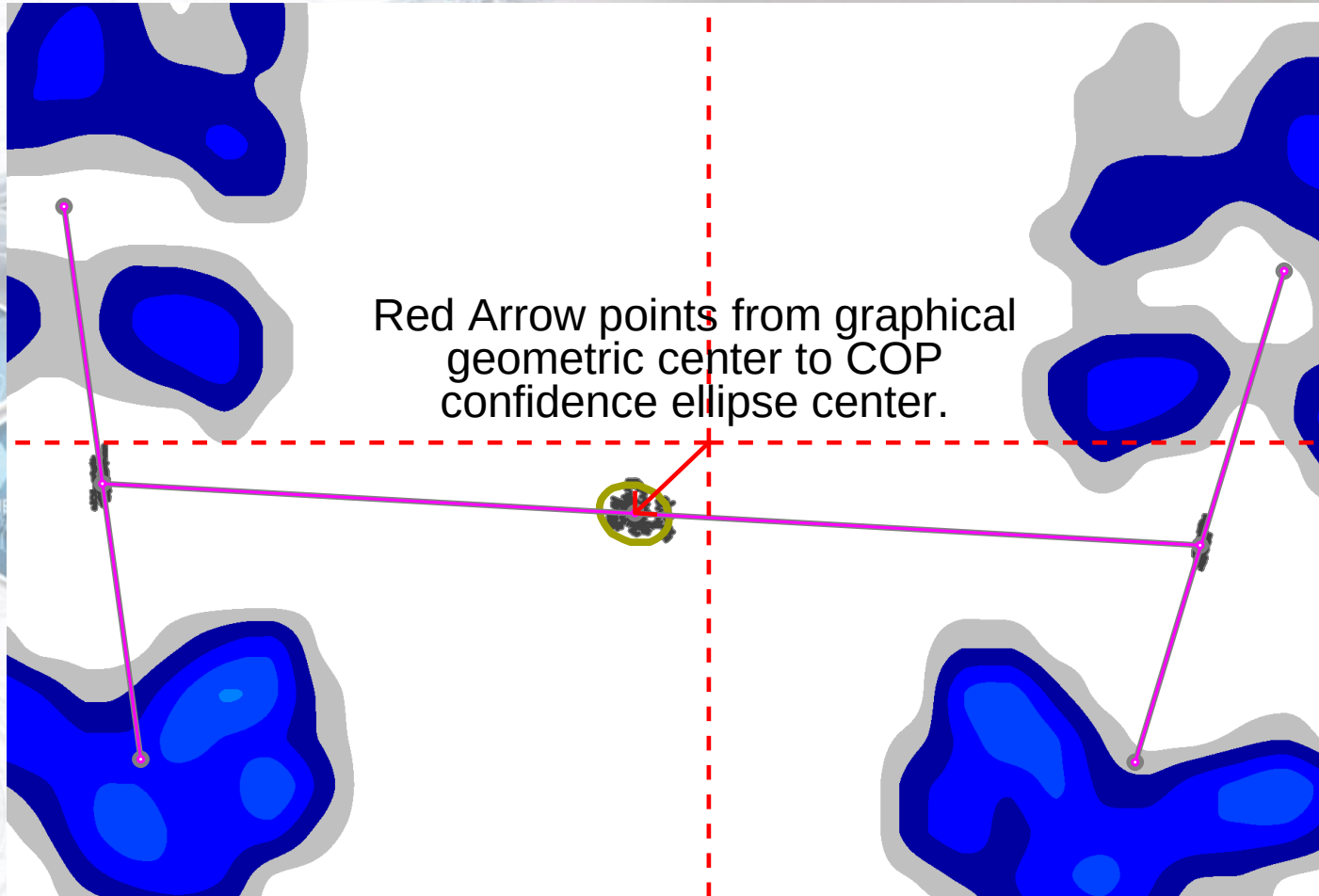


27.10 %

- ☐ The image shows average load distribution under the feet.
- ☐ Each quadrant is quantified by a load distribution %.
- ☐ The pink line is the connecting line of the three COPs (Left foot, Right foot, and Total Body COP).

- ☐ The middle cross is the center of pressure of the whole body.
- ☐ The circle around includes 95% of the COPs.
- ☐ Characteristics of the circle are displayed in a chart.
- ☐ Left and right points or traces are the COPs of the left and right body side.

Direction of Sway

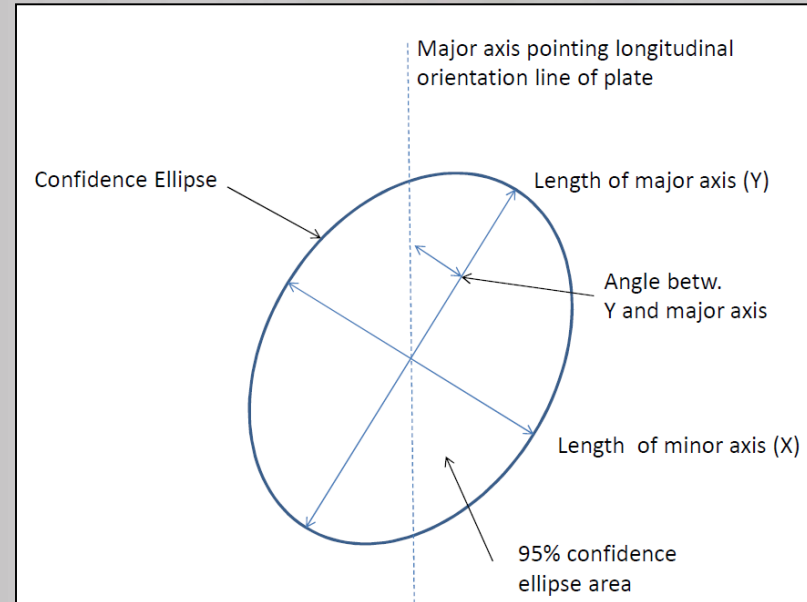
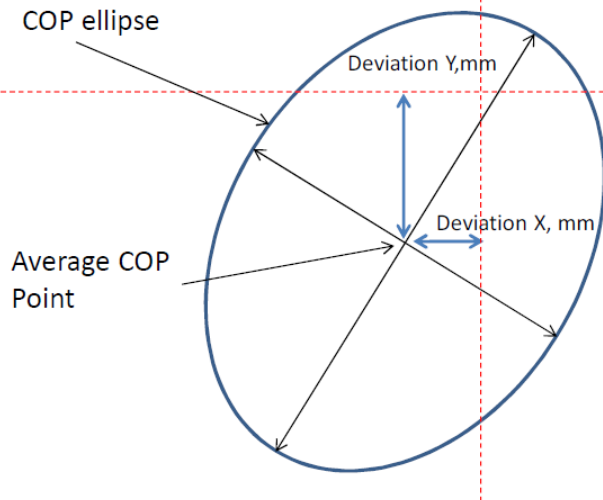


Stance Parameters

COP path length = total length of COP sway in mm

COP average velocity = within sway in mm/sec

Deviation parameters measure the distance of average COP (where y- and x- cross) and the active pressure area center point (defined by the red origin cross)



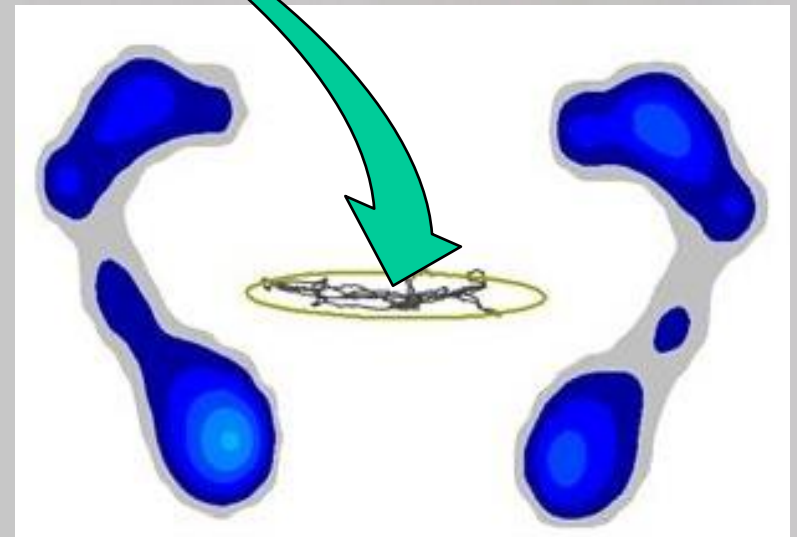
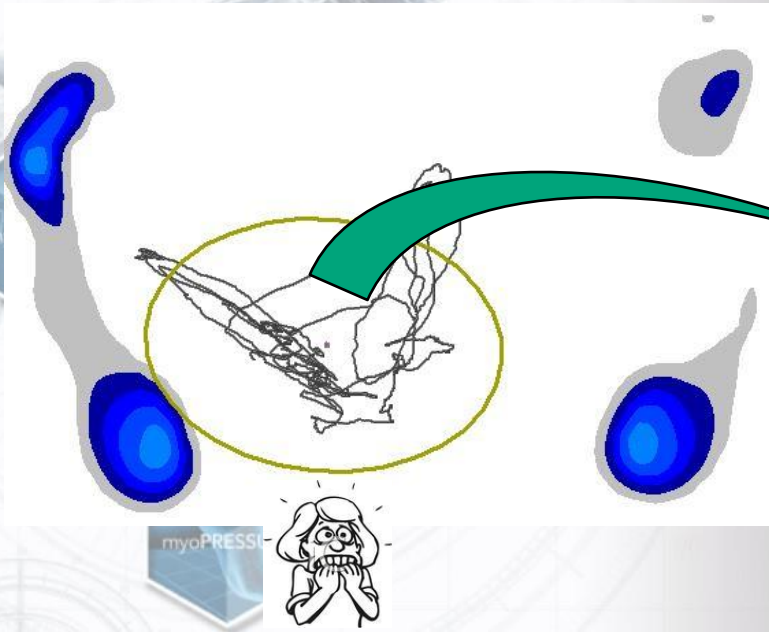
- **Confidence ellipse area:** indicates the area of the ellipse in mm². This area encompasses the path of COP travel.

Parameters

Analysis time, sec	9.5	12 sec
Confidence ellipse area, mm ²	61	80 mm ²
COP path length, mm	27	40 mm
COP average velocity, mm/sec	3	4 mm/sec

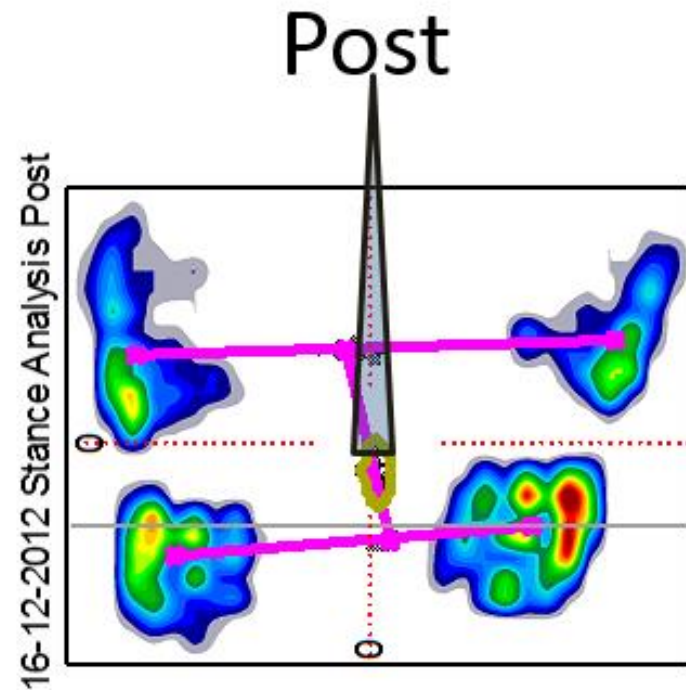
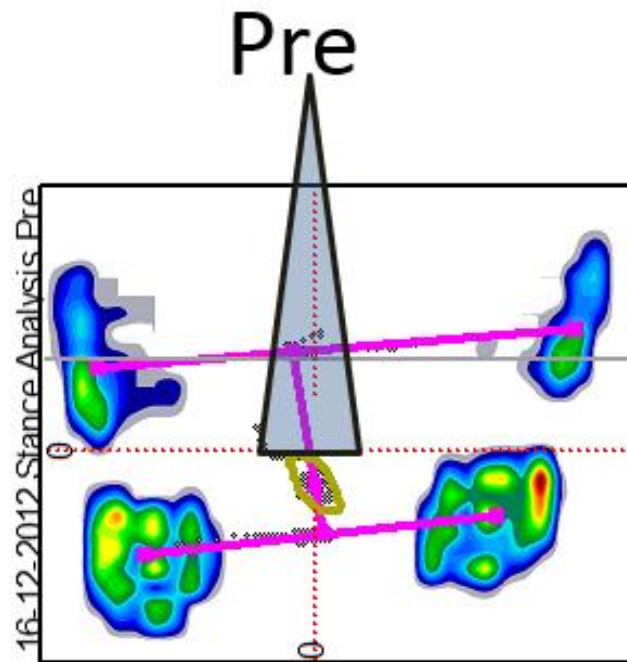
Static/Balance Parameters

Pre and Post Balance and Stability Training

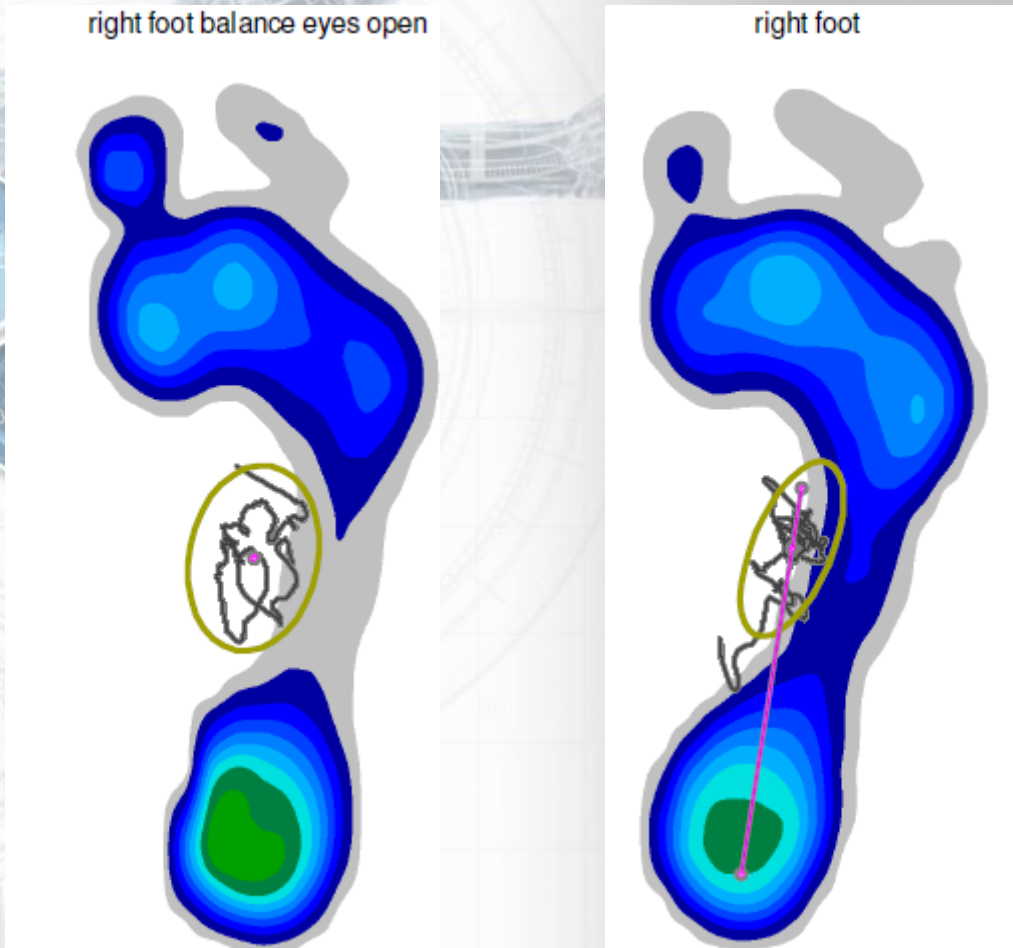


Static/Balance Parameters

Shod

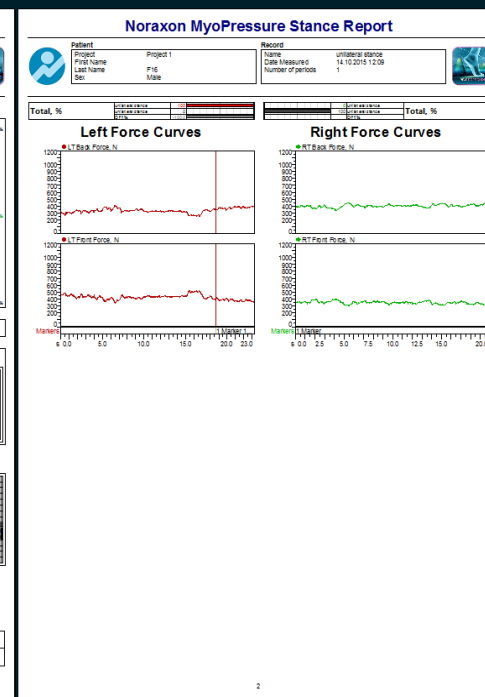
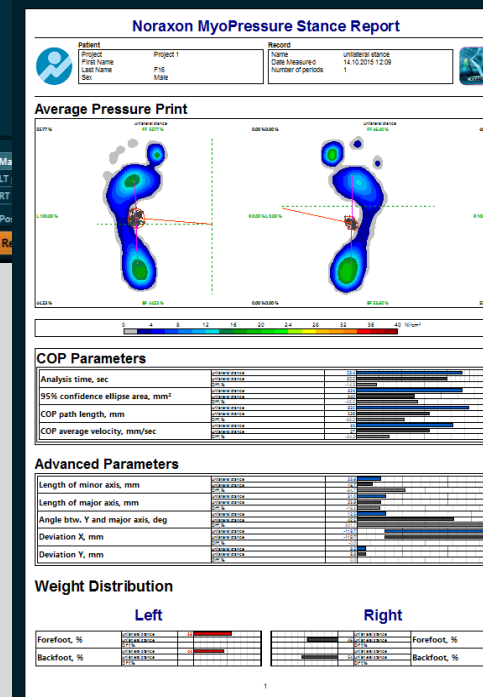
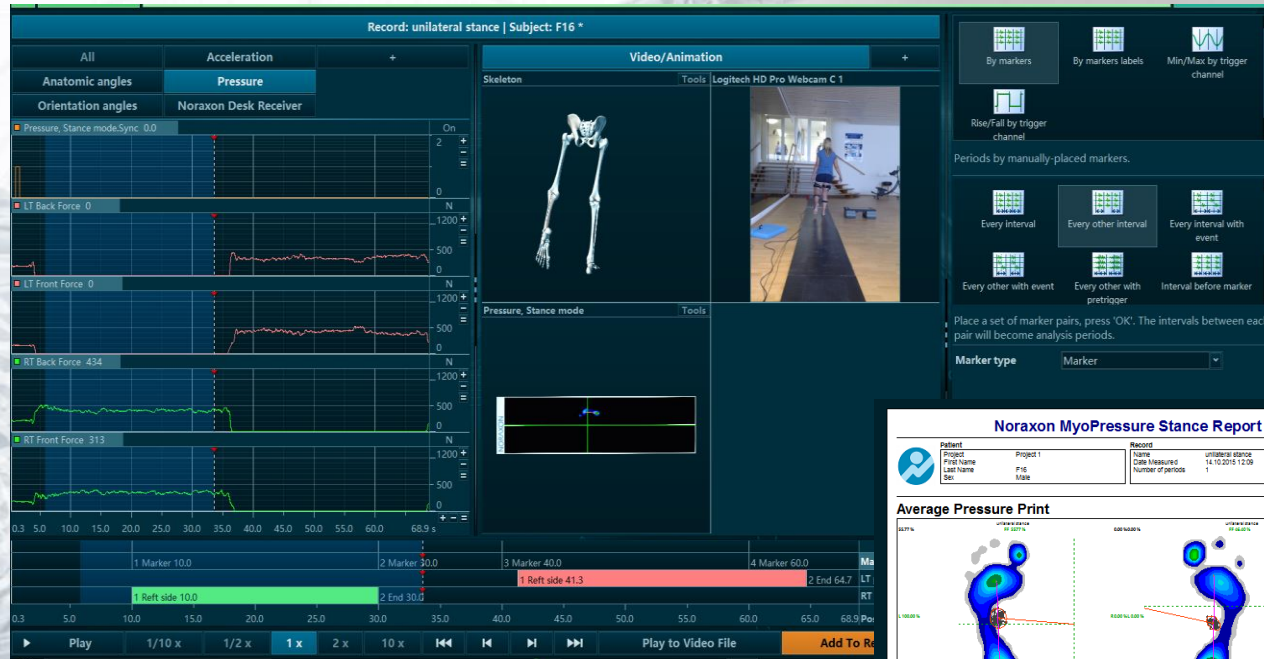


Uni-pedal Balance: Eyes open and Eyes Closed



- Anything less than 8:1 increase in COP area is considered normal (confidence area increases when eyes are closed).

Static/Balance Report Comparison: Left to Right or Eyes Open to Eyes Closed

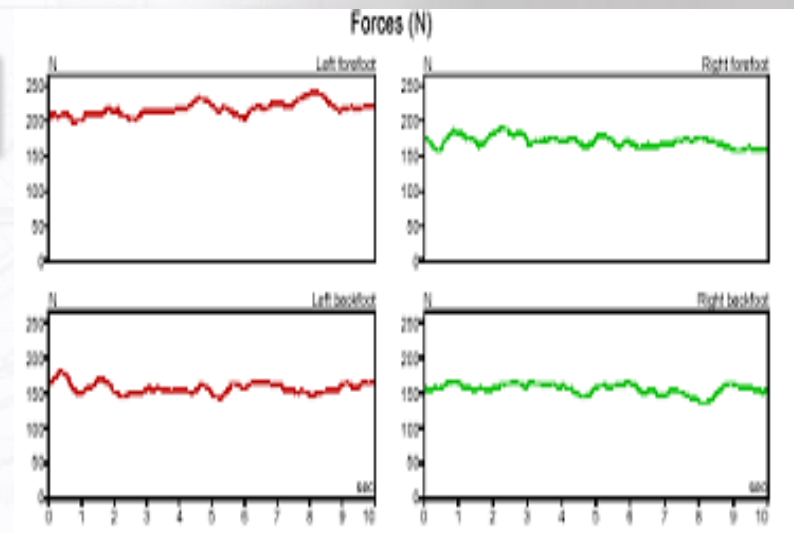


- Combines protocols into one smooth workflow
- One recording and report
- One all-encompassing report

Forces

The diagram “Forces (N)” displays in four force/time diagrams the vertical floor reaction force of the left/right forefoot and the left/right backfoot in relation to the measuring time.

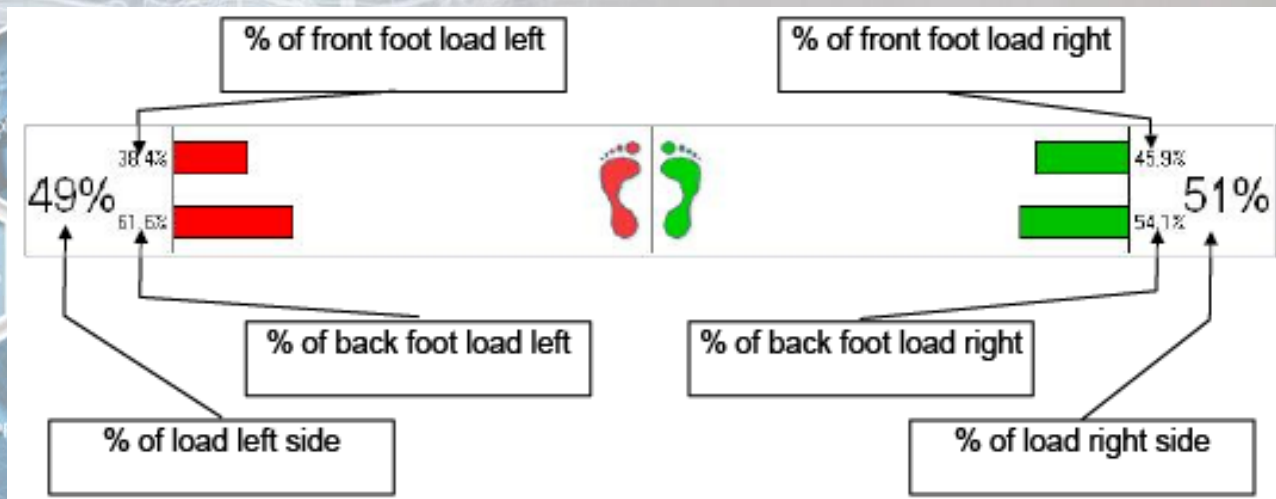
You can see where asymmetrical shifts or balance corrections were made and connect them to the causative action.



✓ Baseline stability and quality of controlled movement or balance.

Symmetry

The histogram represents the average % pressure distribution of the left and right fore and back foot and the % pressure distribution between left and right foot.



- ✓ This load distribution chart can be utilized for uni-pedal or other stance positions as well. The separations are based on the four quadrants of the area in contact.

“Normal” load distribution during upright standing without shoes.

- A perfect distribution is an equal distribution of 50 % for the right and left contact area.
- Ideally, the connecting line of left and right CoP (pink line) runs parallel to the heel contacts.
- The load distribution between back foot and fore foot **the fore foot should be loaded with approx. 1/3 (33%) and the back foot with approx. 2/3 (66%).**
- The **maximal load should not exceed 15 N/cm².**
- For sensitive feet (e.g. diabetics, etc.) the pressure should be less than 11 N/cm² in order to avoid any damage to the soles of the feet.

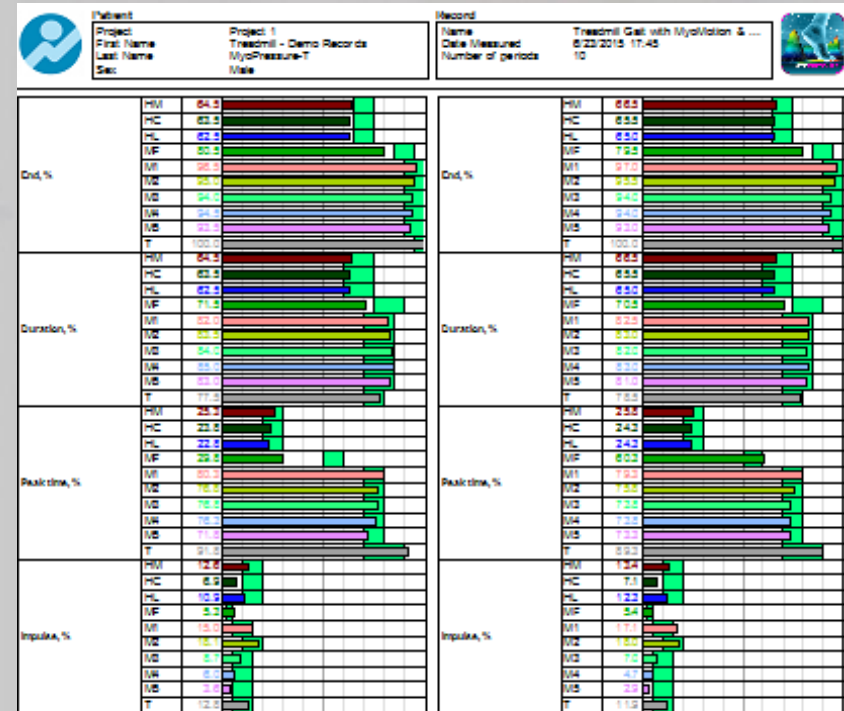
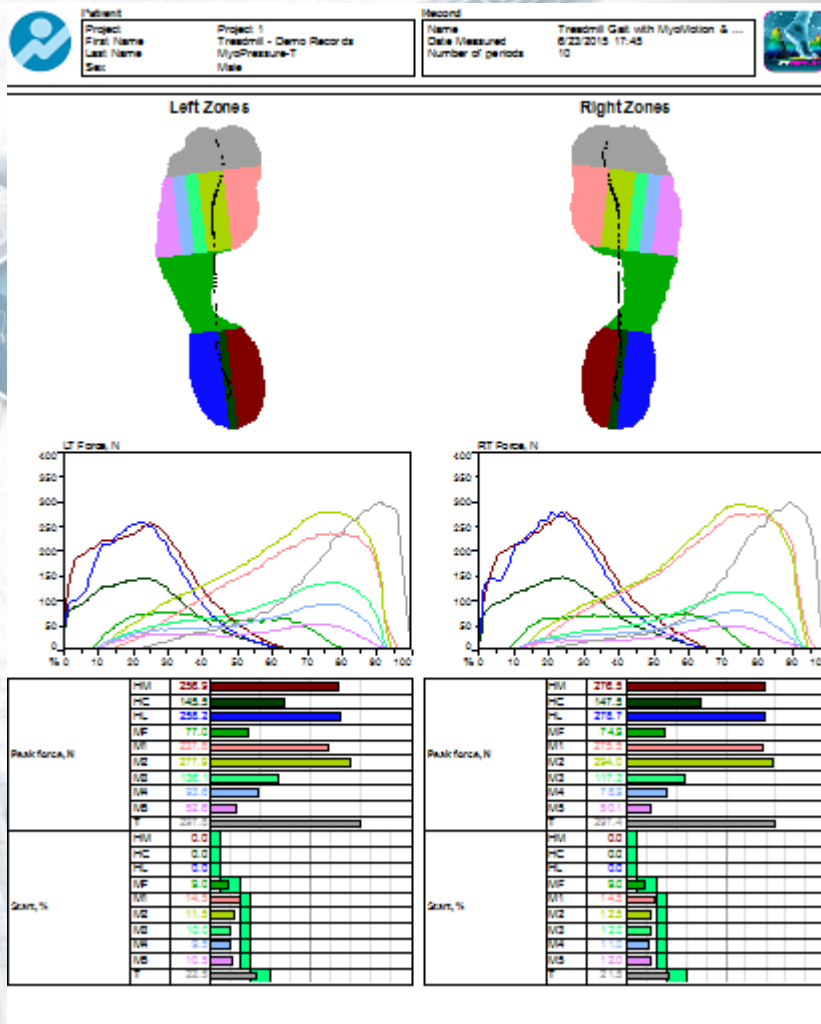


What is next?

DYNAMIC GAIT ADVANCED REPORT



Application examples: Why should we care?





3/7/10 zones

The 3/7/10 zones and new objective metrics are giving you the essential details about the dynamic movement your patients can't always describe or quantify.



3, 7, and 10 Zones

HM	Heel Medial	
HC	Heel Center	
HL	Heel Lateral	
MF	MidFoot	
M1	Met 1	
M2	Met 2	
M3	Met 3	
M4	Met 4	
M5	Met 5	
T	Toes	

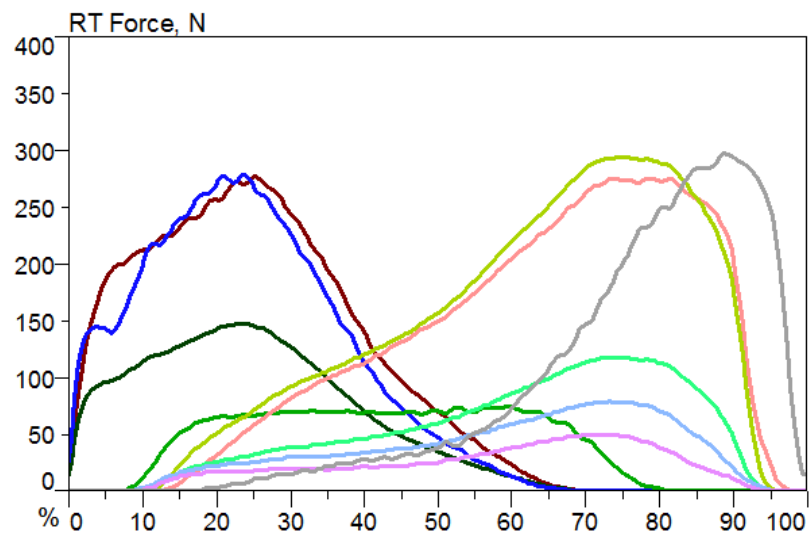
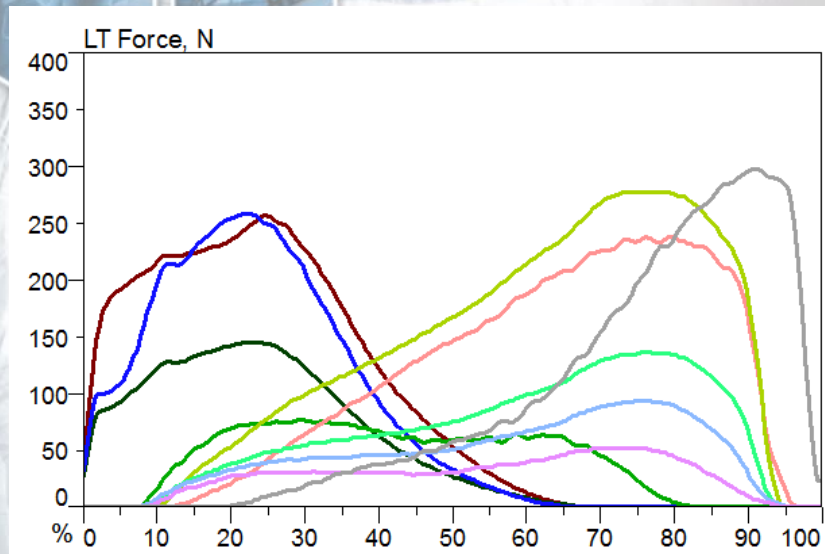
10 Zone Labeling

Zone separations based on active area and mathematically divided to assist in understanding impact specific locations, impact ordering and gait line characteristics as part of 100% stance.



3/7/10 zones

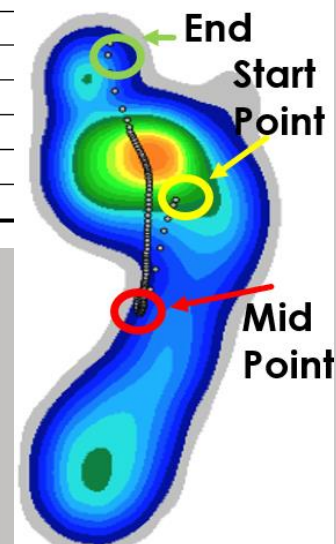
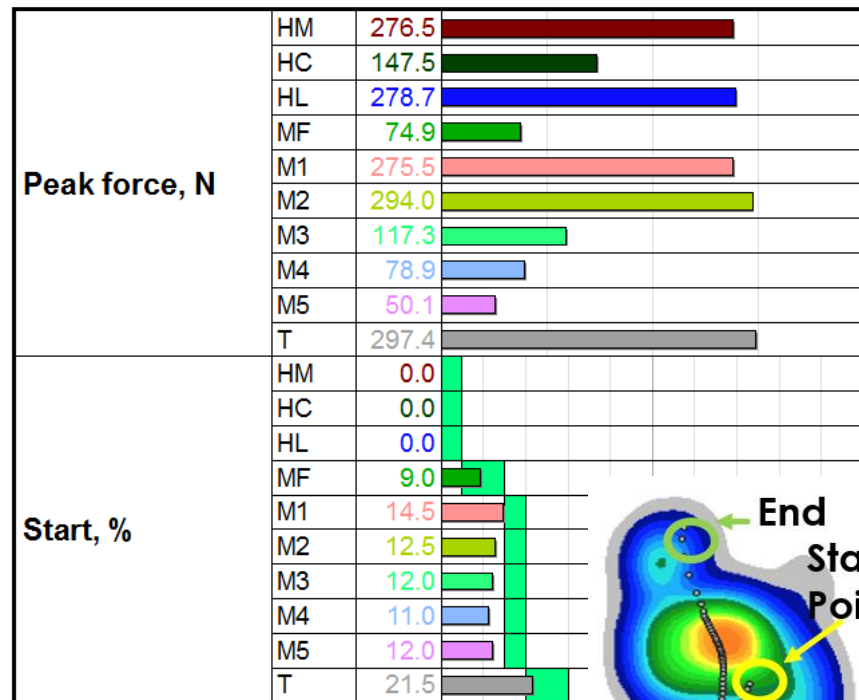
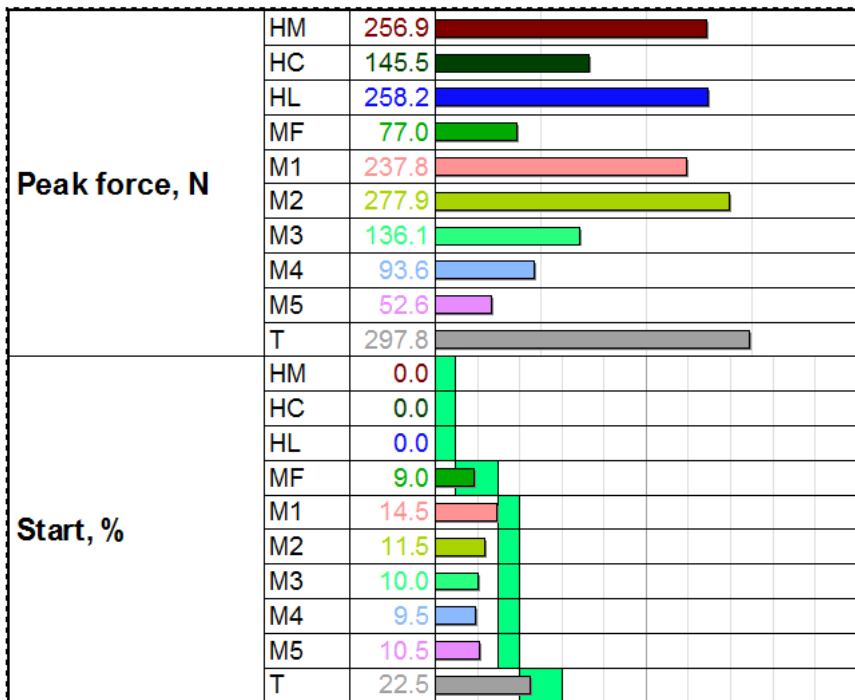
- Start times, End times, Durations, Peak time, and Impulses are available. The order of events in gait and the magnitudes of pressure and loading involved are all based on 100% of the stance phase.



By double clicking on the report element graph the amount of zones/curves can be changed between 3, 7 and 10



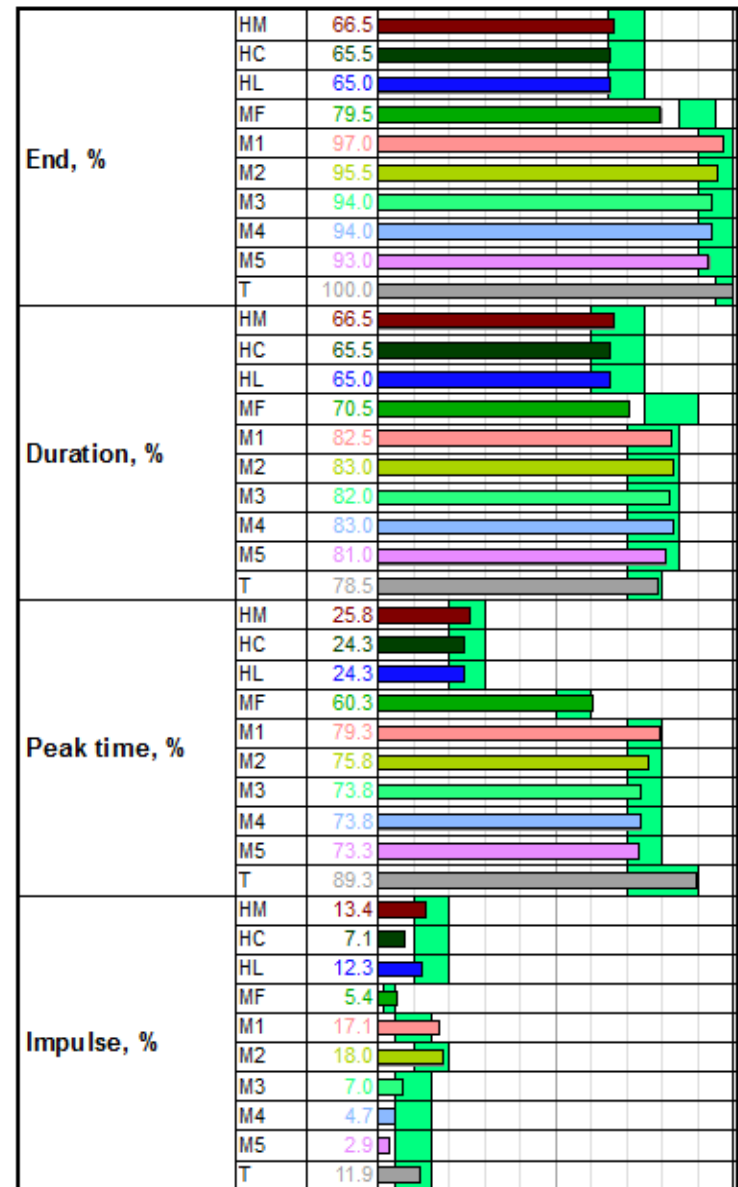
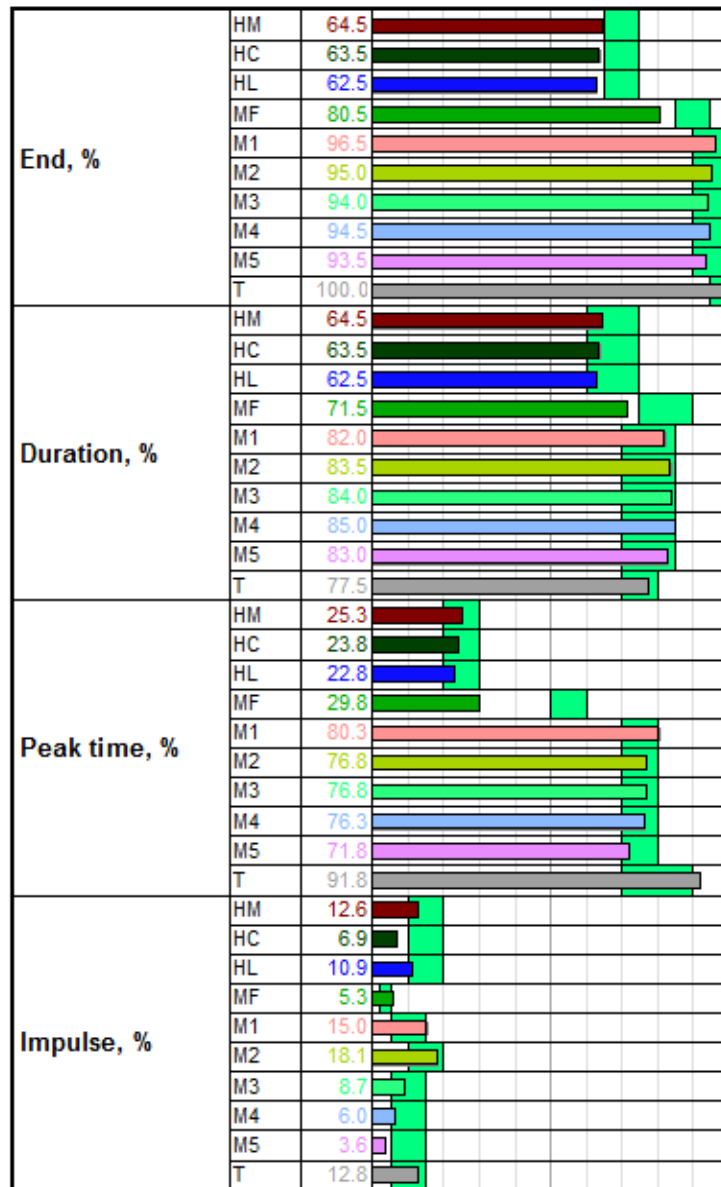
3/7/10 zones



- Where is the peak force?
- Where is the terminal force greatest?
- Could this be a problem?



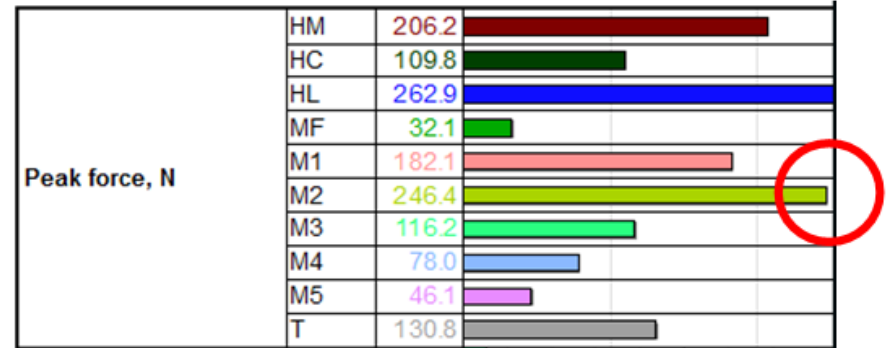
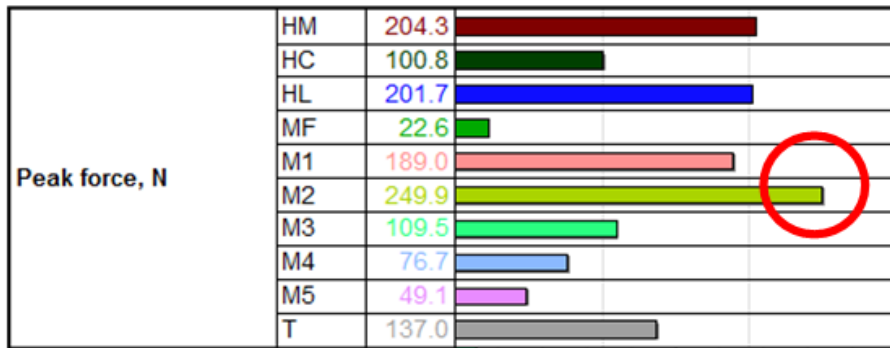
3/7/10 zones





Why Should We Care?

- How well patients' contacts with the ground are providing a stable base of support in stance phase
- What their forward progression of their mass over the ground contacts looks like
- Whether there is minimum energy expenditure, and lastly,
- Are they employing appropriate mechanisms for shock absorption and dissipation of forces?



Higher medial pressures and higher “peak” pressures under the metatarsals also are measured with larger “peak” forces in the medial zones (e.g. Metatarsals 1, 2, 3 and Toe)



3 Zone Applications

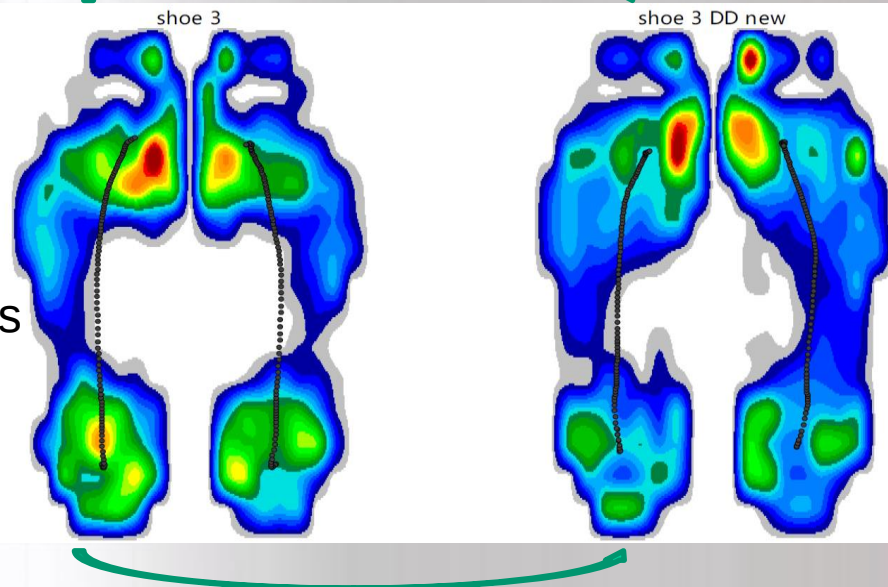
Less force in the mid-foot corresponds to the minimized time in mid-stance "tearing position". Shift to more symmetry left to right based on that area of the heel

Peak force, N	H	Record 1	583.6						
		Record 2	643.9						
	MF	Record 1	322.2						
		Record 2	418.1						
	F	Record 1	793.4						
		Record 2	771.9						

Peak force, N	H	Record 1	356.6						
		Record 2	370.1						
	MF	Record 1	241.4						
		Record 2	254.8						
	F	Record 1	575.3						
		Record 2	587.5						



Midstance forces 100N less





3 Zone Running Technique

225N decrease in the midfoot landing!!

Record 1



Record 2



Left Contact Training Results

Peak force, N	H	Record 1	187.0	
		Record 2	195.4	
	MF	Record 1	324.7	
		Record 2	425.2	
	F	Record 1	1206.5	
		Record 2	1118.2	
Impulse, %	H	Record 1	3.9	
		Record 2	3.8	
	MF	Record 1	16.1	
		Record 2	24.5	
	F	Record 1	80.0	
		Record 2	71.6	

Record 1



Record 2



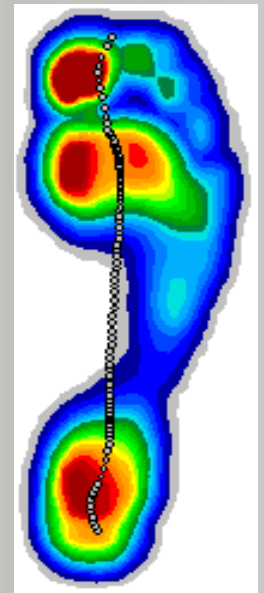
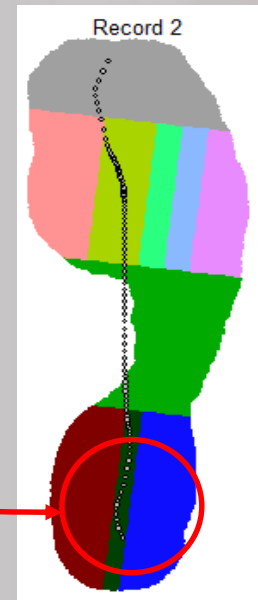
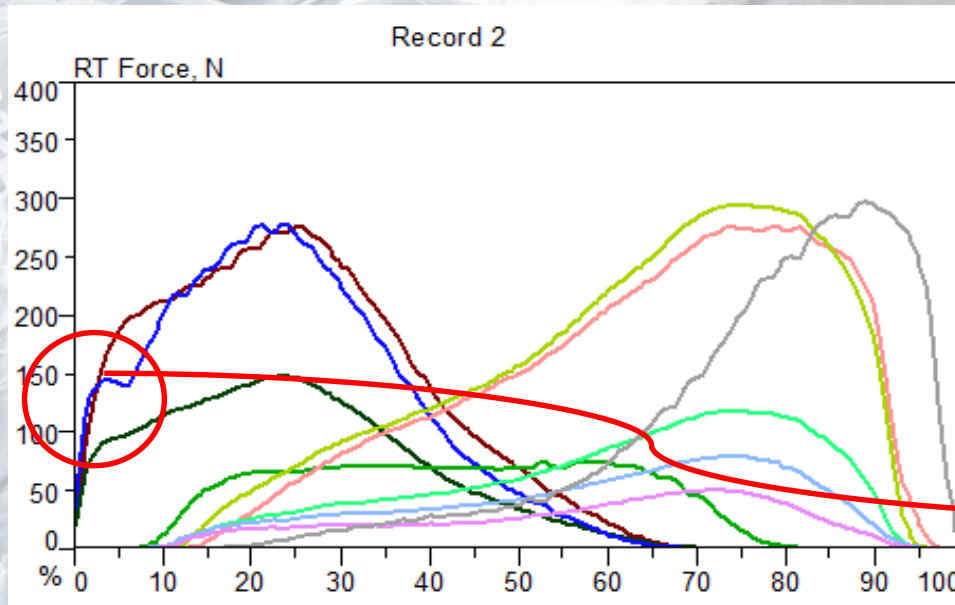
Right Contact Training Results

Peak force, N	H	Record 1	211.9	
		Record 2	225.8	
	MF	Record 1	275.4	
		Record 2	401.0	
	F	Record 1	1235.1	
		Record 2	1157.6	
Impulse, %	H	Record 1	4.7	
		Record 2	4.7	
	MF	Record 1	13.9	
		Record 2	22.6	
	F	Record 1	81.4	
		Record 2	72.6	



10 zone Application

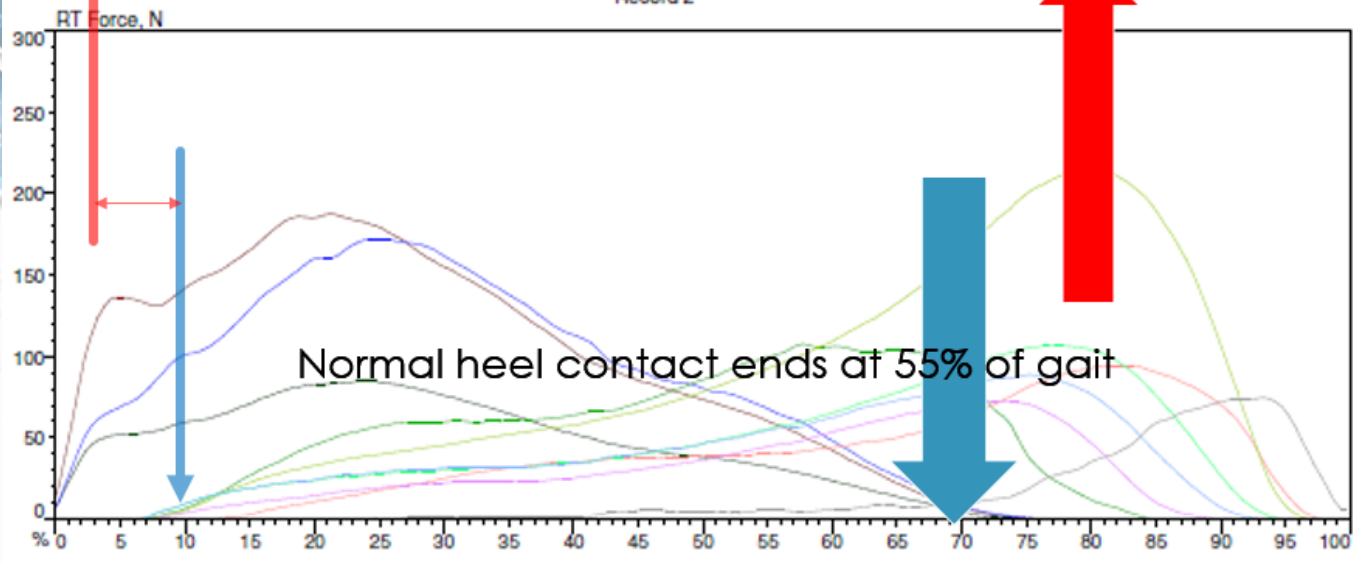
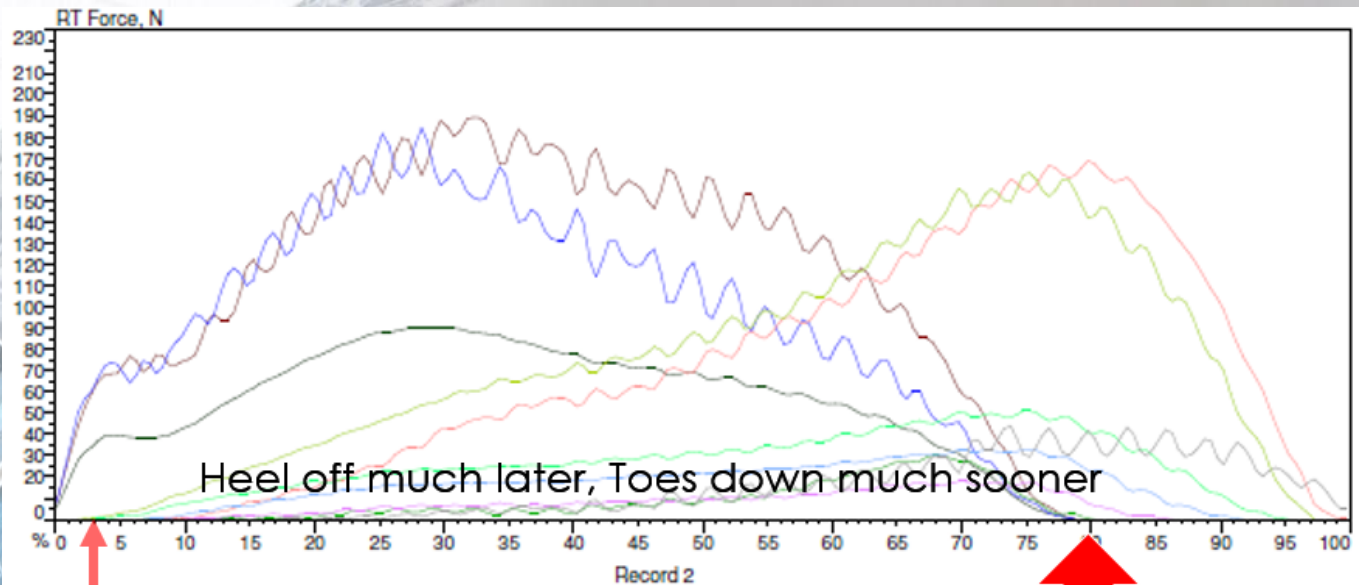
- Loading Rates, Rate of Deformations, Rate of Pronation



Contact beginning in the blue zone is consider landing in a **supinated position**, striking in the brown zone is a **pronated landing** and the central zone initial impact represents a **neutral landing**.



10 zone Application

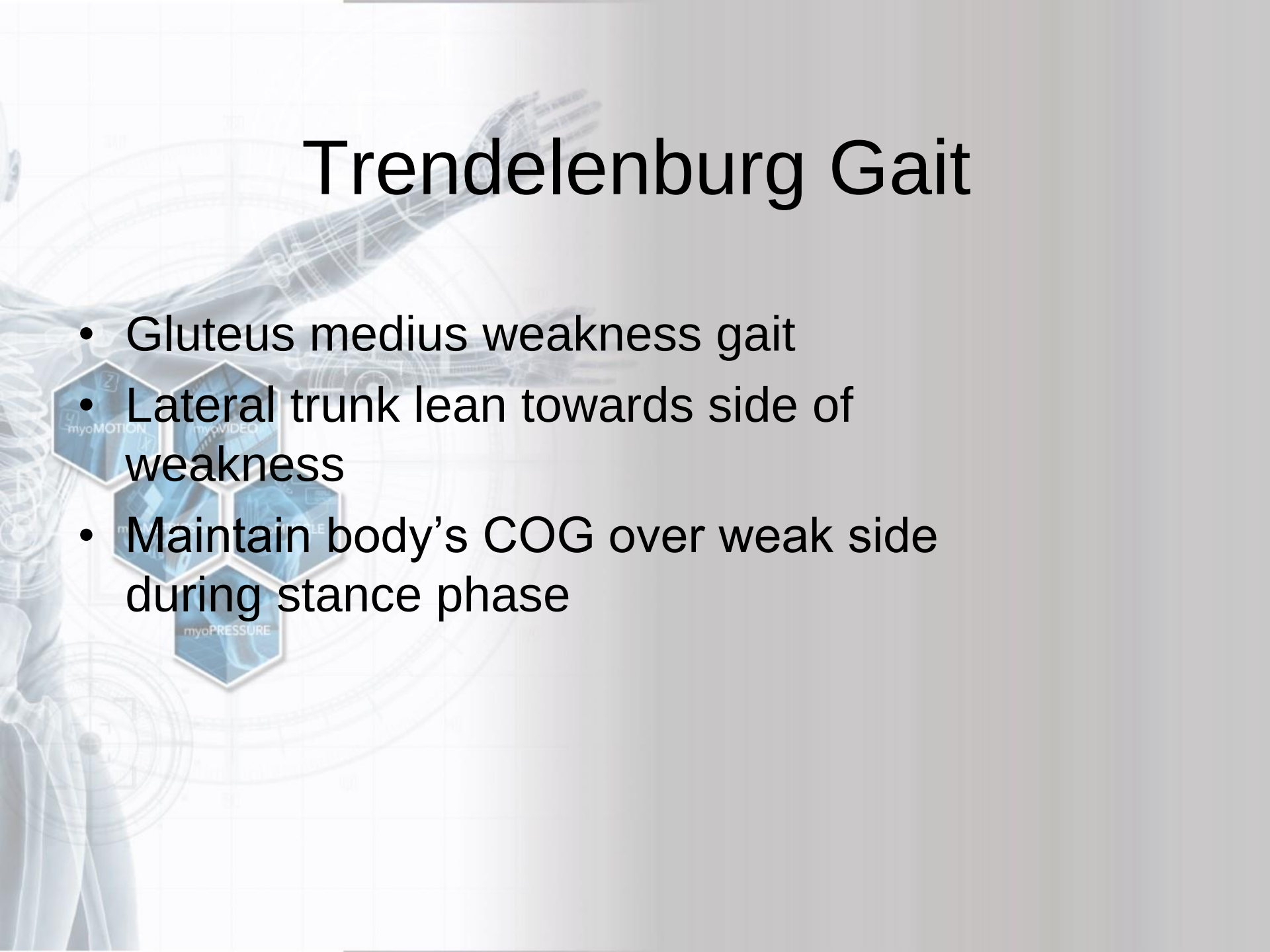


COMMON GAIT ABNORMALITIES:
A. Antalgic Gait

- Gait pattern in which stance phase on affected side is shortened
- Corresponding increase in stance on unaffected side
- Common causes: OA, Fx, tendinitis

Trendelenburg Gait

- Gluteus medius weakness gait
- Lateral trunk lean towards side of weakness
- Maintain body's COG over weak side during stance phase

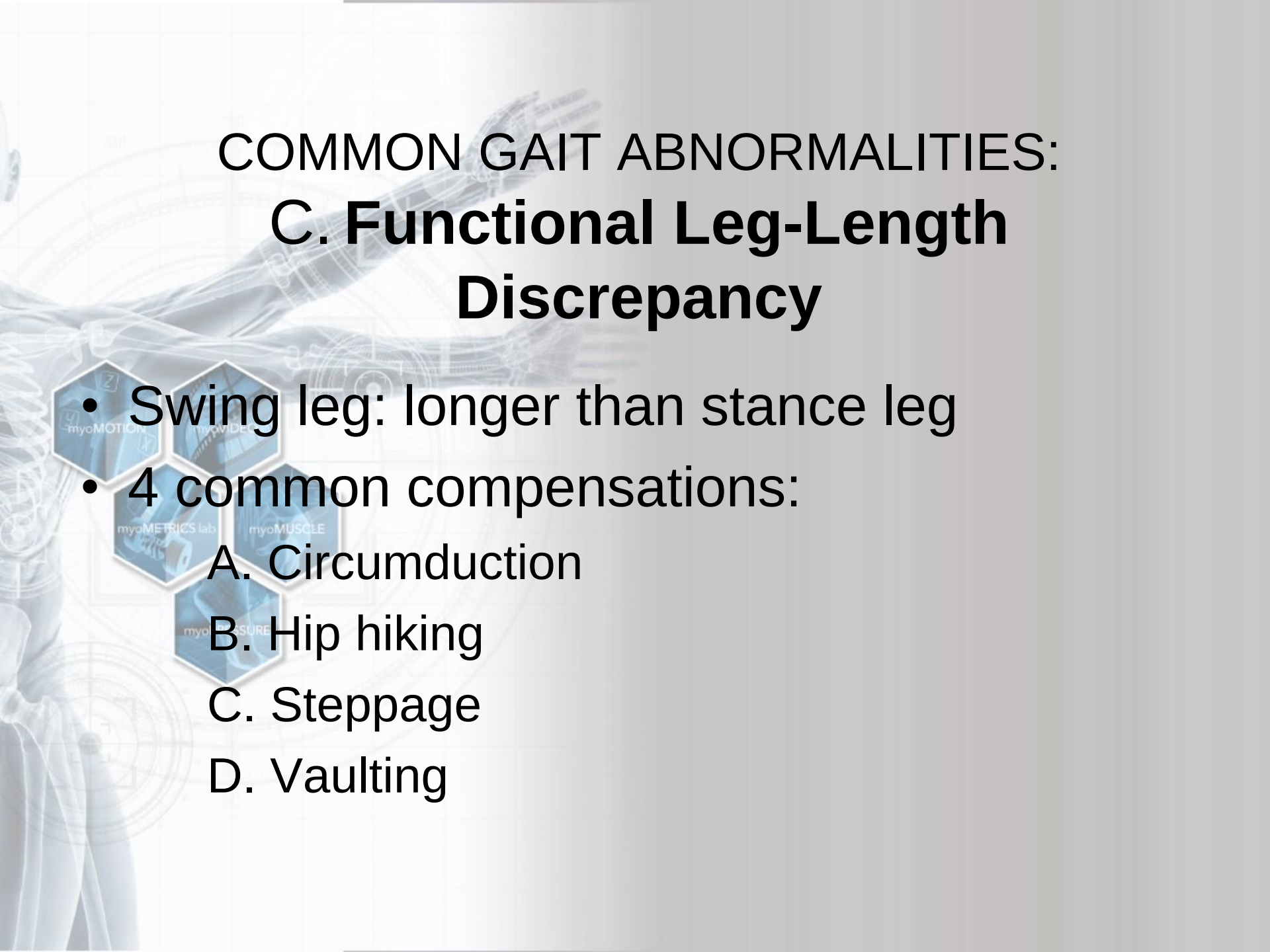


Flexed Knee Gait

- Flexed knees
- Flexed trunk posture
- No arm swing
- No initial contact
- No Toe-off
- No hip extension
- Short step
- Shortened stride
- COG stays within BOS
- Common in elderly with fear of falling

Antalgic Gait

- Painful leg gait
- Decreased stance time on painful leg
- Increased swing time on painful leg
- Decreased swing time on non-painful leg
- Increased stance time on non-painful leg



COMMON GAIT ABNORMALITIES:

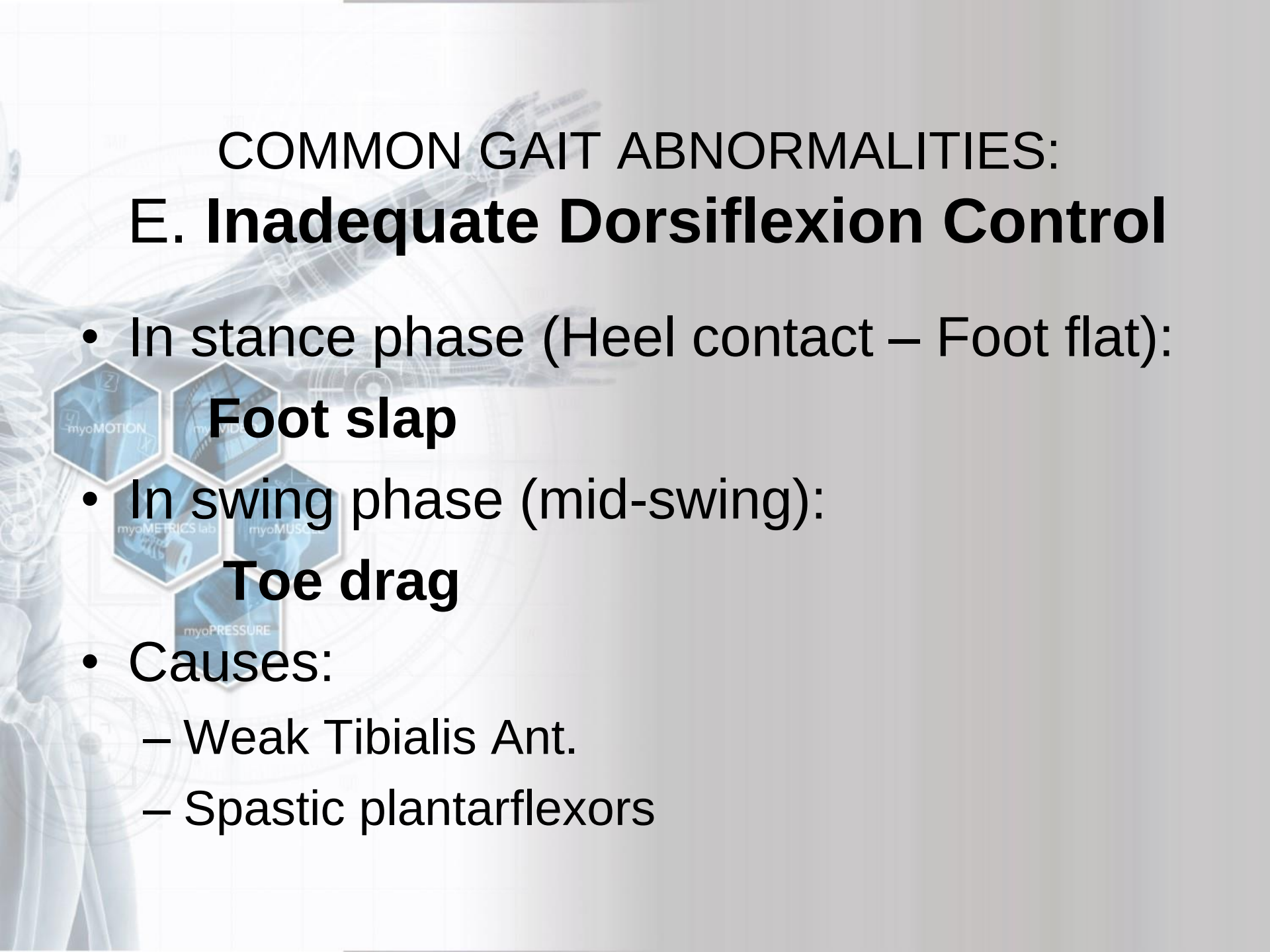
C. Functional Leg-Length Discrepancy

- Swing leg: longer than stance leg
- 4 common compensations:
 - A. Circumduction
 - B. Hip hiking
 - C. Steppage
 - D. Vaulting

COMMON GAIT ABNORMALITIES:

D. Increased Walking Base

- Normal walking base: 5-10 cm
- Common causes:
 - Deformities
 - Abducted hip
 - Valgus knee
 - Instability
 - Cerebellar ataxia
 - Proprioception deficits



COMMON GAIT ABNORMALITIES:

E. Inadequate Dorsiflexion Control

- In stance phase (Heel contact – Foot flat):

Foot slap

- In swing phase (mid-swing):

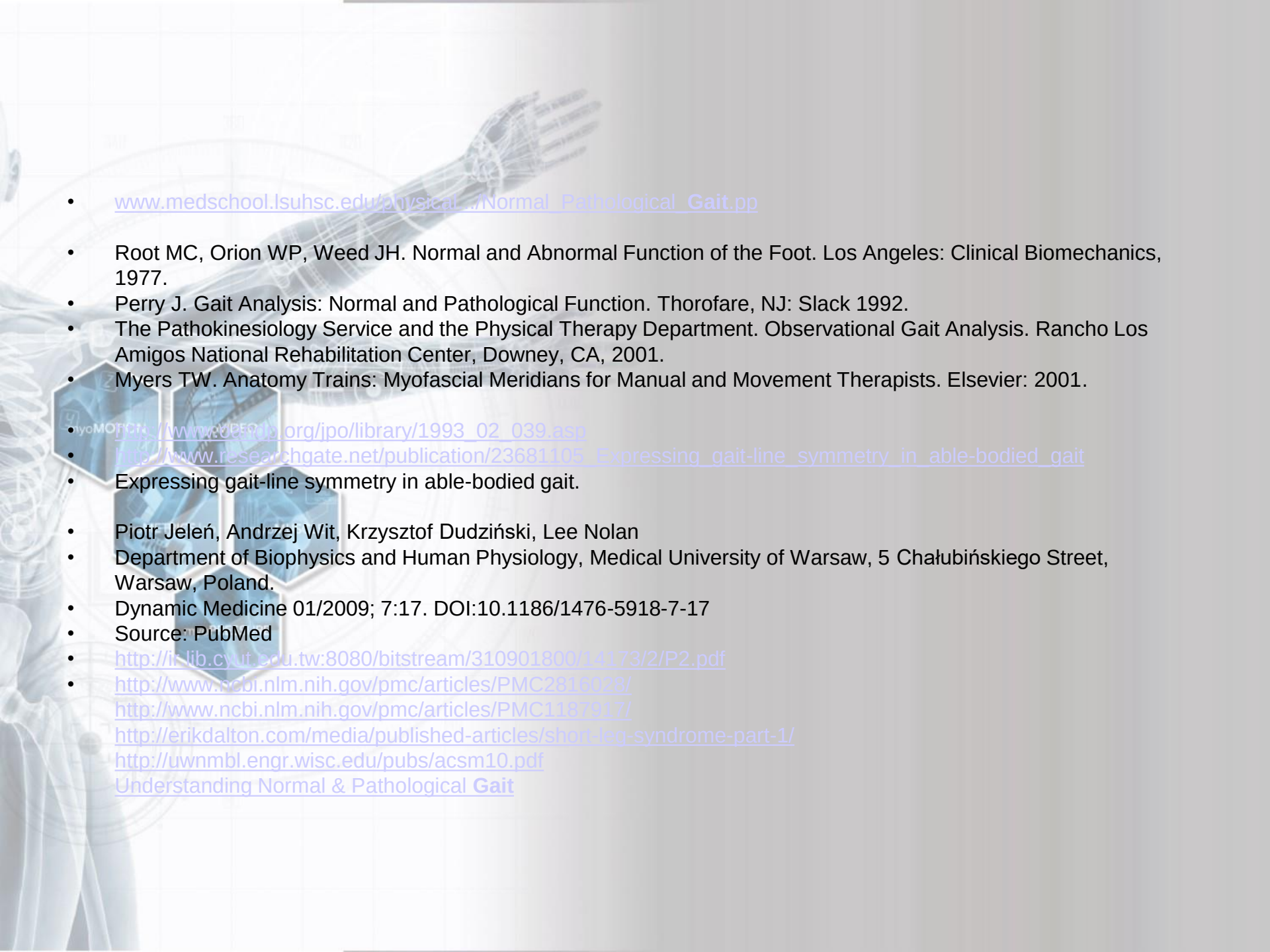
Toe drag

- Causes:
 - Weak Tibialis Ant.
 - Spastic plantarflexors

COMMON GAIT ABNORMALITIES:

F. **Excessive knee extension**

- Loss of normal knee flexion during stance phase
- Knee may go into hyperextension
- *Genu recurvatum*: hyperextension deformity of knee
- Common causes:
 - Quadriceps weakness (mid-stance)
 - Quadriceps spasticity (mid-stance)
 - Knee flexor weakness (end-stance)

- 
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 - <http://uwnmbi.engr.wisc.edu/pubs/acsm10.pdf>
 - [Understanding Normal & Pathological Gait](#)

- **Heel Strike-Early Flatfoot**

- The anterior compartment is most active, which means that the tibialis anterior muscle, the extensor hallucis longus and the extensor digitorum longus work to gently lower the foot onto the ground. If these muscles do not work such as would be the case in someone with a [drop foot](#), the foot will tend to slap onto the ground when it lands.

- **Late Flatfoot-Heel Rise**

- As the body's center of gravity passes over the foot the posterior compartment muscles begin to contract. This contraction of the calf muscle serves to control the body movement as it goes forward so that the body does not fall forward. During this phase of gait (late flatfoot) the calf muscle is strongly contracting and lengthening. This is called an "eccentric" muscle contraction and it serves to generate an extraordinary amount of internal force within the calf muscle and Achilles tendon. This is why most [Achilles tendon ruptures](#) and [calf tears](#) occur during this stage gait.

- **Heel Rise-Toe Off**

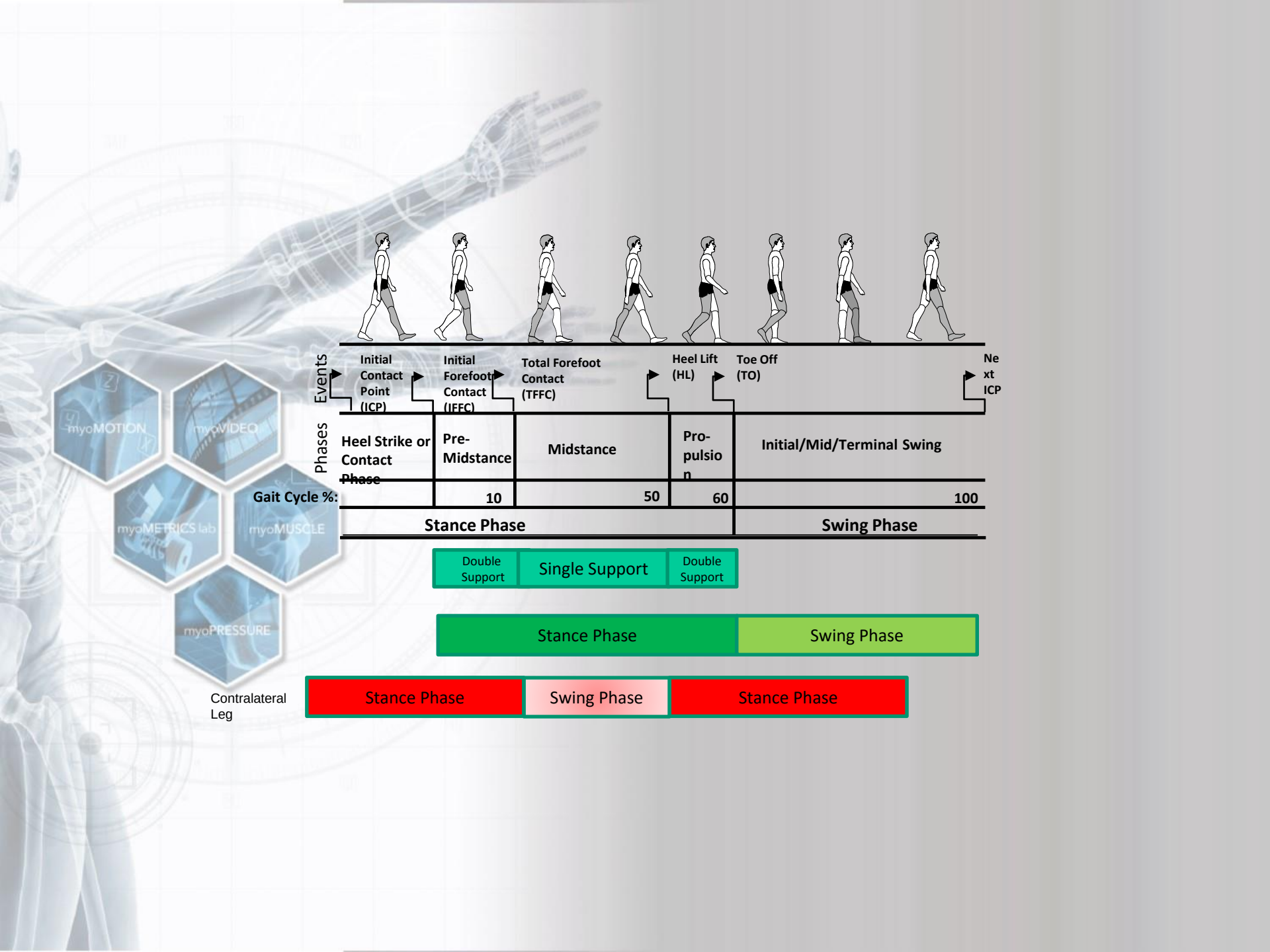
- During the heel rise phase, the calf muscle continues to contract, but is now shortening rather than lengthening (performing a concentric contraction). This lasts until the toe off phase.

- **Antalgic Gait (painful walking)**

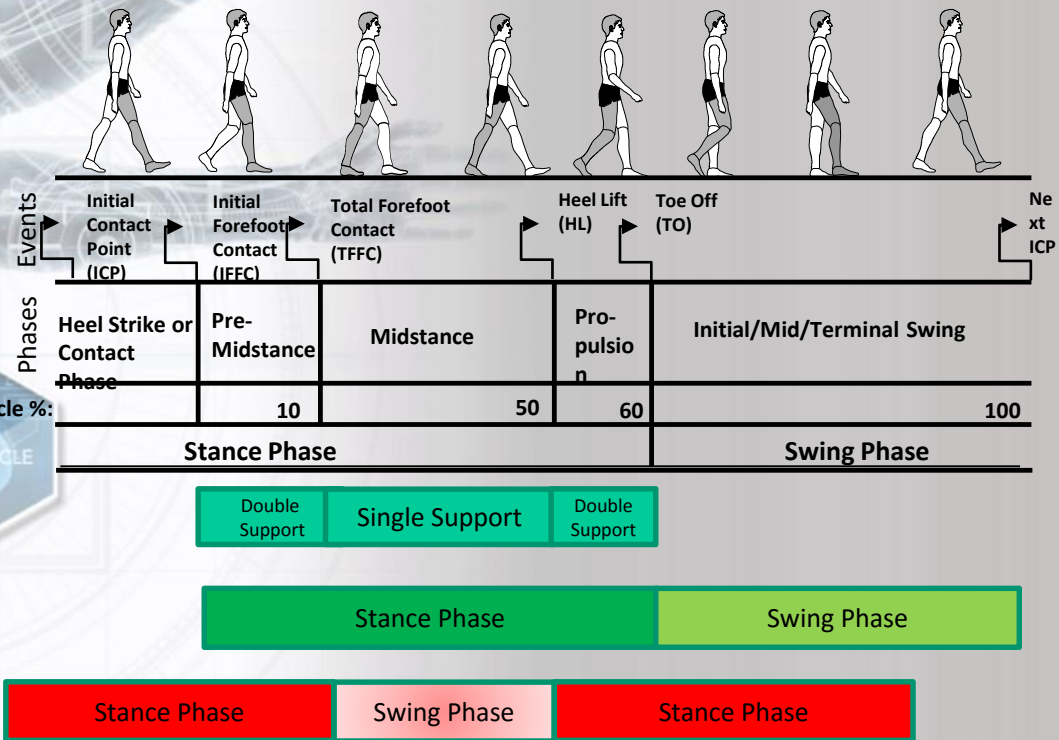
- The primary sign of an antalgic or painful gait is the reduced amount of time spent in the stance phase. This is because people do not want to spend any more time than necessary on a foot that is causing them pain. While the stance phase is usually divided equally between the two legs, someone with a painful foot will spend substantially less "time" on the injured foot, perhaps only 20-30% of their gait rather than 50%. Another sign of painful gait is a decreased stride length, which results from patients not wanting to push off from their painful foot as powerfully as normal. So, one stride tends to be much longer than the other.

- **High Steppage Gait**

- Another example of an abnormal gait is a high-steppage gait pattern. This form of walking is seen in patients whose anterior compartment muscles do not function normally (ex. Patients with a [drop foot](#)). This lack of anterior muscle compartment functioning causes the foot to slap onto the ground during the heel strike phase of walking. Patients normally respond to this problem by



Contralateral Leg





Events	Heel strike					Toe off		2. Heel strike	
	Initial Contact	Loding Response	Mid-Stance	Terminal Stance	Pre-Swing	Initial Swing	Mid-Swing	Terminal Swing	
Phases	Gait Cylce %:	12	31	50	62	75	87	100	
Stances Period						Swing Period			

